MATHEMATICS 7

Solved Exercises

Balochistan Textbook Board

Complete Book - All 12 Units

Balochistan Textbook Board (BTBB)

Unit 1: Rational Numbers

Exercise 1.1 (p.12)

1. Represent the following numbers on a number line.

i. 5, 10, and 15

Draw a number line with equal steps of 5 units. Mark 0, then count 5, 10, 15 to the right of 0 and label each mark.

ii. 8, 12, and 20

Draw a number line with equal steps of 4 units. Mark 0, then 8, 12, and 20 to the right of 0 and label each mark.

iii. 3, 6, 9, and 12

Draw a number line with equal steps of 3 units. Mark 0, then 3, 6, 9, and 12 to the right of 0 and label each mark.

iv. -3, 0, and 4

Draw a number line with steps of 1 unit. Mark 0, then 4 to the right of 0 and -3 to the left of 0, and label each mark.

v. -5, -2, 3, and 6

Draw a number line with steps of 1 unit. Mark 0, then 3 and 6 to the right of 0, and -5 and -2 to the left of 0, and label each mark.

vi. -10, -6, -2, 2, and 8

Draw a number line with equal steps of 2 units. Mark 0, then 2 and

8 to the right of 0, and -10, -6, and -2 to the left of 0, and label each mark.

vii. (printed as a second vi.) 1/2, 3/2, and 5/2

Draw a number line from 0 to 3 with each whole unit divided into 2 equal parts. Mark 1/2 after 0, 3/2 (1 1/2) after 1, and 5/2 (2 1/2) after 2, and label each mark. Note: the book prints this item as a second vi.; it is the seventh item in order.

viii. 1/4, 3/4, and 5/4

Draw a number line from 0 to 2 with each whole unit divided into 4 equal parts. Mark 1/4, then 3/4, then 5/4 (1 1/4) after 1, and label each mark.

ix. -3/5, -1/5, 1/5, and 3/5

Draw a number line from -1 to 1 with each whole unit divided into 5 equal parts. Mark -3/5 and -1/5 to the left of 0, and 1/5 and 3/5 to the right of 0, and label each mark.

x. 0.5, 1.5, and 2.5

Draw a number line from 0 to 3 with each whole unit divided into 2 equal parts (each part = 0.5). Mark 0.5, 1.5, and 2.5 and label each mark.

xi. -1.2, 0, and 2.4

Draw a number line with each whole unit divided into 5 equal parts (each part = 0.2). Mark 0, then 2.4 to the right of 0 and -1.2 to the left of 0, and label each mark.

xii. -2.5, -1.5, -0.5, 0.5

Draw a number line from -3 to 1 with each whole unit divided into 2 equal parts (each part = 0.5). Mark -2.5, -1.5, and -0.5 to the left of 0, and 0.5 to the right of 0, and label each mark.

2. Find the place value of each digit from the following numbers.

i. 87,364

8 is in the ten thousands place = 8 x 10000 = 80000.

7 is in the thousands place = 7 x 1000 = 7000.

3 is in the hundreds place = 3 x 100 = 300.

6 is in the tens place = 6 x 10 = 60.

4 is in the ones place = 4 x 1 = 4.

Place values: 80000, 7000, 300, 60, 4.

ii. 504,321

5 is in the hundred thousands place = 5 x 100000 = 500000.

0 is in the ten thousands place = 0 x 10000 = 0.

4 is in the thousands place = 4 x 1000 = 4000.

3 is in the hundreds place = 3 x 100 = 300.

2 is in the tens place = 2 x 10 = 20.

1 is in the ones place = 1 x 1 = 1.

Place values: 500000, 0, 4000, 300, 20, 1.

iii. 2,385

2 is in the thousands place = 2 x 1000 = 2000.

3 is in the hundreds place = 3 x 100 = 300.

8 is in the tens place = 8 x 10 = 80.

5 is in the ones place = 5 x 1 = 5.

Place values: 2000, 300, 80, 5.

iv. 875.34

8 is in the hundreds place = 8 x 100 = 800.

7 is in the tens place = 7 x 10 = 70.

5 is in the ones place = 5 x 1 = 5.

3 is in the tenths place = 3 x 1/10 = 0.3.

4 is in the hundredths place = 4 x 1/100 = 0.04.

Place values: 800, 70, 5, 0.3, 0.04.

v. 680.984

6 is in the hundreds place = 6 x 100 = 600.

8 is in the tens place = 8 x 10 = 80.

0 is in the ones place = 0 x 1 = 0.

9 is in the tenths place = 9 x 1/10 = 0.9.

8 is in the hundredths place = 8 x 1/100 = 0.08.

4 is in the thousandths place = 4 x 1/1000 = 0.004.

Place values: 600, 80, 0, 0.9, 0.08, 0.004.

3. Find the place value of the indicated (underlined) digit in the following numbers.

i. 4537 (underlined digit 5)

The underlined digit 5 is in the hundreds place.

Place value = 5 x 100 = 500.

500.

ii. 2683 (underlined digit 6)

The underlined digit 6 is in the hundreds place.

Place value = 6 x 100 = 600.

600.

iii. 9247 (underlined digit 2)

The underlined digit 2 is in the hundreds place.

Place value = 2 x 100 = 200.

200.

iv. 52013 (underlined digit 5)

The underlined digit 5 is in the ten thousands place.

Place value = 5 x 10000 = 50000.

50000.

v. 75912 (underlined digit 5)

The underlined digit 5 is in the thousands place.

Place value = 5 x 1000 = 5000.

5000.

Exercise 1.2 (p.17)

1. Round off the following whole numbers to the given degree of accuracy.

(i) 4567 to 2 significant figures

First two significant digits are 4 and 5.

The digit after them is 6, which is 5 or more, so increase the second digit by 1: 5 + 1 = 6.

Replace the remaining digits with zeros.

4600.

(ii) 98765 to 3 significant figures

First three significant digits are 9, 8, and 7.

The digit after them is 6, which is 5 or more, so increase the third digit by 1: 7 + 1 = 8.

Replace the remaining digits with zeros.

98800.

(iii) 3402 to 2 significant figures

First two significant digits are 3 and 4.

The digit after them is 0, which is less than 5, so keep the second digit as it is.

Replace the remaining digits with zeros.

3400.

(iv) 123456 to 3 significant figures

First three significant digits are 1, 2, and 3.

The digit after them is 4, which is less than 5, so keep the third digit as it is.

Replace the remaining digits with zeros.

123000.

(v) 789 to 2 significant figures

First two significant digits are 7 and 8.

The digit after them is 9, which is 5 or more, so increase the second digit by 1: 8 + 1 = 9.

Replace the remaining digit with zero.

790.

(vi) 1568 to 3 significant figures

First three significant digits are 1, 5, and 6.

The digit after them is 8, which is 5 or more, so increase the third

digit by 1: 6 + 1 = 7.

Replace the remaining digit with zero.

1570.

2. Round off the following integers to the given degree of accuracy.

(i) -8234 to 2 significant figures

Ignore the minus sign while rounding; it is restored at the end.

First two significant digits are 8 and 2.

The digit after them is 3, which is less than 5, so keep the second digit as it is.

Replace the remaining digits with zeros and restore the minus sign.

-8200.

(ii) 54768 to 3 significant figures

First three significant digits are 5, 4, and 7.

The digit after them is 6, which is 5 or more, so increase the third digit by 1: 7 + 1 = 8.

Replace the remaining digits with zeros.

54800.

(iii) -235 to 2 significant figures

Ignore the minus sign while rounding; it is restored at the end.

First two significant digits are 2 and 3.

The digit after them is 5, which is 5 or more, so increase the second digit by 1: 3 + 1 = 4.

Replace the remaining digit with zero and restore the minus sign.

-240.

(iv) 67890 to 3 significant figures

First three significant digits are 6, 7, and 8.

The digit after them is 9, which is 5 or more, so increase the third digit by 1: 8 + 1 = 9.

Replace the remaining digits with zeros.

67900.

(v) 4123 to 2 significant figures

First two significant digits are 4 and 1.

The digit after them is 2, which is less than 5, so keep the second digit as it is.

Replace the remaining digits with zeros.

4100.

(vi) 76321 to 3 significant figures

First three significant digits are 7, 6, and 3.

The digit after them is 2, which is less than 5, so keep the third digit as it is.

Replace the remaining digits with zeros.

76300.

3. Round off the following rational numbers to the stated significant figures or decimal places.

(i) 7/2 to 2 significant figures

Convert to a decimal: 7/2 = 3.5.

3.5 already has 2 significant figures.

3.5.

(ii) 13/4 to 3 significant figures

Convert to a decimal: 13/4 = 3.25.

3.25 already has 3 significant figures.

3.25.

(iii) 23/9 to 3 significant figures

Convert to a decimal: 23/9 = 2.555...

First three significant digits are 2, 5, and 5.

The digit after them is 5, which is 5 or more, so increase the third digit by 1: 5 + 1 = 6.

2.56.

(iv) 10/3 to 2 decimal places

Convert to a decimal: 10/3 = 3.333...

Keep 2 decimal places: 3.33.

The digit after the second decimal place is 3, which is less than 5, so the second decimal digit stays 3.

3.33.

(v) 25/8 to 3 decimal places

Convert to a decimal: 25/8 = 3.125.

3.125 already has 3 decimal places.

3.125.

(vi) 9/4 to 2 decimal places

Convert to a decimal: 9/4 = 2.25.

2.25 already has 2 decimal places.

2.25.

4. Round off the following decimals to the given degree of accuracy.

(i) 17.8569 to 2 significant figures

First two significant digits are 1 and 7.

The digit after them is 8, which is 5 or more, so increase the second digit by 1: 7 + 1 = 8.

Drop the remaining digits.

18.

(ii) 0.004567 to 3 significant figures

Leading zeros are not significant; the first significant digit is 4.

First three significant digits are 4, 5, and 6.

The digit after them is 7, which is 5 or more, so increase the third digit by 1: 6 + 1 = 7.

0.00457.

(iii) 9.87654 to 3 significant figures

First three significant digits are 9, 8, and 7.

The digit after them is 6, which is 5 or more, so increase the third digit by 1: 7 + 1 = 8.

9.88.

(iv) 0.004567 to 3 decimal places

Keep 3 decimal places: 0.004.

The digit after the third decimal place is 5, which is 5 or more, so increase the third decimal digit by 1: 4 + 1 = 5.

So 0.004 becomes 0.005.

0.005.

(v) 123.987 to 2 decimal places

Keep 2 decimal places: 123.98.

The digit after the second decimal place is 7, which is 5 or more, so increase the second decimal digit by 1: 8 + 1 = 9.

123.99.

(vi) 0.032145 to 2 decimal places

Keep 2 decimal places: 0.03.

The digit after the second decimal place is 2, which is less than 5, so the second decimal digit stays 3.

0.03.

Note on Exercise 1.2 Q4: the book's answer key as scanned gives different values for Q4 (i), (iii), (iv), and (vi); the answers above follow the unit's own rounding rules step by step (for example 9.87654 to 3 significant figures is 9.88, and 0.032145 to 2 decimal places is 0.03).

Exercise 1.3 (pp.23-24)

1. Find the HCF of 36 and 48

Factors of 36: 1, 2, 3, 4, 6, 9, 12, 18, 36.

Factors of 48: 1, 2, 3, 4, 6, 8, 12, 16, 24, 48.

Common factors: 1, 2, 3, 4, 6, 12.

The highest common factor is 12.

HCF = 12.

2. Find the LCM of 15 and 20

Multiples of 15: 15, 30, 45, 60, 75, ...

Multiples of 20: 20, 40, 60, 80, ...

The first common multiple is 60.

LCM = 60.

3. Find the HCF and LCM of 42 and 56

Prime factors: 42 = 2 x 3 x 7; 56 = 2 x 2 x 2 x 7.

HCF uses the common prime factors with the lowest powers: 2 x 7 = 14.

LCM uses every prime factor with the highest power: 2 x 2 x 2 x 3 x 7 = 168.

HCF = 14, LCM = 168.

4. Identify the following fractions (Proper, Improper, or Mixed).

(i) 5/8

Proper fraction: the numerator (5) is smaller than the denominator (8).

(ii) 9/17

Proper fraction: the numerator (9) is smaller than the denominator (17).

(iii) 12/5

Improper fraction: the numerator (12) is greater than the

denominator (5).

(iv) 28/11

Improper fraction: the numerator (28) is greater than the denominator (11).

(v) 24/32

Proper fraction: the numerator (24) is smaller than the denominator (32).

(vi) 5 2/9

Mixed fraction: a whole number (5) with a proper fraction (2/9).

(vii) 3 7/4

Mixed fraction in form: a whole number (3) with a fraction (7/4). Note that 7/4 itself is improper, so 3 7/4 = 4 3/4 in lowest mixed form.

(viii) 3/16

Proper fraction: the numerator (3) is smaller than the denominator (16).

5. Convert fractions as directed.

(i) 17/6 into a mixed fraction

Divide 17 by 6: 17 = 6 x 2 + 5.

The quotient 2 is the whole part and the remainder 5 stays over 6.

2 5/6.

(ii) 2 3/4 into an improper fraction

Multiply the whole part by the denominator and add the numerator:

2 x 4 + 3 = 11.

Keep the denominator 4.

11/4.

(iii) 4 7/10 into an improper fraction

Multiply the whole part by the denominator and add the numerator:

4 x 10 + 7 = 47.

Keep the denominator 10.

47/10.

(iv) 8/5 into a decimal

Divide 8 by 5: 8/5 = 1.6.

1.6.

(v) 18/13 into a mixed fraction

Divide 18 by 13: 18 = 13 x 1 + 5.

The quotient 1 is the whole part and the remainder 5 stays over 13.

1 5/13.

6. Compare using symbols (=, <, >, <=, >=).

(i) 8423 and 7489

Both have 4 digits; compare the thousands digits: 8 > 7.

8423 > 7489.

(ii) 13567 and 13657

Both have 5 digits and agree to the hundreds place (13_); compare the tens digits: 6 < 5 is false, so compare digit by digit: 13567 vs 13657 differ at the hundreds digit: 5 < 6.

13567 < 13657.

(iii) -12 and -8

On the number line, -12 lies to the left of -8, so -12 is smaller.

-12 < -8.

(iv) -5 and 0

Every negative number is smaller than 0.

-5 < 0.

(v) 15 and -15

Every positive number is greater than every negative number.

15 > -15.

(vi) 3/5 and 5/8

Convert to decimals: 3/5 = 0.6 and 5/8 = 0.625.

0.6 < 0.625.

3/5 < 5/8.

(vii) -7/9 and -5/9

Same denominator (9); compare numerators as negatives: -7 < -5.

-7/9 < -5/9.

(viii) -4/5 and -8/5

Same denominator (5); compare numerators as negatives: -4 > -8.

-4/5 > -8/5.

(ix) 0.80 and 0.750

0.80 = 0.800 and 0.750 = 0.750.

0.800 > 0.750.

0.80 > 0.750.

(x) 2.35 and 2.305

Compare digit by digit: ones 2 = 2, tenths 3 = 3, hundredths 5 > 0.

2.35 > 2.305.

7. Arrange the following in ascending order.

(i) 342, 154, 876, 569

Compare hundreds digits: 154 < 342 < 569 < 876.

154, 342, 569, 876.

(ii) 9234, 4218, 5723, 3142

Compare thousands digits: 3142 < 4218 < 5723 < 9234.

3142, 4218, 5723, 9234.

(iii) -7, -2, 4, -5, 0

Negatives come first, most negative first: -7 < -5 < -2 < 0 < 4.

-7, -5, -2, 0, 4.

(iv) -25, -30, -15, -5, -10

Order from the most negative: -30 < -25 < -15 < -10 < -5.

-30, -25, -15, -10, -5.

(v) 2/3, -1/2, 3/4, -2/3

Convert to decimals: 2/3 = 0.667, -1/2 = -0.5, 3/4 = 0.75, -2/3 = -0.667.

Order smallest first: -0.667 < -0.5 < 0.667 < 0.75.

-2/3, -1/2, 2/3, 3/4.

(vi) 3/5, 2/3, 1/2, 4/5

Convert to decimals: 3/5 = 0.6, 2/3 = 0.667, 1/2 = 0.5, 4/5 = 0.8.

Order smallest first: 0.5 < 0.6 < 0.667 < 0.8.

1/2, 3/5, 2/3, 4/5.

(vii) 1.23, 1.1, 1.13, 1.2

Write with equal decimal places: 1.23, 1.10, 1.13, 1.20.

Order smallest first: 1.10 < 1.13 < 1.20 < 1.23.

1.1, 1.13, 1.2, 1.23.

(viii) 2.34, 1.56, 3.45, 2.15

Compare ones digits, then tenths: 1.56 < 2.15 < 2.34 < 3.45.

1.56, 2.15, 2.34, 3.45.

8. Arrange the following in descending order (p.24).

(i) 435, 278, 899, 643

Compare hundreds digits: 899 > 643 > 435 > 278.

899, 643, 435, 278.

(ii) 54321, 43210, 67890, 12345

Compare ten-thousands digits: 67890 > 54321 > 43210 > 12345.

67890, 54321, 43210, 12345.

(iii) -7, 3, -2, 8, 0

Order largest first: 8 > 3 > 0 > -2 > -7.

8, 3, 0, -2, -7.

(iv) -50, -45, -60, -30, -35

Order largest first: -30 > -35 > -45 > -50 > -60.

-30, -35, -45, -50, -60.

(v) 7/10, 1/2, 4/5, 2/3

Convert to decimals: 7/10 = 0.7, 1/2 = 0.5, 4/5 = 0.8, 2/3 = 0.667.

Order largest first: 0.8 > 0.7 > 0.667 > 0.5.

4/5, 7/10, 2/3, 1/2.

(vi) -2/3, 1/3, 3/4, -1/4

Convert to decimals: -2/3 = -0.667, 1/3 = 0.333, 3/4 = 0.75, -1/4 = -0.25.

Order largest first: 0.75 > 0.333 > -0.25 > -0.667.

3/4, 1/3, -1/4, -2/3.

(vii) 4.75, 3.9, 5.2, 4.15

Compare ones digits, then tenths: 5.2 > 4.75 > 4.15 > 3.9.

5.2, 4.75, 4.15, 3.9.

(viii) 0.125, 0.25, 0.375, 0.0625

Compare thousandths: 0.375 > 0.25 > 0.125 > 0.0625.

0.375, 0.25, 0.125, 0.0625.

Exercise 1.4 (p.29)

1. Verify Associative Property w.r.t. Addition and Multiplication.

Flag: Exercise 1.4 Q1 (i) and (ii) as printed do not balance, because the first term on the right-hand side is printed without its minus sign (verified at 300 dpi). The verifications below restore the minus sign on the right-hand side, which is the only reading under which the associative property is being tested; the book's own worked Example 1.45 (p.24) drops the minus on both sides and so balances at 13/12.

(i) (-1/2 + 1/3) + 1/4 = -1/2 + (1/3 + 1/4) [minus restored on the right]

LHS: -1/2 + 1/3 = -3/6 + 2/6 = -1/6.

LHS: -1/6 + 1/4 = -2/12 + 3/12 = 1/12.

RHS: 1/3 + 1/4 = 4/12 + 3/12 = 7/12.

RHS: -1/2 + 7/12 = -6/12 + 7/12 = 1/12.

LHS = RHS = 1/12, so the associative property is verified.

Verified: both sides equal 1/12. As printed (right-hand side +1/2), the right side would be 13/12 and the equation would not balance.

(ii) (-1/5 + 2/7) + 3/8 = -1/5 + (2/7 + 3/8) [minus restored on the right]
LHS: -1/5 + 2/7 = -7/35 + 10/35 = 3/35.

LHS: 3/35 + 3/8 = 24/280 + 105/280 = 129/280.

RHS: 2/7 + 3/8 = 16/56 + 21/56 = 37/56.

RHS: -1/5 + 37/56 = -56/280 + 185/280 = 129/280.

LHS = RHS = 129/280, so the associative property is verified.

Verified: both sides equal 129/280. As printed (right-hand side +1/5), the right side would be 241/280 and the equation would not balance.

(iii) (3/4 + 2/5) + 1/6 = 3/4 + (2/5 + 1/6)
LHS: 3/4 + 2/5 = 15/20 + 8/20 = 23/20.

LHS: 23/20 + 1/6 = 69/60 + 10/60 = 79/60.

RHS: 2/5 + 1/6 = 12/30 + 5/30 = 17/30.

RHS: 3/4 + 17/30 = 45/60 + 34/60 = 79/60.

LHS = RHS = 79/60, so the associative property is verified.

Verified: both sides equal 79/60.

(iv) (1/2 x 2/3) x 3/4 = 1/2 x (2/3 x 3/4)

LHS: 1/2 x 2/3 = 2/6 = 1/3.

LHS: 1/3 x 3/4 = 3/12 = 1/4.

RHS: 2/3 x 3/4 = 6/12 = 1/2.

RHS: 1/2 x 1/2 = 1/4.

LHS = RHS = 1/4, so the associative property is verified.

Verified: both sides equal 1/4.

(v) (2/3 x 3/4) x 1/5 = 2/3 x (3/4 x 1/5)

LHS: 2/3 x 3/4 = 6/12 = 1/2.

LHS: 1/2 x 1/5 = 1/10.

RHS: 3/4 x 1/5 = 3/20.

RHS: 2/3 x 3/20 = 6/60 = 1/10.

LHS = RHS = 1/10, so the associative property is verified.

Verified: both sides equal 1/10.

(vi) (2/5 x 3/7) x 1/2 = 2/5 x (3/7 x 1/2)

LHS: 2/5 x 3/7 = 6/35.

LHS: 6/35 x 1/2 = 6/70 = 3/35.

RHS: 3/7 x 1/2 = 3/14.

RHS: 2/5 x 3/14 = 6/70 = 3/35.

LHS = RHS = 3/35, so the associative property is verified.

Verified: both sides equal 3/35.

2. Verify Commutative Property w.r.t. Addition and Multiplication.

(i) 2/5 + 3/7 = 3/7 + 2/5

LHS: 2/5 + 3/7 = 14/35 + 15/35 = 29/35.

RHS: 3/7 + 2/5 = 15/35 + 14/35 = 29/35.

LHS = RHS, so the commutative property is verified.

Verified: both sides equal 29/35.

(ii) -4/9 + 5/6 = 5/6 + -4/9

LHS: -4/9 + 5/6 = -8/18 + 15/18 = 7/18.

RHS: 5/6 + -4/9 = 15/18 - 8/18 = 7/18.

LHS = RHS, so the commutative property is verified.

Verified: both sides equal 7/18.

(iii) 7/8 + 11/12 = 11/12 + 7/8

LHS: 7/8 + 11/12 = 21/24 + 22/24 = 43/24.

RHS: 11/12 + 7/8 = 22/24 + 21/24 = 43/24.

LHS = RHS, so the commutative property is verified.

Verified: both sides equal 43/24.

(iv) 2/15 x 7/10 = 7/10 x 2/15

LHS: 2/15 x 7/10 = 14/150 = 7/75.

RHS: 7/10 x 2/15 = 14/150 = 7/75.

LHS = RHS, so the commutative property is verified.

Verified: both sides equal 7/75.

(v) 4/9 x 7/10 = 7/10 x 4/9

LHS: 4/9 x 7/10 = 28/90 = 14/45.

RHS: 7/10 x 4/9 = 28/90 = 14/45.

LHS = RHS, so the commutative property is verified.

Verified: both sides equal 14/45.

(vi) -3/8 x 11/15 = 11/15 x -3/8

LHS: -3/8 x 11/15 = -33/120 = -11/40.

RHS: 11/15 x -3/8 = -33/120 = -11/40.

LHS = RHS, so the commutative property is verified.

Verified: both sides equal -11/40.

3. Verify distributive property of rational numbers w.r.t. multiplication over addition/subtraction.

(i) 1/2 x (1/3 + 1/4) = 1/2 x 1/3 + 1/2 x 1/4

LHS: 1/3 + 1/4 = 4/12 + 3/12 = 7/12.

LHS: 1/2 x 7/12 = 7/24.

RHS: 1/2 x 1/3 = 1/6 and 1/2 x 1/4 = 1/8.

RHS: 1/6 + 1/8 = 4/24 + 3/24 = 7/24.

LHS = RHS, so the distributive property is verified.

Verified: both sides equal 7/24.

(ii) 2/3 x (3/4 + 2/5) = 2/3 x 3/4 + 2/3 x 2/5

LHS: 3/4 + 2/5 = 15/20 + 8/20 = 23/20.

LHS: 2/3 x 23/20 = 46/60 = 23/30.

RHS: 2/3 x 3/4 = 6/12 = 1/2 and 2/3 x 2/5 = 4/15.

RHS: 1/2 + 4/15 = 15/30 + 8/30 = 23/30.

LHS = RHS, so the distributive property is verified.

Verified: both sides equal 23/30.

(iii) 3/4 x (1/2 + 3/7) = 3/4 x 1/2 + 3/4 x 3/7

LHS: 1/2 + 3/7 = 7/14 + 6/14 = 13/14.

LHS: 3/4 x 13/14 = 39/56.

RHS: 3/4 x 1/2 = 3/8 and 3/4 x 3/7 = 9/28.

RHS: 3/8 + 9/28 = 21/56 + 18/56 = 39/56.

LHS = RHS, so the distributive property is verified.

Verified: both sides equal 39/56.

(iv) 1/2 x (1/3 - 1/4) = 1/2 x 1/3 - 1/2 x 1/4

LHS: 1/3 - 1/4 = 4/12 - 3/12 = 1/12.

LHS: 1/2 x 1/12 = 1/24.

RHS: 1/2 x 1/3 = 1/6 and 1/2 x 1/4 = 1/8.

RHS: 1/6 - 1/8 = 4/24 - 3/24 = 1/24.

LHS = RHS, so the distributive property is verified.

Verified: both sides equal 1/24.

(v) 2/3 x (3/4 - 2/5) = 2/3 x 3/4 - 2/3 x 2/5

LHS: 3/4 - 2/5 = 15/20 - 8/20 = 7/20.

LHS: 2/3 x 7/20 = 14/60 = 7/30.

RHS: 2/3 x 3/4 = 6/12 = 1/2 and 2/3 x 2/5 = 4/15.

RHS: 1/2 - 4/15 = 15/30 - 8/30 = 7/30.

LHS = RHS, so the distributive property is verified.

Verified: both sides equal 7/30.

(vi) 3/4 x (1/2 - 3/7) = 3/4 x 1/2 - 3/4 x 3/7

LHS: 1/2 - 3/7 = 7/14 - 6/14 = 1/14.

LHS: 3/4 x 1/14 = 3/56.

RHS: 3/4 x 1/2 = 3/8 and 3/4 x 3/7 = 9/28.

RHS: 3/8 - 9/28 = 21/56 - 18/56 = 3/56.

LHS = RHS, so the distributive property is verified.

Verified: both sides equal 3/56.

Exercise 1.5 (p.31) - Word Problems

1. Sarah walks 2/5 kilometers in the morning and 3/7 kilometers in the evening. How much distance does she walk in total?

Total distance = 2/5 + 3/7.

LCM of 5 and 7 is 35.

2/5 = 14/35 and 3/7 = 15/35.

14/35 + 15/35 = 29/35.

Sarah walks 29/35 kilometers in total.

2. A tank is 3/8 filled with water. Another 1/4 of the tank is filled. What fraction of the tank is now filled?

Fraction now filled = 3/8 + 1/4.

1/4 = 2/8.

3/8 + 2/8 = 5/8.

5/8 of the tank is now filled.

3. A ribbon is 7/8 meters long. If 3/8 meters is cut off, how much ribbon remains?

Remaining ribbon = 7/8 - 3/8.

7/8 - 3/8 = 4/8 = 1/2.

1/2 meter of ribbon remains.

4. A jug contains 3/4 litres of water. 2/5 litres are used. How much water is left in the jug?

Water left = 3/4 - 2/5.

LCM of 4 and 5 is 20.

3/4 = 15/20 and 2/5 = 8/20.

15/20 - 8/20 = 7/20.

7/20 litres of water is left in the jug.

5. A piece of marble is 2/3 metres wide and 3/4 metres long. What is its area?

Area = length x width = 3/4 x 2/3.

3/4 x 2/3 = 6/12 = 1/2.

The area is 1/2 square metres.

6. A car consumes 1/8 litre of fuel per kilometer. How much fuel will it consume for 3/5 kilometers?

Fuel consumed = 1/8 x 3/5.

1/8 x 3/5 = 3/40.

The car will consume 3/40 litre of fuel.

7. A chocolate cake weighs 3/4 kilograms. If each serving is 1/8 kilograms, how many servings are in the chocolate cake?

Number of servings = 3/4 divided by 1/8.

3/4 divided by 1/8 = 3/4 x 8/1 = 24/4 = 6.

There are 6 servings in the chocolate cake.

Review Exercise 1 (pp.31-32)

1. Choose the correct options.

i. Round off 42.567 to 2 decimal places:

- 42.56
- ✓ 42.57
- 42.5
- 43.00

ii. Arrange 345, 125, 678, 456 in ascending order:

- ✓ 125, 345, 456, 678
- 678, 456, 345, 125
- 456, 125, 345, 678
- 125, 678, 345, 456

iii. Arrange 1.45, 2.3, 0.89, 1.89 in descending order:

- ✓ 2.3, 1.89, 1.45, 0.89
- 0.89, 1.45, 1.89, 2.3
- 1.45, 0.89, 2.3, 1.89
- 1.89, 2.3, 1.45, 0.89

iv. Which of the following is the largest rational number?

- 3/5
- 4/7
- 5/8
- ✓ 2/3

v. 7/4 into a mixed fraction is:

- ✓ 1 3/4
- 2 3/4

3 3/4

1 1/4

vi. LCM of 12 and 18 is:

24

✓ 36

48

54

vii. Two bells ring at intervals of 4 minutes and 6 minutes. After how many minutes will they ring together again?

10 minutes

✓ 12 minutes

14 minutes

24 minutes

viii. HCF of 42 and 56 is:

7

✓ 14

28

42

ix. 3/4 + 1/8 =

1/8

3/8

4/8

✓ 7/8

x. What is the place value of 5 in the number 356,789?

50

500

5,000

✓ 50,000

2. In 0.4567, what is the place value of 6?

In 0.4567, the digits after the decimal point are: 4 tenths, 5 hundredths, 6 thousandths, 7 ten-thousandths.

The digit 6 is in the thousandths place.

Place value = 6 x 1/1000 = 0.006.

0.006.

3. In 89.0563, what is the place value of 5 and 6?

In 89.0563, the digits after the decimal point are: 0 tenths, 5 hundredths, 6 thousandths, 3 ten-thousandths.

The digit 5 is in the hundredths place: 5 x 1/100 = 0.05.

The digit 6 is in the thousandths place: 6 x 1/1000 = 0.006.

Place value of 5 is 0.05 and place value of 6 is 0.006.

4. Identify the place value of 3 in 23,465.

In 23,465, the digit 3 is in the thousands place.

Place value = 3 x 1000 = 3000.

3000.

5. Round off 654,987 to the nearest thousand.

The digit in the thousands place is 4.

The digit to its right (hundreds place) is 9, which is 5 or more, so increase the thousands digit by 1: 4 + 1 = 5.

Replace the digits to the right with zeros.

655,000.

6. Round off 93,425 to the nearest ten thousand.

The digit in the ten thousands place is 9.

The digit to its right (thousands place) is 3, which is less than 5, so keep the ten thousands digit as it is.

Replace the digits to the right with zeros.

90,000.

7. A water tank is 7/8 full. If 5/12 of the water is used for irrigation, how much water remains?

Water remaining = 7/8 - 5/12.

LCM of 8 and 12 is 24.

7/8 = 21/24 and 5/12 = 10/24.

21/24 - 10/24 = 11/24.

11/24 of the water remains.

8. A rectangular field is 3/5 km long and 4/7 km wide. What is the area of the field?

Area = length x width = 3/5 x 4/7.

3/5 x 4/7 = 12/35.

The area of the field is 12/35 square kilometers.

Unit 2: Rate, Ratio and Proportion

Exercise 2.1 (p.37) - Average Rate

1. A factory produces 130 units in 4 hours. What is its rate?

Rate = units divided by hours = 130/4.

130/4 = 32.5.

The rate is 32.5 units per hour.

2. Jamal types 80 words in 2 minutes. What is its rate?

Rate = words divided by minutes = 80/2.

80/2 = 40.

The rate is 40 words per minute.

3. A factory produces 600 boxes in 8 hours. Find the rate of production per hour.

Rate = boxes divided by hours = 600/8.

600/8 = 75.

The rate of production is 75 boxes per hour.

4. A plant grows 10 cm in 5 days. What is its average rate of growth

per day?

Average rate = growth divided by days = 10/5.

10/5 = 2.

The average rate of growth is 2 cm per day.

5. A student working part-time earns Rs. 15290 for working 16 days and Rs. 10500 for working another 14 days. Find his average wage rate per day.

First daily rate = 15290/16 = 955.625 rupees per day.

Second daily rate = 10500/14 = 750 rupees per day.

Average wage rate = (955.625 + 750)/2 = 1705.625/2 = 852.8125.

The average wage rate is Rs. 852.8125 per day (the book's answer-key method, the average of the two daily rates). Note: total earnings divided by total days would give (15290 + 10500)/(16 + 14) = 25790/30 = Rs. 859.67 per day; the two methods differ because the two periods have different lengths.

6. Zia's scores in Maths first, second and third term exams are 80, 70 and 90. What is his average score?

Total score = 80 + 70 + 90 = 240.

Number of exams = 3.

Average score = 240/3 = 80.

His average score is 80 marks.

7. The first day, Riaz reads 50 pages of a novel in 3 hours and 70 pages in 4 hours the next day. What is the average rate of his reading per hour?

First day rate = 50/3 = 16.67 pages per hour.

Second day rate = 70/4 = 17.5 pages per hour.

Average rate = (16.67 + 17.5)/2 = 34.17/2 = 17.08 pages per hour (the book's average-of-rates method, as in Q5 and Q8).

The average rate of reading is about 17.08 pages per hour. Note: total pages divided by total hours would give 120/7 = 17.14 pages per hour; the answer key as scanned reads 17.83, which matches neither method and is treated as a key misprint.

8. Nawaz is going to Karachi in a car. He drives 450 km in 5 hours and after a stay started his journey and completed 236 km in 4 hours. What is the average rate of his car speed per hour?

First part speed = 450/5 = 90 km per hour.

Second part speed = 236/4 = 59 km per hour.

Average rate = (90 + 59)/2 = 149/2 = 74.5 km per hour (the book's average-of-rates method, matching the answer key).

The average rate of his car speed is 74.5 km per hour. Note: total distance divided by total time would give 686/9 = 76.22 km per hour.

Exercise 2.2 (p.39) - Increasing and Decreasing Quantities

1. Find the new increasing and decreasing quantity in the following:

(i) Increase 45 min in the ratio 4:3.

New quantity = 4/3 x 45.

4/3 x 45 = 4 x 15 = 60 min.

Increase = 60 - 45 = 15 min (the answer-key figure).

The new quantity is 60 minutes (an increase of 15 minutes).

(ii) Increase 650 cm in the ratio 6:5.

New quantity = 6/5 x 650.

6/5 x 650 = 6 x 130 = 780 cm.

Increase = 780 - 650 = 130 cm.

The new quantity is 780 cm (an increase of 130 cm).

(iii) Increase 855 km in the ratio 7:5.

New quantity = 7/5 x 855.

7/5 x 855 = 7 x 171 = 1197 km.

Increase = 1197 - 855 = 342 km (the answer-key figure).

The new quantity is 1197 km (an increase of 342 km).

(iv) Decrease 75 m in the ratio 3:5.

New quantity = 3/5 x 75.

3/5 x 75 = 3 x 15 = 45 m.

Decrease = 75 - 45 = 30 m (the answer-key figure).

The new quantity is 45 m (a decrease of 30 m).

(v) Decrease 660 m in the ratio 5:6.

New quantity = 5/6 x 660.

5/6 x 660 = 5 x 110 = 550 m.

Decrease = 660 - 550 = 110 m (the answer-key figure).

The new quantity is 550 m (a decrease of 110 m).

(vi) Decrease 430 kg in the ratio 3:4.

New quantity = 3/4 x 430.

3/4 x 430 = 1290/4 = 322.5 kg.

Decrease = 430 - 322.5 = 107.5 kg (the answer-key figure).

The new quantity is 322.5 kg (a decrease of 107.5 kg).

2. The price of a shirt rises from Rs. 1500 to Rs. 2500. In what ratio does the price increase?

The ratio of the old price to the new price is 1500:2500.

Divide both by 500: 1500/500 = 3 and 2500/500 = 5.

So the ratio is 3:5 (old:new), matching the answer key.

The price increases in the ratio 3:5 (old price:new price; new:old is 5:3).

3. The cost of a laptop decreases from Rs. 80000 to Rs. 70000. In what ratio does the cost decrease?

The ratio of the old cost to the new cost is 80000:70000.

Divide both by 10000: 80000/10000 = 8 and 70000/10000 = 7.

So the ratio is 8:7 (old:new), matching the answer key.

The cost decreases in the ratio 8:7 (old cost:new cost; new:old is 7:8).

4. Huzaifa and Musfira are two siblings; they obtained marks in the

ratio 8:7. If Musfira obtained 72 marks, find the marks of Huzaifa.

Huzaifa's marks : Musfira's marks = 8:7.

So Huzaifa's marks = 8/7 x 72 = 576/7 = 82.29 marks.

Huzaifa obtained 576/7 = 82.29 marks as printed. Flag: this is not a whole number of marks, and the book's answer key gives 63, which is 7/8 x 72 - the value for the reversed ratio 7:8. The printed ratio (8:7) and marks (72) are inconsistent with the key; neither has been silently changed.

5. Ratio of the height of father to the height of his son is 3:2. If the height of the son is 116 cm, find the height of the father.

Father's height : son's height = 3:2.

Father's height = 3/2 x 116.

3/2 x 116 = 3 x 58 = 174 cm.

The height of the father is 174 cm.

6. Riaz has an IT shop, in which the selling ratio of desktop to laptop is 2:5. If he sold 35 laptops, find the number of desktops sold.

Desktops : laptops = 2:5.

Number of desktops = 2/5 x 35.

2/5 x 35 = 2 x 7 = 14.

14 desktops were sold.

Exercise 2.3 (p.43) - Proportion

1. Check whether the following are proportions or not.

(i) 2:3 and 8:12

Product of extremes = 2 x 12 = 24.

Product of means = 3 x 8 = 24.

The products are equal.

Yes, it is a proportion.

(ii) 4:7 and 8:14

Product of extremes = 4 x 14 = 56.

Product of means = 7 x 8 = 56.

The products are equal.

Yes, it is a proportion.

(iii) 14/16 = 5/9

Product of extremes = 14 x 9 = 126.

Product of means = 16 x 5 = 80.

The products are not equal.

No, it is not a proportion.

(iv) 3/5 = 6/8

Product of extremes = 3 x 8 = 24.

Product of means = 5 x 6 = 30.

The products are not equal.

No, it is not a proportion.

(v) 5:7 and 10:14

Product of extremes = 5 x 14 = 70.

Product of means = 7 x 10 = 70.

The products are equal.

Yes, it is a proportion.

(vi) 7:21 and 4:12

Product of extremes = 7 x 12 = 84.

Product of means = 21 x 4 = 84.

The products are equal.

Yes, it is a proportion.

2. Find the unknown value in the following.

(i) x:2 and 4:8

Product of extremes = product of means: x x 8 = 2 x 4.

8x = 8.

x = 8/8 = 1.

x = 1.

(ii) x:6 and 8:12

Product of extremes = product of means: x x 12 = 6 x 8.

12x = 48.

x = 48/12 = 4.

x = 4.

(iii) 3:x and 9:12

Product of extremes = product of means: 3 x 12 = x x 9.

36 = 9x.

x = 36/9 = 4.

x = 4.

(iv) 7:x and 14:28

Product of extremes = product of means: 7 x 28 = x x 14.

196 = 14x.

x = 196/14 = 14.

x = 14.

(v) 10:x and 3:15

Product of extremes = product of means: 10 x 15 = x x 3.

150 = 3x.

x = 150/3 = 50.

x = 50.

(vi) 9:x and 12:15

Product of extremes = product of means: 9 x 15 = x x 12.

135 = 12x.

x = 135/12 = 11.25.

x = 11.25.

(vii) 5:8 and x:10

Product of extremes = product of means: 5 x 10 = 8 x x.

50 = 8x.

x = 50/8 = 6.25.

x = 6.25.

(viii) 8:9 and x:12

Product of extremes = product of means: 8 x 12 = 9 x x.

96 = 9x.

x = 96/9 = 32/3 = 10.67.

x = 10.67 (32/3).

(ix) 15:5 and x:20

Product of extremes = product of means: 15 x 20 = 5 x x.

300 = 5x.

x = 300/5 = 60.

x = 60.

(x) 9:25 and x:18

Product of extremes = product of means: 9 x 18 = 25 x x.

162 = 25x.

x = 162/25 = 6.48.

x = 6.48.

(xi) 35:7 and 5:x

Product of extremes = product of means: 35 x x = 7 x 5.

35x = 35.

x = 35/35 = 1.

x = 1.

(xii) 24:16 and 6:x

Product of extremes = product of means: 24 x x = 16 x 6.

24x = 96.

x = 96/24 = 4.

x = 4.

3. A painter paints 300 square feet in 5 hours. How many square feet will he paint in 3 hours at the same rate?

Rate = 300/5 = 60 square feet per hour.

In 3 hours: 60 x 3 = 180 square feet.

He will paint 180 square feet in 3 hours.

4. A car travels 240 miles in 4 hours. How far will it travel in 2 hours at the same rate?

Rate = 240/4 = 60 miles per hour.

In 2 hours: 60 x 2 = 120 miles.

It will travel 120 miles in 2 hours.

5. A student can write 400 words per hour. How long will it take to write 1200 words at the same rate?

Time = words divided by rate = 1200/400.

1200/400 = 3.

It will take 3 hours to write 1200 words.

6. Thirty workers build a house in 18 days. How many days will 25 workers working at the same rate take to build the same house?

This is inverse proportion: fewer workers take more days.

Total work = 30 x 18 = 540 worker-days.

Days for 25 workers = 540/25 = 21.6 days.

25 workers will take 21.6 days (21 days and about 14 hours).

7. Four pipes can fill a tank in 60 minutes. How long will it take to fill the same tank if 8 pipes are used?

This is inverse proportion: more pipes take less time.

Total work = 4 x 60 = 240 pipe-minutes.

Time for 8 pipes = 240/8 = 30 minutes.

It will take 30 minutes with 8 pipes.

8. A farmer has enough feed for 1040 cattle for 50 days. How many cattle can be sold so that the same feed will be used for 75 days?

Total feed = 1040 x 50 = 52000 cattle-days.

Cattle that the feed lasts 75 days for = 52000/75 = 693.33.

So about 693 cattle can be kept.

Cattle that can be sold = 1040 - 693 = 347.

About 347 cattle can be sold (693 cattle kept; 52000/75 is not a

whole number, so the answer is approximate).

Review Exercise 2 (p.44)

1. Choose the correct option.

i. If an eraser costs Rs. 5 and a pencil Rs. 10, how much will 2 erasers and 2 pencils cost?

- Rs. 10
- Rs. 20
- ✓ Rs. 30
- Rs. 40

ii. A biker can travel 120 km in 2 hours. How many kilometers will it travel in 3 hours at the same rate?

- ✓ 180 km
- 240 km
- 360 km
- 380 km

iii. The ratio of 45 and 15 is:

- ✓ 3:1
- 3:10
- 1:3
- 10:3

iv. If Arsalan reads 10, 15 and 5 pages in 3 hours, what is the average rate?

- 30
- 25

✓ 10

5

v. What is the value of x in 2:x = 3:6?

2

3

✓ 4

5

2. Find the unknown value in the following.

(a) 3:6 and 9:x

Product of extremes = product of means: 3 x x = 6 x 9.

3x = 54.

x = 54/3 = 18.

x = 18.

(b) 7:14 and x:21

Product of extremes = product of means: 7 x 21 = 14 x x.

147 = 14x.

x = 147/14 = 10.5.

x = 10.5.

(c) 10:25 and 50:x

Product of extremes = product of means: 10 x x = 25 x 50.

10x = 1250.

x = 1250/10 = 125.

x = 125.

(d) 4:8 and x:12

Product of extremes = product of means: 4 x 12 = 8 x x.

48 = 8x.

x = 48/8 = 6.

x = 6.

3. A tap fills 30 tonnes of water in 5 minutes. What is the flow rate of water per minute?

Flow rate = 30/5 = 6.

The flow rate is 6 tonnes per minute.

4. A house in Karachi pays Rs. 3,500 for 200 units of electricity. How much will the bill be for 500 units?

Cost per unit = 3500/200 = Rs. 17.5.

Cost for 500 units = 17.5 x 500 = Rs. 8750.

The bill for 500 units will be Rs. 8,750.

5. A printing press prints 100 books in 5 hours. How many books will it print in 20 hours?

Rate = 100/5 = 20 books per hour.

In 20 hours: 20 x 20 = 400 books.

It will print 400 books in 20 hours.

6. A 2 GB file downloads in 20 minutes with a speed of 5 Mbps. How long will it take if the speed is increased to 10 Mbps?

This is inverse proportion: double the speed takes half the time.

Total work = 5 x 20 = 100.

Time at 10 Mbps = 100/10 = 10 minutes.

It will take 10 minutes at 10 Mbps.

7. A school bus takes 50 minutes to reach its destination at a speed of 40 km/h. How long will it take if the speed is increased to 80 km/h?

This is inverse proportion: double the speed takes half the time.

Total work = 40 x 50 = 2000.

Time at 80 km/h = 2000/80 = 25 minutes.

It will take 25 minutes at 80 km/h.

Unit 3: Financial Arithmetic

Exercise 3.1 (pp.49-50) - Cost Price, Selling Price, Profit, Loss and Discount

1. Identify and differentiate the cost price and selling price in the following questions.

(i) A bookstore buys a used novel for Rs. 250 and gives it to a customer for Rs. 350.

Cost price is what the seller paid to buy the item: Rs. 250.

Selling price is what the buyer paid: Rs. 350.

Profit = selling price - cost price = 350 - 250 = Rs. 100.

Cost price = Rs. 250 (what the bookstore paid); selling price = Rs. 350 (what the customer paid). The bookstore makes a profit of Rs. 100.

(ii) A grocery store buys a box of cereal for Rs. 60 per unit from a wholesaler. He stocks the cereal on the shelf with a price tag of Rs. 80 per unit.

Cost price is what the store paid: Rs. 60 per unit.

Selling price is the price tag: Rs. 80 per unit.

Profit = 80 - 60 = Rs. 20 per unit.

Cost price = Rs. 60 per unit (what the store paid the wholesaler); selling price = Rs. 80 per unit (the shelf price tag). The store makes a profit of Rs. 20 per unit.

(iii) A car dealer buys a used car for Rs. 400,000. They clean and detail the car before putting it on the showroom floor with a price tag of Rs. 450,000.

Cost price is what the dealer paid: Rs. 400,000.

Selling price is the showroom price tag: Rs. 450,000.

Profit = 450,000 - 400,000 = Rs. 50,000.

Cost price = Rs. 400,000 (what the dealer paid); selling price = Rs. 450,000 (the showroom price tag). The dealer makes a profit of Rs. 50,000.

2. Mirza bought a carton of milk for Rs. 80 from the wholesale market. He sold it for Rs. 90. How much profit or loss did he make?

Cost price = Rs. 80, selling price = Rs. 90.

Selling price is greater than cost price, so it is a profit.

Profit = 90 - 80 = Rs. 10.

Mirza made a profit of Rs. 10.

3. It is your birthday, and your favourite store is offering a 20% discount on all clothes. A t-shirt you really want normally costs Rs. 300. How much money will you save with the discount?

Discount = 20% of Rs. 300.

20% of 300 = 20/100 x 300 = Rs. 60.

You will save Rs. 60.

4. An electronics store offers a discount of Rs. 1,000 on a smartphone originally priced at Rs. 25,000. Calculate the sale price after the discount.

Original price = Rs. 25,000, discount = Rs. 1,000.

Sale price = 25,000 - 1,000 = Rs. 24,000.

The sale price after the discount is Rs. 24,000.

5. A television has a marked price of Rs. 60,000, and it is sold for Rs. 54,000. Calculate the discount amount.

Discount = marked price - selling price.

Discount = 60,000 - 54,000 = Rs. 6,000.

The discount amount is Rs. 6,000.

6. A laptop has a marked price of Rs. 50,000 and is sold for Rs. 45,000. Determine the discount amount.

Discount = marked price - selling price.

Discount = 50,000 - 45,000 = Rs. 5,000.

The discount amount is Rs. 5,000.

7. A shopkeeper bought a purse for Rs. 1500 and sold it for Rs. 2000. What is the profit percentage?

Profit = 2000 - 1500 = Rs. 500.

Profit percentage = profit/cost price x 100% = 500/1500 x 100%.

500/1500 x 100% = 33.33%.

The profit percentage is 33.33%.

8. Waseem bought a laptop for Rs. 60000 and sold it for Rs. 75000. What is the profit percentage?

Profit = 75,000 - 60,000 = Rs. 15,000.

Profit percentage = 15,000/60,000 x 100% = 25%.

The profit percentage is 25%.

9. A dealer bought a blanket for Rs. 16000 and sold it for Rs. 20000. What is the profit percentage?

Profit = 20,000 - 16,000 = Rs. 4,000.

Profit percentage = 4,000/16,000 x 100% = 25%.

The profit percentage is 25%.

10. Ali bought a waistcoat for Rs. 5000 and sold it for Rs. 4000. What is the loss percentage?

Loss = 5000 - 4000 = Rs. 1,000.

Loss percentage = loss/cost price x 100% = 1,000/5,000 x 100% = 20%.

The loss percentage is 20%.

11. Khuzaima bought a LED TV for Rs. 75000 and sold it for Rs. 60000. What is the loss percentage?

Loss = 75,000 - 60,000 = Rs. 15,000.

Loss percentage = 15,000/75,000 x 100% = 20%.

The loss percentage is 20%.

12. Shazia bought a tea set for Rs. 35000 and sold it for Rs. 30000. What is the loss percentage?

Loss = 35,000 - 30,000 = Rs. 5,000.

Loss percentage = 5,000/35,000 x 100% = 14.29%.

The loss percentage is 14.29%.

Exercise 3.2 (p.57) - Tax, GST, VAT and Commission

Note on Exercise 3.2 numbering: the book prints the VAT items out of order (an item numbered 11 before an item numbered 10, and two items numbered 11). The items below are numbered in printed order with the print numbers noted.

1. Ali has a monthly salary of Rs. 80,000. Calculate his annual income tax liability based on the 2024-2025 tax slabs.

Annual income = 80,000 x 12 = Rs. 960,000.

The 2024-2025 slab for income above Rs. 600,000 up to Rs. 1,200,000 is 5% of the amount exceeding Rs. 600,000.

Amount exceeding Rs. 600,000 = 960,000 - 600,000 = Rs. 360,000.

Tax = 5% of 360,000 = Rs. 18,000.

His annual income tax liability is Rs. 18,000.

2. Sara earns a monthly salary of Rs. 150,000. Calculate her annual income tax liability based on the 2024-2025 tax slabs.

Annual income = 150,000 x 12 = Rs. 1,800,000.

The slab for income above Rs. 1,200,000 up to Rs. 2,200,000 is Rs. 30,000 + 15% of the amount exceeding Rs. 1,200,000.

Amount exceeding Rs. 1,200,000 = 1,800,000 - 1,200,000 = Rs. 600,000.

Tax = 30,000 + 15% of 600,000 = 30,000 + 90,000 = Rs. 120,000.

Her annual income tax liability is Rs. 120,000.

3. A business owner, Ahmed, earns a monthly income of Rs. 80,000 from his business. Calculate his annual income tax liability based on the 2024-2025 tax slabs.

Annual income = 80,000 x 12 = Rs. 960,000.

Slab: 5% of the amount exceeding Rs. 600,000.

Amount exceeding Rs. 600,000 = Rs. 360,000.

Tax = 5% of 360,000 = Rs. 18,000.

His annual income tax liability is Rs. 18,000 (same slab computation as Q1; the unit gives one slab table for individuals and businesses).

4. A house in Quetta has an annual rental value of Rs. 500,000. If the property tax rate is 15%, calculate the annual property tax.

Property tax = annual rental value x tax rate.

Property tax = 500,000 x 15% = 500,000 x 15/100 = Rs. 75,000.

The annual property tax is Rs. 75,000.

5. The property tax on a commercial building is Rs. 120,000 per year. If the tax rate is 12%, find the assessed value of the property.
Assessed value = property tax / tax rate.

Assessed value = 120,000 / 12% = 120,000 x 100/12 = Rs. 1,000,000.

The assessed value of the property is Rs. 1,000,000.

6. A plot of land is valued at Rs. 2,000,000, and the property tax rate is 5%. Calculate the annual property tax.
Property tax = 2,000,000 x 5% = 2,000,000 x 5/100 = Rs. 100,000.

The annual property tax is Rs. 100,000.

7. Zara buys a laptop for Rs. 120,000. If the GST rate is 18%, calculate the GST amount and the total price.
GST amount = 18% x 120,000 = Rs. 21,600.

Total price = 120,000 + 21,600 = Rs. 141,600.

The GST amount is Rs. 21,600 and the total price is Rs. 141,600.

8. Naseem buys a designer dress for Rs. 25,000. If the GST rate is 18%, calculate the GST amount and the total price he has to pay.
GST amount = 18% x 25,000 = Rs. 4,500.

Total price = 25,000 + 4,500 = Rs. 29,500.

The GST amount is Rs. 4,500 and the total price he has to pay is Rs. 29,500.

9. (printed 11) A manufacturer sells goods to a wholesaler for Rs. 300,000. If the VAT rate is 18%, calculate the VAT amount and the total price the wholesaler pays.
VAT amount = 18% x 300,000 = Rs. 54,000.

Total price = 300,000 + 54,000 = Rs. 354,000.

The VAT amount is Rs. 54,000 and the total price the wholesaler pays is Rs. 354,000.

10. (printed 10) If the total amount paid including 18% VAT is Rs. 118,000, what is the original price before VAT?
Original price = total price / (1 + VAT rate).

Original price = 118,000 / 1.18 = Rs. 100,000.

The original price before VAT is Rs. 100,000.

11. (printed 10, second item so numbered) A real estate agent sells a house for Rs. 8,000,000. If the commission rate is 3%, calculate the commission earned by the agent.
Commission = sale price x commission rate.

Commission = 8,000,000 x 3% = Rs. 240,000.

The commission earned by the agent is Rs. 240,000.

12. (printed 11, second item so numbered) If a financial advisor earns a 2% commission on an investment of Rs. 400,000, how much commission does the advisor make?
Commission = 400,000 x 2% = Rs. 8,000.

The advisor makes Rs. 8,000 commission.

Exercise 3.3 (pp.60-61) - Zakat and Ushr
1. Mariam has Rs. 900,000 in her bank account, which she has held for over a year. The nisab threshold for zakat is Rs. 87,480. Calculate the amount of zakat Mariam needs to pay.
Her savings (Rs. 900,000) exceed the nisab (Rs. 87,480), so zakat is due.

Zakat rate = 2.5%.

Zakat = 900,000 x 2.5% = Rs. 22,500.

Mariam needs to pay Rs. 22,500 as zakat.

2. Imran owns 150 grams of gold, which he has held for over a year. The nisab for gold is 87.48 grams. Calculate the amount of zakat Imran needs to pay.
His gold (150 grams) exceeds the nisab (87.48 grams), so zakat is due.

Zakat rate = 2.5%.

Zakat in gold = 150 x 2.5% = 3.75 grams.

Imran needs to pay zakat of 3.75 grams of gold (its rupee value at the current market price). Flag: the question gives no gold price, so no rupee amount can be computed from the question alone; 2.5% of 150 grams is 3.75 grams.

3. Laila paid Rs. 15,000 as zakat on her savings. Calculate the total savings amount Laila had if the zakat rate is 2.5%.
Savings x 2.5% = 15,000.

Savings = 15,000 / 2.5% = 15,000 x 100/2.5 = Rs. 600,000.

Laila had Rs. 600,000 in total savings.

4. Nadia has Rs. 700,000 in savings and 40 grams of gold. The current market value of gold is Rs. 8,000 per gram. The nisab threshold for zakat is Rs. 87,480. Calculate the total amount of zakat Nadia needs to pay on her combined assets.
Value of gold = 40 x 8,000 = Rs. 320,000.

Total wealth = 700,000 + 320,000 = Rs. 1,020,000.

Total wealth exceeds the nisab, so zakat is due.

Zakat = 1,020,000 x 2.5% = Rs. 25,500.

Nadia needs to pay Rs. 25,500 as zakat on her combined assets. Note: the question prints the nisab as Rs. 87,480, while Example 3.19 uses Rs. 874,800 for the gold-based nisab; either way her wealth exceeds the nisab, so the zakat amount is unaffected.

5. Khadija owns a date palm grove that is naturally irrigated by a nearby river. This year, she harvested 2,500 kilograms of dates. Calculate the ushr Khadija needs to pay.
Naturally irrigated crops pay ushr at 10%.

Ushr = 2,500 x 10% = 250 kilograms.

Khadija needs to pay 250 kilograms of dates as ushr.

6. Yusuf owns a farm where he grows various vegetables. His farm is irrigated by a combination of rainwater and a pump. This year, he harvested 1,000 kilograms of vegetables from the rain-fed part and 2,000 kilograms from the pump-irrigated part. Calculate the total ushr Yusuf needs to pay.
Rain-fed part: 1,000 x 10% = 100 kilograms.

Pump-irrigated part: 2,000 x 5% = 100 kilograms.

Total ushr = 100 + 100 = 200 kilograms.

Yusuf needs to pay 200 kilograms of vegetables as total ushr.

7. Bilal has a farm where he grows vegetables. Part of his farm is naturally irrigated by rain, and the rest is irrigated by a canal. This year, he harvested 3,000 kilograms of vegetables from the naturally irrigated part and 4,000 kilograms from the artificially irrigated part. Calculate the total ushr he needs to pay.

Naturally irrigated part: 3,000 x 10% = 300 kilograms.

Artificially irrigated part: 4,000 x 5% = 200 kilograms.

Total ushr = 300 + 200 = 500 kilograms.

Bilal needs to pay 500 kilograms of vegetables as ushr. Note: this question repeats Example 3.21 of the unit word for word (same figures, same answer).

Review Exercise 3 (pp.61-62)

1. Choose the correct option.

i. An item is sold for Rs. 750 with a profit of Rs. 150. What was the cost price?

- Rs. 900
- ✓ Rs. 600
- Rs. 750
- Rs. 150

ii. If the cost price of an item is Rs. 500 and the profit is 20%, what is the selling price?

✓ Rs. 600

Rs. 550

Rs. 625

Rs. 500

iii. An item is sold at Rs. 450 with a loss of Rs. 50. What was the cost price?

Rs. 400

✓ Rs. 500

Rs. 350

Rs. 450

iv. If the cost price is Rs. 800 and the selling price is Rs. 1000, what is the profit percentage?

✓ 25%

20%

10%

15%

v. If the cost price of an item is Rs. 500 and it is sold for Rs. 450, what is the loss percentage?

5%

✓ 10%

15%

20%

vi. The marked price of an item is Rs. 500. If a discount of 10% is given, what is the selling price?

✓ Rs. 450

Rs. 400

Rs. 500

Rs. 350

vii. If an individual's annual income is Rs. 1,500,000 and the tax rate is 10%, what is the income tax amount?

✓ Rs. 150,000

Rs. 120,000

Rs. 90,000

Rs. 75,000

viii. A property is valued at Rs. 5,000,000 and the property tax rate is 2%. What is the annual property tax amount?

Rs. 50,000

✓ Rs. 100,000

Rs. 75,000

Rs. 250,000

ix. If the selling price of an item is Rs. 1,000 and the GST rate is 18%, what is the total price including GST?

✓ Rs. 1,180

Rs. 1,200

Rs. 1,150

Rs. 1,100

x. If a real estate agent sells a house for Rs. 5,000,000 with a commission rate of 3%, what is the commission amount?

✓ Rs. 150,000

Rs. 100,000

Rs. 200,000

Rs. 250,000

2. If the selling price of an item is Rs. 1400 and the profit is Rs. 200, what is the cost price?

Cost price = selling price - profit.

Cost price = 1400 - 200 = Rs. 1,200.

The cost price was Rs. 1,200.

3. If the cost price is Rs. 500 and the selling price is Rs. 600, what is the profit percentage?

Profit = 600 - 500 = Rs. 100.

Profit percentage = 100/500 x 100% = 20%.

The profit percentage is 20%.

4. If the cost price of an item is Rs. 1000 and it is sold for Rs. 800, what is the loss percentage?

Loss = 1000 - 800 = Rs. 200.

Loss percentage = 200/1000 x 100% = 20%.

The loss percentage is 20%.

5. An item with a marked price of Rs. 2000 is sold at a 20% discount. What is the selling price?

Discount = 20% of 2000 = Rs. 400.

Selling price = 2000 - 400 = Rs. 1,600.

The selling price is Rs. 1,600.

6. If an individual's annual income is Rs. 3,000,000 and the tax rate is 12%, what is the income tax amount?

Income tax = 12% of 3,000,000 = Rs. 360,000.

The income tax amount is Rs. 360,000.

7. If the annual property tax for a commercial building is Rs. 90,000 and the tax rate is 2%, what is the value of the property?

Value = property tax / tax rate = 90,000 / 2%.

90,000 x 100/2 = Rs. 4,500,000.

The value of the property is Rs. 4,500,000.

8. If the selling price of an item is Rs. 800 and the GST rate is 18%, what is the total price including GST?

GST amount = 18% of 800 = Rs. 144.

Total price = 800 + 144 = Rs. 944.

The total price including GST is Rs. 944.

9. A financial advisor earns a 1.5% commission on investments totaling Rs. 2,000,000. How much commission does the advisor earn?

Commission = 1.5% of 2,000,000 = Rs. 30,000.

The advisor earns Rs. 30,000 commission.

10. Sara paid Rs. 25,000 as zakat. What was her total savings amount if the zakat rate is 2.5%?

Savings x 2.5% = 25,000.

Savings = 25,000 / 2.5% = 25,000 x 100/2.5 = Rs. 1,000,000.

Her total savings amount was Rs. 1,000,000.

11. Zainab's farm is irrigated by a canal and she harvested 4,000 kilograms of maize. How much ushr is due at a rate of 5%?

Canal irrigation is artificial irrigation, so the ushr rate is 5%.

Ushr = 4,000 x 5% = 200 kilograms.

The ushr due is 200 kilograms of maize.

Unit 4: Square and Square Root

Exercise 4.1 (p.69) - Square of a Number

1. Calculate the square of the following:

25 squared

25 x 25 = 625.

625.

29 squared

29 x 29 = 841.

841.

46 squared

46 x 46 = 2116.

2116.

77 squared

77 x 77 = 5929.

5929.

116 squared

116 x 116 = 13456.

13456.

176 squared

176 x 176 = 30976.

30976.

207 squared

207 x 207 = 42849.

42849.

318 squared

318 x 318 = 101124.

101124.

2. Identify which of the following are perfect squares or not.

324

Perfect square: 18 x 18 = 324, so the square root of 324 is 18.

441

Perfect square: 21 x 21 = 441, so the square root of 441 is 21.

676

Perfect square: 26 x 26 = 676, so the square root of 676 is 26.

1024

Perfect square: 32 x 32 = 1024, so the square root of 1024 is 32.

1369

Perfect square: 37 x 37 = 1369, so the square root of 1369 is 37.

2401

Perfect square: 49 x 49 = 2401, so the square root of 2401 is 49.

289

Perfect square: 17 x 17 = 289, so the square root of 289 is 17.

484

Perfect square: 22 x 22 = 484, so the square root of 484 is 22.

Note: every number in Exercise 4.1 Q2 is a perfect square.

3. A bookshelf has 5 shelves, each measuring 3 feet by 3 feet. What is the total area of the shelves?

Area of one shelf = 3 x 3 = 9 square feet.

Total area = 5 x 9 = 45 square feet.

The total area of the shelves is 45 square feet.

4. What is the area of a square with a perimeter of 256 meters?

Perimeter of a square = 4 x side, so side = 256/4 = 64 meters.

Area = side x side = 64 x 64 = 4096 square meters.

The area is 4096 square meters.

5. The length of one side of a square-shaped table is 215 cm. Find the area of the table.

Area = 215 x 215.

215 x 215 = 46225 square cm.

The area of the table is 46225 square cm.

Exercise 4.2 (p.72) - Square Root by Prime Factorization

1. Find the square root of the following by the prime factorization method.

Square root of 36

Prime factorization: 36 = 2 x 2 x 3 x 3.

Take one factor from each pair and multiply.

The square root of 36 is 6.

Square root of 289

Prime factorization: 289 = 17 x 17.

Take one factor from each pair and multiply.

The square root of 289 is 17.

Square root of 441

Prime factorization: 441 = 3 x 3 x 7 x 7.

Take one factor from each pair and multiply.

The square root of 441 is 21.

Square root of 529

Prime factorization: 529 = 23 x 23.

Take one factor from each pair and multiply.

The square root of 529 is 23.

Square root of 625

Prime factorization: 625 = 5 x 5 x 5 x 5.

Take one factor from each pair and multiply.

The square root of 625 is 25.

Square root of 841

Prime factorization: 841 = 29 x 29.

Take one factor from each pair and multiply.

The square root of 841 is 29.

Square root of 961

Prime factorization: 961 = 31 x 31.

Take one factor from each pair and multiply.

The square root of 961 is 31.

Square root of 1225

Prime factorization: 1225 = 5 x 5 x 7 x 7.

Take one factor from each pair and multiply.

The square root of 1225 is 35.

2. Find the square root of the following fractions by the prime factorization method.

Square root of 64/196

Square root of 64 is 8 and square root of 196 is 14.

So the square root is 8/14 = 4/7 in lowest terms.

The square root of 64/196 is 4/7.

Square root of 225/324

Square root of 225 is 15 and square root of 324 is 18.

So the square root is 15/18 = 5/6 in lowest terms.

The square root of 225/324 is 5/6.

Square root of 256/400

Square root of 256 is 16 and square root of 400 is 20.

So the square root is 16/20 = 4/5 in lowest terms.

The square root of 256/400 is 4/5.

Square root of 484/676

Square root of 484 is 22 and square root of 676 is 26.

So the square root is 22/26 = 11/13 in lowest terms.

The square root of 484/676 is 11/13.

Square root of 1681/784

Square root of 1681 is 41 and square root of 784 is 28.

So the square root is 41/28.

The square root of 1681/784 is 41/28.

Square root of 2116/2209

Square root of 2116 is 46 and square root of 2209 is 47.

So the square root is 46/47.

The square root of 2116/2209 is 46/47.

Square root of 2401/625

Square root of 2401 is 49 and square root of 625 is 25.

So the square root is 49/25.

The square root of 2401/625 is 49/25.

Square root of 2209/1024

Square root of 2209 is 47 and square root of 1024 is 32.

So the square root is 47/32.

The square root of 2209/1024 is 47/32.

Square root of 7921/7225

Square root of 7921 is 89 and square root of 7225 is 85.

So the square root is 89/85.

The square root of 7921/7225 is 89/85.

3. Find the square root of the following decimals by the prime factorization method.

Square root of 3.61

Check: 1.9 x 1.9 = 3.61.

The square root of 3.61 is 1.9.

Square root of 5.76

Check: 2.4 x 2.4 = 5.76.

The square root of 5.76 is 2.4.

Square root of 77.44

Check: 8.8 x 8.8 = 77.44.

The square root of 77.44 is 8.8.

Square root of 0.2916

Check: 0.54 x 0.54 = 0.2916.

The square root of 0.2916 is 0.54.

Square root of 13.69

Check: 3.7 x 3.7 = 13.69.

The square root of 13.69 is 3.7.

Square root of 70.56

Check: 8.4 x 8.4 = 70.56.

The square root of 70.56 is 8.4.

Exercise 4.3 (p.75) - Square Root by Division Method

1. Find the square root by division method.

Square root of 81 by division method

Group the digits in pairs from the right: 81.

Follow the division method step by step; the quotient comes out 9 with remainder 0.

Check: 9 x 9 = 81.

The square root of 81 is 9.

Square root of 441 by division method

Group the digits in pairs from the right: 441.

Follow the division method step by step; the quotient comes out 21 with remainder 0.

Check: 21 x 21 = 441.

The square root of 441 is 21.

Square root of 676 by division method

Group the digits in pairs from the right: 676.

Follow the division method step by step; the quotient comes out 26 with remainder 0.

Check: 26 x 26 = 676.

The square root of 676 is 26.

Square root of 2116 by division method

Group the digits in pairs from the right: 2116.

Follow the division method step by step; the quotient comes out 46 with remainder 0.

Check: 46 x 46 = 2116.

The square root of 2116 is 46.

Square root of 289/729 by division method

Square root of 289 is 17 and square root of 729 is 27.

The square root of 289/729 is 17/27.

Square root of 121/169 by division method

Square root of 121 is 11 and square root of 169 is 13.

The square root of 121/169 is 11/13.

Square root of 4/121 by division method

Square root of 4 is 2 and square root of 121 is 11.

The square root of 4/121 is 2/11.

Square root of 4/169 by division method

Square root of 4 is 2 and square root of 169 is 13.

The square root of 4/169 is 2/13.

Note: Exercise 4.3 Q1 prints the last two fraction items both as (vii); the second is treated here as (viii).

2. Find the square root up to 2 decimal places by division.

Square root of 65.87 up to 2 decimal places

Apply the division method to 2 decimal places.

8.12 x 8.12 = 65.9344, and 8.11 x 8.11 = 65.7721, so 65.87 lies between them and rounds to 8.12.

The square root of 65.87 is about 8.12.

Square root of 0.986 up to 2 decimal places

Apply the division method to 2 decimal places.

0.99 x 0.99 = 0.9801 and 0.993 x 0.993 = 0.986049, so the root rounds to 0.99.

The square root of 0.986 is about 0.99.

Square root of 21.45 up to 2 decimal places

Apply the division method to 2 decimal places.

4.63 x 4.63 = 21.4369 and 4.64 x 4.64 = 21.5296, so the root

rounds to 4.63.

The square root of 21.45 is about 4.63.

Square root of 0.576 up to 2 decimal places
Apply the division method to 2 decimal places.

0.76 x 0.76 = 0.5776 and 0.759 x 0.759 = 0.576081, so the root rounds to 0.76.

The square root of 0.576 is about 0.76.

2. (word problem) A farmer wants to plant 7921 trees in his orchard in such a way that the number of trees in one row is equal to the number of rows. How many trees will be in each row?
Trees per row = square root of 7921.

89 x 89 = 7921.

There will be 89 trees in each row.

3. An urban planner wants to design a square grid of streets in a new development, with 1600 intersections. How many streets will be in each direction?
Streets in each direction = square root of 1600.

40 x 40 = 1600.

There will be 40 streets in each direction.

4. An engineer wants to build a square foundation for a building with 50625 square feet. How many feet will each side be?

Side = square root of 50625.

225 x 225 = 50625.

Each side will be 225 feet.

5. A landscaper wants to plant 1600 flowers in a square garden bed. How many flowers will be in each row?

Flowers per row = square root of 1600.

40 x 40 = 1600.

There will be 40 flowers in each row.

Review Exercise 4 (p.76)

1. Choose the correct option.

i. What is the square root of 225?

14

✓ 15

16

17

ii. What is the square root of 400?

19

✓ 20

21

22

iii. What is the square root of 225/256?

✓ 15/16

16/17

17/18

18/19

iv. What is the square root of 3.24?

1.7

✓ 1.8

1.9

2.0

v. What is the square root of 2.25?

1.2

1.3

1.4

✓ 1.5

vi. A square garden has an area of 49 square meters. What is the length of each side of the garden?

5 m

6 m

✓ 7 m

8 m

vii. A rectangular field has a length of 6 meters and a width of 8 meters. What is the length of the diagonal across the field?

- ✓ 10 m
- 11 m
- 12 m
- 14 m

2. Check whether the following numbers are perfect squares or not.

3025

Perfect square: 55 x 55 = 3025.

2704

Perfect square: 52 x 52 = 2704.

725

Not a perfect square: 26 x 26 = 676 and 27 x 27 = 729, and 725 lies between them, so no whole number squared gives 725.

3. Find the square root of the following:

Square root of 1681

41 x 41 = 1681.

The square root of 1681 is 41.

Square root of 900

30 x 30 = 900.

The square root of 900 is 30.

Square root of 625

25 x 25 = 625.

The square root of 625 is 25.

Square root of 36/49

Square root of 36 is 6 and square root of 49 is 7.

The square root of 36/49 is 6/7.

Square root of 121/49

Square root of 121 is 11 and square root of 49 is 7.

The square root of 121/49 is 11/7.

Square root of 169/289

Square root of 169 is 13 and square root of 289 is 17.

The square root of 169/289 is 13/17.

Square root of 81

9 x 9 = 81.

The square root of 81 is 9.

4. A contractor wants to pave a square parking lot with 28900

square feet of blacktop. How many feet will each side be?

Side = square root of 28900.

170 x 170 = 28900.

Each side will be 170 feet.

5. In a school assembly 1024 students stand in rows in such a way that the number of rows is equal to the number of students in a row. How many students are there in each row?

Students per row = square root of 1024.

32 x 32 = 1024.

There are 32 students in each row.

6. A farmer wants 3600 trees in his orchard in a square pattern. How many trees will be in each row?

Trees per row = square root of 3600.

60 x 60 = 3600.

There will be 60 trees in each row.

Unit 5: Set

Note on Unit 5: Venn diagrams are described in words in these notes, because the builder draws only single shapes (circle, square, rectangle, triangle) and has no overlapping-circle Venn renderer. Each description states the rectangle (universal set), the circles, and which elements go in each region, so the diagram can be drawn exactly as described.

Exercise 5.1 (p.82) - Forms of a Set

1. Write the following sets in descriptive form.

(i) {1, 2, 3, 4, 5, 6}

The set of the first six natural numbers (counting numbers from 1 to 6).

(ii) {x | x is a factor of 24}

The set of all factors of 24, that is {1, 2, 3, 4, 6, 8, 12, 24}.

(iii) {5, 7, 11, 17, 19}

The set of prime numbers 5, 7, 11, 17 and 19. Flag: 13 is also a prime number between 5 and 19 but is missing from the printed list, so the list as printed is not the full set of primes in that range.

(iv) {x | x is an even number}

The set of all even numbers.

(v) {3, 6, 9, 12, 15, 18}

The set of the first six multiples of 3 (multiples of 3 up to 18).

2. Write the following sets in tabular form.

(i) {x | x is a multiple of 7 between 35 and 63}

{42, 49, 56}.

(ii) The set of even numbers between 20 and 32.

{22, 24, 26, 28, 30}.

(iii) {x | x is a multiple of 4 between 16 and 32}

{20, 24, 28}.

(iv) {x | x is a prime number between 20 and 30}

{23, 29}.

(v) {x | x is a factor of 48}

{1, 2, 3, 4, 6, 8, 12, 16, 24, 48}.

(vi) The set of odd numbers between 50 and 70.

{51, 53, 55, 57, 59, 61, 63, 65, 67, 69}.

3. Write the following sets in set builder form.

(i) {11, 13, 15, 17, 19}

{x | x is an odd number between 10 and 20}.

(ii) {2, 3, 5, 7, 11}

{x | x is a prime number less than 12}.

(iii) The set of perfect squares less than 30.

{x | x is a perfect square and x < 30} = {1, 4, 9, 16, 25}.

(iv) {-1, -2, -3, -4, -5}

{x | x is a negative integer greater than -6}.

(v) The set of multiples of 5 less than 30.

{x | x is a multiple of 5 and x < 30} = {5, 10, 15, 20, 25}.

Exercise 5.2 (p.87) - Venn Diagrams, Subsets and Supersets

1. Represent sets through Venn diagrams, with U = {1, 2, 3, 4, 5, 6, 7, 8} as the universal set.

(i) A = {1, 2, 3, 4, 5}

Draw a rectangle for U. Inside it draw one circle for A containing 1, 2, 3, 4, 5. Write 6, 7, 8 inside the rectangle but outside the circle.

(ii) A = {1, 2, 3}, B = {1, 2, 3, 4, 5}

Draw a rectangle for U. Draw a circle for B containing 1, 2, 3, 4, 5, and inside it a smaller circle for A containing 1, 2, 3 (A is a subset of B). Write 4, 5 inside B but outside A, and 6, 7, 8 inside the rectangle but outside B.

(iii) A = {2, 4, 6, 8}, B = {1, 3, 5, 7}

Draw a rectangle for U. Draw two separate (non-overlapping) circles: one for A containing 2, 4, 6, 8 and one for B containing 1, 3, 5, 7. No element of U lies outside the two circles.

(iv) A = {1, 2, 3, 4, 5, 6}, B = {4, 5, 6, 7, 8}

Draw a rectangle for U. Draw two overlapping circles: in the A-only part write 1, 2, 3; in the overlapping part write 4, 5, 6; in the B-only part write 7, 8. No element of U lies outside the two circles.

2. Identify and differentiate the subset and superset in the following sets.

(i) A = {1, 2, 3, 4, 5}, B = {3, 4, 5}

Every element of B (3, 4, 5) is in A, so B is a subset of A (B is contained in A) and A is a superset of B.

(ii) C = {x | x is a whole number, 4 <= x <= 20} = {4, 5, ..., 20}; D = set of the first 6 multiples of 3 = {3, 6, 9, 12, 15, 18}

Neither is a subset of the other: 3 is in D but not in C, and 4 is in C but not in D. The sets overlap (common elements 6, 9, 12, 15, 18).

(iii) E = set of prime numbers less than 25 and greater than 2 = {3, 5, 7, 11, 13, 17, 19, 23}; F = set of odd numbers

Every element of E is odd (every prime greater than 2 is odd), so E is a subset of F and F is a superset of E.

(iv) G = {-3, -2, -1, 0, 1, 2, 3, 4}; H = set of integers

Every element of G is an integer, so G is a subset of H and H is a superset of G.

(v) I = set of natural numbers greater than 3 and less than 6 = {4, 5}; J = set of factors of 20 = {1, 2, 4, 5, 10, 20}

Every element of I (4, 5) is a factor of 20, so I is a subset of J and J is a superset of I.

3. Identify proper and improper subsets in the following sets and differentiate them.

(i) U = {1, 2, 3, 4, 5, 6}; A = {2, 3}, B = {4, 5, 6}, C = {1, 2, 3, 4, 5, 6}

A is a proper subset of U (A is contained in U but A is not equal to U). B is a proper subset of U for the same reason. C is equal to U, so C is an improper subset of U. A proper subset leaves out at least one element of the parent set; an improper subset is the set itself.

(ii) U = {x | x is an integer, 1 < x <= 15} = {2, 3, ..., 15}; P = {x | x is prime, 2 < x < 15} = {3, 5, 7, 11, 13}; R = {x | x is a whole number, 3x <= 15}; S = {x | x is an integer greater than 1 and less than or equal to 15}

P is a proper subset of U. S is equal to U, so S is an improper subset of U. R = {0, 1, 2, 3, 4, 5} (whole numbers with x <= 5); R is NOT a subset of U at all, because 0 and 1 are in R but not in U. Flag: R's whole-number reading includes 0, which falls outside U.

4. Write all subsets of the following sets and identify proper and improper subsets.

T = {a, e, i}

All 8 subsets: {} (empty set), {a}, {e}, {i}, {a, e}, {a, i}, {e, i}, {a, e, i}. The 7 subsets other than {a, e, i} itself are proper subsets; {a, e, i} itself is the improper subset.

S = {9, 10, 11}

All 8 subsets: {}, {9}, {10}, {11}, {9, 10}, {9, 11}, {10, 11}, {9,

10, 11}. The 7 subsets other than {9, 10, 11} itself are proper subsets; {9, 10, 11} itself is the improper subset.

V = {Quetta, Pishin, Loralai, Khuzdar}
All 16 subsets: {}; the 4 singletons {Quetta}, {Pishin}, {Loralai}, {Khuzdar}; the 6 pairs {Quetta, Pishin}, {Quetta, Loralai}, {Quetta, Khuzdar}, {Pishin, Loralai}, {Pishin, Khuzdar}, {Loralai, Khuzdar}; the 4 triples {Quetta, Pishin, Loralai}, {Quetta, Pishin, Khuzdar}, {Quetta, Loralai, Khuzdar}, {Pishin, Loralai, Khuzdar}; and {Quetta, Pishin, Loralai, Khuzdar} itself. The 15 subsets other than V itself are proper subsets; V itself is the improper subset.

5. Separate equal and equivalent sets in the following sets.
(i) A = {2, 3, 5}, B = {a, b, c}
Equivalent but not equal: both sets have 3 elements, but their elements are different.

(ii) C = {7, 9, 10}, D = {10, 9, 7}
Equal and equivalent: both sets have exactly the same elements (order does not matter), and both have 3 elements.

(iii) D = {Saturday, Sunday, Monday}, E = {March, April, May}
Equivalent but not equal: both sets have 3 elements, but their elements are different.

(iv) F = {1, 2, 3, 4, 5, 6, 7}, G = {7, 6, 5, 4, 3, 2, 1}
Equal and equivalent: both sets have exactly the same elements, and both have 7 elements.

Exercise 5.3 (p.96) - Union, Intersection, Complement and De Morgan's Laws

1. Find the union of the following sets and represent in a Venn diagram.

(i) A = {i, j, k, l, m, n}, B = {p, q, r, s}

A union B = {i, j, k, l, m, n, p, q, r, s}. Venn diagram: two separate circles (no common elements), one containing i, j, k, l, m, n and the other p, q, r, s; the union is everything inside either circle.

(ii) C = {x | x is a factor of 8}, D = {x | x is a natural number, x <= 7}

C = {1, 2, 4, 8} (positive factors of 8) and D = {1, 2, 3, 4, 5, 6, 7}. C union D = {1, 2, 3, 4, 5, 6, 7, 8}. Venn diagram: two overlapping circles; overlap {1, 2, 4}; C-only {8}; D-only {3, 5, 6, 7}.

(iii) E = {x | x is a multiple of 5, x < 30}, F = {x | x is a multiple of 6, x <= 30}

E = {5, 10, 15, 20, 25} and F = {6, 12, 18, 24, 30}. E union F = {5, 6, 10, 12, 15, 18, 20, 24, 25, 30}. Venn diagram: two separate circles (no common elements).

(iv) P = {7, 11, 15, 19, 23, 27}, Q = {15, 16, 17, 18, 19, 20, 21, 22, 23}

P union Q = {7, 11, 15, 16, 17, 18, 19, 20, 21, 22, 23, 27}. Venn diagram: two overlapping circles; overlap {15, 19, 23}; P-only {7, 11, 27}; Q-only {16, 17, 18, 20, 21, 22}.

2. Find the intersection of the following sets and represent in a Venn diagram.

(i) P = {1, 2, 3, 4, 5, 6, 7, 11, 15}, Q = {5, 6, 7, 8, 9, 10}

P intersection Q = {5, 6, 7}. Venn diagram: two overlapping circles with 5, 6, 7 in the overlap.

(ii) R = {x | x is a factor of 12}, S = {x | x is a composite number <= 12}

R = {1, 2, 3, 4, 6, 12} and S = {4, 6, 8, 9, 10, 12}. R intersection S = {4, 6, 12}. Venn diagram: two overlapping circles with 4, 6, 12 in the overlap.

(iii) T = {x | x is prime, 7 <= x <= 18} = {7, 11, 13, 17}, U = {4, 6, 8, 10, 12, 14}

T intersection U = {} (the empty set): no element is common to both sets. Venn diagram: two separate circles with nothing in the overlap.

(iv) X = {1, 2, 3, ..., 18}, Y = {x | x is divisible by 3, x <= 20}

Y = {3, 6, 9, 12, 15, 18}. X intersection Y = {3, 6, 9, 12, 15, 18}. Venn diagram: two overlapping circles with 3, 6, 9, 12, 15, 18 in the overlap.

3. Represent the following in Venn diagrams and identify disjoint and overlapping sets.

(i) A = {1, 2, 3}, B = {3, 4, 5}

Overlapping sets: they share the element 3. Venn diagram: two overlapping circles with 3 in the overlap.

(ii) C = {1, 3, 5, 7, 9, 11}, D = {2, 3, 5, 7, 11}

Overlapping sets: they share 3, 5, 7 and 11. Venn diagram: two overlapping circles with 3, 5, 7, 11 in the overlap.

(iii) E = set of odd numbers, F = set of even numbers

Disjoint sets: no number is both odd and even. Venn diagram: two separate circles with nothing in the overlap.

(iv) G = set of names of days of the week, H = set of names of months

Disjoint sets: no name is both a day and a month. Venn diagram: two separate circles with nothing in the overlap.

4. Find the complement of the given set.

(i) U = {3, 6, 9, 12, 15, 18, 21}, A = {12, 15, 18}. Find A'

A' = {3, 6, 9, 21}: the elements of U that are not in A.

(ii) U = set of natural numbers, B = set of even numbers. Find B'

B' = the set of odd natural numbers: every natural number that is not even is odd.

(iii) U = {x | x is a perfect square less than 30} = {1, 4, 9, 16, 25}; C = {x | x is a multiple of 4, x <= 30}. Find C' (relative to U)

Within U, the multiples of 4 are 4 and 16, so C' = {1, 9, 25}: the elements of U that are not multiples of 4.

(iv) U = {1, 2, 5, 10, 17}, D = factors of 10. Find D'

D = {1, 2, 5, 10}, so D' = {17}: the element of U that is not a factor of 10.

5. Verify De Morgan's laws.

(i) U = {8, 9, 10, 11, 12, 13, 14, 15, 16}; A = {5, 6, 7, 8, 9}; B = {14, 15, 16, 17, 18}

Flag: A and B as printed contain elements outside U (5, 6, 7 in A; 17, 18 in B). Complements are taken relative to U, so the laws still verify. First law: A union B = {5, 6, 7, 8, 9, 14, 15, 16, 17, 18}; (A union B)' = {10, 11, 12, 13}. A' = {10, 11, 12, 13, 14, 15, 16} and B' = {8, 9, 10, 11, 12, 13}; A' intersection B' = {10, 11, 12, 13}. So (A union B)' = A' intersection B'. Second law: A intersection B = {} (empty set); (A intersection B)' = U = {8, 9, 10, 11, 12, 13, 14, 15, 16}. A' union B' = {8, 9, 10, 11, 12, 13, 14, 15, 16} = U. So (A intersection B)' = A' union B'. Both laws verified.

(ii) U = {-3, -2, -1, 0, 1, 2, 3, 4}; A = {0, 1, 2, 3}; B = {-2, -1, 0, 1}

First law: A union B = {-2, -1, 0, 1, 2, 3}; (A union B)' = {-3, 4}. A' = {-3, -2, -1, 4} and B' = {-3, 2, 3, 4}; A' intersection B' = {-3, 4}. So (A union B)' = A' intersection B'. Second law: A intersection B = {0, 1}; (A intersection B)' = {-3, -2, -1, 2, 3, 4}. A' union B' = {-3, -2, -1, 2, 3, 4}. So (A intersection B)' = A' union B'. Both laws verified.

Review Exercise 5 (p.97)

1. Tick the correct option.

i. Which of the following is a set?

✓ {1, 2, 3, 4}

(1, 2, 3, 4)

[1, 2, 3, 4]

1, 2, 3, 4

ii. If A = {1, 2, 3} and B = {2, 3, 4}, what is A intersection B?

{1, 2}

✓ {2, 3}

{3, 4}

{1, 4}

iii. Which set is an example of an infinite set?

{1, 2, 3, 4, 5}

{0}

✓ {x | x is a natural number}

{a, e, i, o, u}

iv. Which set is a proper subset of {1, 2, 3}?

{1, 2, 3}

{1, 2, 4}

✓ {1, 2}

{2, 3, 4}

v. What is the complement of set A = {1, 2} in the universal set U = {1, 2, 3, 4}?

{1, 2}

✓ {3, 4}

{1, 2, 3, 4}

{2, 3}

vi. Which set is an example of a disjoint set when compared with {1, 2, 3}?

{3, 4, 5}

{2, 3, 4}

✓ {4, 5, 6}

{1, 2, 4}

vii. If A is a subset of B and B is a superset of A, then:

✓ A = B

A is not equal to B

A - B

B - A

viii. If A is a superset of B then it is represented as:

A is a subset of B

✓ A is a superset of B

A is a proper subset of B

A is a proper superset of B

ix. Which of the following is an example of overlapping sets?

{1, 2} and {3, 4}

✓ {a, b} and {b, c}

{x, y} and {z, u}

{p, q} and {r, s}

x. If every element of set A is also an element of set B, then A is a:

superset

✓ subset

union

intersection

Unit 6: Polynomials

Exercise 6.1 (p.105) - Patterns and Sequences

1. Use the first three terms in the pattern to find the next three terms.

(i) 22, 29, 36, ...

The difference is 29 - 22 = 7 and 36 - 29 = 7, so add 7 each time.

Next terms: 36 + 7 = 43, 43 + 7 = 50, 50 + 7 = 57.

The next three terms are 43, 50, 57.

(ii) 100, 96, 92, ...

The difference is 96 - 100 = -4 and 92 - 96 = -4, so subtract 4 each time.

Next terms: 92 - 4 = 88, 88 - 4 = 84, 84 - 4 = 80.

The next three terms are 88, 84, 80.

(iii) -9, 9, -9, ...

The signs alternate: -9, then 9, then -9, so the next signs are +, -, +.

Next terms: 9, -9, 9.

The next three terms are 9, -9, 9.

(iv) -24, -22, -20, ...

The difference is -22 - (-24) = 2 and -20 - (-22) = 2, so add 2 each time.

Next terms: -20 + 2 = -18, -18 + 2 = -16, -16 + 2 = -14.

The next three terms are -18, -16, -14.

2. Use the pattern rule to make the pattern.

(i) Start at 5 and add 11 each time.

5, 5 + 11 = 16, 16 + 11 = 27, 27 + 11 = 38, 38 + 11 = 49.

The pattern is 5, 16, 27, 38, 49.

(ii) Start at 8 and multiply by 3 each time.

8, 8 x 3 = 24, 24 x 3 = 72, 72 x 3 = 216, 216 x 3 = 648.

The pattern is 8, 24, 72, 216, 648.

(iii) Start at 16 and subtract 1 each time.

16, 15, 14, 13, 12.

The pattern is 16, 15, 14, 13, 12.

(iv) Start at 243 and divide by 3 each time.

243, 243 / 3 = 81, 81 / 3 = 27, 27 / 3 = 9, 9 / 3 = 3.

The pattern is 243, 81, 27, 9, 3.

3. What number is added or subtracted to make the sequence?

(i) 6, 12, 18, 24, ...

6 is added each time (12 - 6 = 6, 18 - 12 = 6, 24 - 18 = 6).

(ii) 21, 22, 23, 24, ...

1 is added each time.

(iii) 40, 37, 34, 31, ...

3 is subtracted each time (37 - 40 = -3, 34 - 37 = -3, 31 - 34 = -3).

(iv) -3, -5, -7, -9, ...

2 is subtracted each time (-5 - (-3) = -2, -7 - (-5) = -2, -9 - (-7) = -2).

4. What number is multiplied or divided to make the sequence?

Flag: Exercise 6.1 Q4 (i) and (ii) as printed are addition sequences, not multiplication or division sequences, so no multiplication/division rule fits them; their additive rules are stated. Q4 (iii) as printed ends in 3, but dividing by 3 gives 5 after 15, so the printed 3 is treated as a misprint for 5.

(i) 4, 8, 12, 16, ...

No multiplication or division rule fits: the sequence adds 4 each time (8 - 4 = 4, 12 - 8 = 4, 16 - 12 = 4).

(ii) 9, 18, 27, 36, ...

No multiplication or division rule fits: the sequence adds 9 each

time.

(iii) 135, 45, 15, ...

3 is the divisor: 135 / 3 = 45, 45 / 3 = 15, 15 / 3 = 5, so the next term is 5 (the printed 3 after 15 does not follow the division rule).

(iv) 256, 64, 16, 4, ...

4 is the divisor: 256 / 4 = 64, 64 / 4 = 16, 16 / 4 = 4.

Exercise 6.2 (pp.111-112) - Algebraic Expressions

1. Identify variables, constants and coefficients in each of the following algebraic expressions.

3x

Variable: x. Constant: none. Coefficient: 3.

x + 5

Variable: x. Constant: 5. Coefficient: 1 (the coefficient of x).

x + 2y - 4

Variables: x, y. Constant: -4. Coefficients: 1 (of x) and 2 (of y).

3x^2 - 6x - 2

Variable: x. Constant: -2. Coefficients: 3 (of x^2) and -6 (of x).

4x^2 - 5y^2 + 2xy - 7

Variables: x, y. Constant: -7. Coefficients: 4 (of x^2), -5 (of y^2) and 2 (of xy).

2. Write the number of terms of the following algebraic expressions.

3x

1 term.

2x + 3

2 terms.

6x - 7

2 terms.

4x + 5y

2 terms.

2x - 3y - 2

3 terms.

3x/5

1 term.

7xy

1 term.

9

1 term.

3xyz/4

1 term.

3. Identify the like and unlike terms in each of the following algebraic expressions.

(i) 3x - 2x

Like terms: 3x and -2x (same variable x with the same exponent).

Unlike terms: none.

(ii) 5x + 4y

Like terms: none (x and y are different variables). Unlike terms: 5x and 4y.

(iii) 2x - y + 5

Like terms: none. Unlike terms: 2x, -y and 5 (all different).

(iv) x + 3y - 5y

Like terms: 3y and -5y. Unlike terms: x (the remaining term).

(v) 2x - 7y + x + 2y + 1

Like terms: 2x and x; -7y and 2y. Unlike term: 1 (the constant, which has no like partner).

4. Identify the open and closed sentences in the following.

x + 2 = 3

Open sentence: it contains the variable x, so its truth depends on the value of x.

x - 4 < 3

Open sentence: it contains the variable x.

$4 - y > 5$

Open sentence: it contains the variable y.

$8 - 5 = 3$

Closed sentence: it contains no variable, and it is true.

$7 - 2 = 6$

Closed sentence: it contains no variable, and it is false ($7 - 2 = 5$).

$6y > 4y$

Open sentence: it contains the variable y.

$x + 8 = 8 + x$

Open sentence: it contains the variable x (it is true for every x, but it is still an open sentence).

$2x < 4x$

Open sentence: it contains the variable x.

$5x < 4x$

Open sentence: it contains the variable x.

$x < 3x$

Open sentence: it contains the variable x.

5. Identify and write the names of the algebraic expressions, equations and inequalities in the following.

$3x + 1$

Algebraic expression (no equality or inequality sign).

$3x + 1 = 2$

Equation (it has an = sign).

$3x + 1 > 2$

Inequality (it has a > sign).

$(2x - 3)/3 = 5$

Equation (it has an = sign).

$5x - 2 = 7$

Equation (it has an = sign).

$6x < 2x + 1$

Inequality (it has a < sign).

$5x > 3x$

Inequality (it has a > sign).

$x - 1 < 2x + 3$

Inequality (it has a < sign).

$6x - 2 = 4x - 1$

Equation (it has an = sign).

$4x > 3x$

Inequality (it has a > sign).

6. Shazia is 5 years older than her brother Amjad. Write the ages of Amjad and Shazia in variables and find their ages for various numerical values.

Let Amjad's age be x years.

Then Shazia's age is x + 5 years.

For x = 5: Amjad is 5 years old and Shazia is 10 years old.

For x = 10: Amjad is 10 years old and Shazia is 15 years old.

For x = 15: Amjad is 15 years old and Shazia is 20 years old.

Amjad's age = x years, Shazia's age = x + 5 years; for example x = 5 gives 5 and 10, x = 10 gives 10 and 15, x = 15 gives 15 and 20.

7. Naeem bought a geometry box for Rs. 50 more than Waleed. Write the algebraic expression for it and find their purchases for various numerical values.

Let Waleed's geometry box cost Rs. x.

Then Naeem's geometry box costs Rs. (x + 50).

For x = 100: Waleed Rs. 100, Naeem Rs. 150.

For x = 200: Waleed Rs. 200, Naeem Rs. 250.

Naeem's cost = Rs. (x + 50) where Waleed's cost is Rs. x; for example x = 100 gives Rs. 150, x = 200 gives Rs. 250.

Exercise 6.3 (p.117) - Polynomials

1. Identify whether each of the following algebraic expressions is a polynomial or not, and why.

5x

Polynomial: the variable x has the whole-number exponent 1.

3x + 1

Polynomial: all variable exponents are whole numbers.

x^2 + 2x + 1

Polynomial: all variable exponents are whole numbers.

7x/6 + 1

Polynomial: the coefficient 7/6 is allowed; the exponent of x is 1.

(x + 2)/x

Not a polynomial: (x + 2)/x = 1 + 2/x, and 2/x = 2x^(-1) has a negative exponent.

1/x - 4

Not a polynomial: 1/x = x^(-1) has a negative exponent.

(1/5)x^2 - 2x

Polynomial: all variable exponents are whole numbers.

1 + 3/y

Not a polynomial: 3/y = 3y^(-1) has a negative exponent.

1/y^2 + 4

Not a polynomial: 1/y^2 = y^(-2) has a negative exponent.

x^2 + 5xy + y^2

Polynomial: all variable exponents are whole numbers.

2. Identify the following polynomials as monomials, binomials and trinomials.

-6x

Monomial: one term.

7x - 4x

Binomial as written (two terms); it simplifies to the monomial 3x.

3x^2 + x

Binomial: two terms.

5x^2 + 6x + 2

Trinomial: three terms.

x^3 - 2x^2 + 6x

Trinomial: three terms.

(4x - 2)/2

Binomial: it simplifies to 2x - 1, which has two terms.

14x/7

Monomial: it simplifies to 2x, which has one term.

2x - 5xy

Binomial: two terms.

4x^2 - 12xy + 9y^2

Trinomial: three terms.

x^3 - 9

Binomial: two terms.

3. Identify the degree of the following polynomials.

4x

Degree 1 (the greatest exponent of the variable).

3x + 2

Degree 1 (the greatest exponent of the variable).

2x^3 - 4x^2 + x

Degree 3 (the greatest exponent of the variable).

6x^4 - 5x^2

Degree 4 (the greatest exponent of the variable).

8y - y^6

Degree 6 (the greatest exponent of the variable).

y^7 + 5

Degree 7 (the greatest exponent of the variable).

3y^2 + 5y^3 - 9y^4

Degree 4 (the greatest exponent of the variable).

9y^3 + 4y^5

Degree 5 (the greatest exponent of the variable).

8

Degree 0 (the greatest exponent of the variable).

7x^8

Degree 8 (the greatest exponent of the variable).

4. Add the following polynomials.

5x + 2x

5x + 2x = (5 + 2)x = 7x.

7x

-5x + (-2x)

-5x + (-2x) = (-5 - 2)x = -7x.

-7x

-5x + 2x

-5x + 2x = (-5 + 2)x = -3x.

-3x

2x^2 + 3x

2x^2 and 3x are unlike terms, so they cannot be combined.

2x^2 + 3x

4x^3 + (-2x^2)

4x^3 and -2x^2 are unlike terms, so they cannot be combined.

4x^3 - 2x^2

(5x + 2y) + (3x + 4y)

Add x terms: 5x + 3x = 8x.

Add y terms: 2y + 4y = 6y.

8x + 6y

(-3x - 6y) + (-2x - y)

Add x terms: -3x - 2x = -5x.

Add y terms: -6y - y = -7y.

-5x - 7y

(7x + 6y + 3) + (-5x - 4y - 8)

Add x terms: 7x - 5x = 2x.

Add y terms: 6y - 4y = 2y.

Add constants: 3 - 8 = -5.

2x + 2y - 5

5. Find the sum of the following polynomials.

6x, 8x

6x + 8x = 14x.

14x

x^2 - 5, 3x^2 + 2

Add x^2 terms: x^2 + 3x^2 = 4x^2.

Add constants: -5 + 2 = -3.

4x^2 - 3

6x^2 + x - 4, 2x^2 - 5x + 3

Add x^2 terms: 6x^2 + 2x^2 = 8x^2.

Add x terms: x - 5x = -4x.

Add constants: -4 + 3 = -1.

$8x^2 - 4x - 1$

$x^2 + 2xy + y^2, x^2 - 2xy + y^2$

Add x^2 terms: $x^2 + x^2 = 2x^2$.

Add xy terms: $2xy - 2xy = 0$.

Add y^2 terms: $y^2 + y^2 = 2y^2$.

$2x^2 + 2y^2$

$2x^3 - 7x^2 + 4x - 5, x^3 - x + 2x^2 + 8$

Add x^3 terms: $2x^3 + x^3 = 3x^3$.

Add x^2 terms: $-7x^2 + 2x^2 = -5x^2$.

Add x terms: $4x - x = 3x$.

Add constants: $-5 + 8 = 3$.

$3x^3 - 5x^2 + 3x + 3$

$3a^2 - 5ab - 3b^2, 2a^2 - 3ab + 6b^2$

Add a^2 terms: $3a^2 + 2a^2 = 5a^2$.

Add ab terms: $-5ab - 3ab = -8ab$.

Add b^2 terms: -3b^2 + 6b^2 = 3b^2.

5a^2 - 8ab + 3b^2

6. Subtract the following polynomials.

5x - 4x

5x - 4x = (5 - 4)x = x.

x

-5x - (-4x)

-5x - (-4x) = -5x + 4x = -x.

-x

-5x - 4x

-5x - 4x = (-5 - 4)x = -9x.

-9x

-3x - 2y

-3x and 2y are unlike terms, so the difference stays -3x - 2y (the printed box pairs -3x with 2y as separate terms).

-3x - 2y

(6x + 5y) - (3x + 2y)

6x - 3x = 3x.

5y - 2y = 3y.

3x + 3y

(-7x - 6y) - (4x + 5y)

-7x - 4x = -11x.

-6y - 5y = -11y.

-11x - 11y

(x^3 - 6x^2 - 5y + 8) - (4x^3 + x^2 + 3y + 7)

x^3 - 4x^3 = -3x^3.

-6x^2 - x^2 = -7x^2.

-5y - 3y = -8y.

8 - 7 = 1.

-3x^3 - 7x^2 - 8y + 1

(3x^2 - 5y^2 + 6xy) - (3x^2 - 3y^2 - 4xy)

3x^2 - 3x^2 = 0.

-5y^2 - (-3y^2) = -2y^2.

6xy - (-4xy) = 10xy.

-2y^2 + 10xy

(6x^3 - 7y^3 + 3) - (2x^3 + 4y^3 + 5)

6x^3 - 2x^3 = 4x^3.

-7y^3 - 4y^3 = -11y^3.

3 - 5 = -2.

4x^3 - 11y^3 - 2

7. Find the difference of the following polynomials (first from second expression, that is, second minus first).

-5x^4, 6x^4

6x^4 - (-5x^4) = 6x^4 + 5x^4 = 11x^4.

11x^4

3x^5, 4y^5

4y^5 - 3x^5: the terms are unlike, so they cannot be combined.

-3x^5 + 4y^5

5x^3 + 4x^2, 3x^3 - 8x^2

3x^3 - 5x^3 = -2x^3.

-8x^2 - 4x^2 = -12x^2.

-2x^3 - 12x^2

6x^3 - x + 2x^2, -2x^3 + 4x^2 - 5x

-2x^3 - 6x^3 = -8x^3.

4x^2 - 2x^2 = 2x^2.

-5x - (-x) = -4x.

-8x^3 + 2x^2 - 4x

3x^2 - 5x + 4, 7x^2 - x + 1

7x^2 - 3x^2 = 4x^2.

-x - (-5x) = 4x.

1 - 4 = -3.

4x^2 + 4x - 3

8. Subtract 6a^2 - 3b^2 + 5c^2 from 8a^2 - 2b^2 - 7c^2.

(8a^2 - 2b^2 - 7c^2) - (6a^2 - 3b^2 + 5c^2).

8a^2 - 6a^2 = 2a^2.

-2b^2 - (-3b^2) = b^2.

-7c^2 - 5c^2 = -12c^2.

2a^2 + b^2 - 12c^2.

9. Subtract a^3 - 2b^2 + 3a^2b^2 from 4a^3 - 6b^2 - 8a^2b^2.

(4a^3 - 6b^2 - 8a^2b^2) - (a^3 - 2b^2 + 3a^2b^2).

4a^3 - a^3 = 3a^3.

-6b^2 - (-2b^2) = -4b^2.

-8a^2b^2 - 3a^2b^2 = -11a^2b^2.

3a^3 - 4b^2 - 11a^2b^2.

Exercise 6.4 (p.121) - Multiplication of Polynomials

1. Multiply the following polynomials by Horizontal Method.

-3x by 7

-3x x 7 = -21x.

-21x

5x by -4x

5x x (-4x) = -20x^2.

-20x^2

6x^2 by -8x^3

6x^2 x (-8x^3) = -48x^5 (add exponents: 2 + 3 = 5).

-48x^5

-4x^3 by -2xy

-4x^3 x (-2xy) = 8x^4y (add exponents of x: 3 + 1 = 4).

8x^4y

4x^2y by 5xy^2

4x^2y x 5xy^2 = 20x^3y^3.

20x^3y^3

3x by 2x + 3y

3x x 2x = 6x^2.

3x x 3y = 9xy.

6x^2 + 9xy

5x^2 by x - 4y

5x^2 x x = 5x^3.

5x^2 x (-4y) = -20x^2y.

5x^3 - 20x^2y

-2x^3 by -4x^2 - 5y^2

-2x^3 x (-4x^2) = 8x^5.

-2x^3 x (-5y^2) = 10x^3y^2.

8x^5 + 10x^3y^2

6x by x^2 + y^2 + 2x

6x x x^2 = 6x^3.

6x x y^2 = 6xy^2.

6x x 2x = 12x^2.

6x^3 + 6xy^2 + 12x^2

-5x^3 by x^3 + y^3 + 2x^2y^2

-5x^3 x x^3 = -5x^6.

-5x^3 x y^3 = -5x^3y^3.

-5x^3 x 2x^2y^2 = -10x^5y^2.

-5x^6 - 5x^3y^3 - 10x^5y^2

2. Multiply the following polynomials by Vertical Method.

(a - b)(a - b)

(a - b)(a - b) = a x a - a x b - b x a + b x b = a^2 - 2ab + b^2.

a^2 - 2ab + b^2

(4a - 5)(4a^2 + 3)

4a x 4a^2 = 16a^3.

4a x 3 = 12a.

-5 x 4a^2 = -20a^2.

-5 x 3 = -15.

16a^3 - 20a^2 + 12a - 15

(2a - b)(a^2 - a^2)

Flag: as printed (verified at 300 dpi) the second factor is a^2 - a^2 = 0, so the product is 0. This looks like a print error: if the intended factor were (a - a^2), the product would be 2a^2 - 2a^3 - ab + a^2b.

0

(x - y)(x^2 + y^2 - 2xy)

Note that x^2 + y^2 - 2xy = (x - y)^2, so the product is (x - y)^3.

(x - y)^3 = x^3 - 3x^2y + 3xy^2 - y^3.

x^3 - 3x^2y + 3xy^2 - y^3

(a + b)(3a^2 + 3b^2 - 2ab)

a x 3a^2 = 3a^3; a x 3b^2 = 3ab^2; a x (-2ab) = -2a^2b.

b x 3a^2 = 3a^2b; b x 3b^2 = 3b^3; b x (-2ab) = -2ab^2.

Combine: 3a^3 + 3b^3 + (-2a^2b + 3a^2b) + (3ab^2 - 2ab^2) = 3a^3 + 3b^3 + a^2b + ab^2.

3a^3 + 3b^3 + a^2b + ab^2

(5a + 3b)(2a - 3b - 4c)

5a x 2a = 10a^2; 5a x (-3b) = -15ab; 5a x (-4c) = -20ac.

3b x 2a = 6ab; 3b x (-3b) = -9b^2; 3b x (-4c) = -12bc.

Combine: 10a^2 + (-15ab + 6ab) - 20ac - 9b^2 - 12bc = 10a^2 - 9ab - 20ac - 9b^2 - 12bc.

10a^2 - 9ab - 20ac - 9b^2 - 12bc

(a + b)(a^2 - ab + b^2)

a x a^2 = a^3; a x (-ab) = -a^2b; a x b^2 = ab^2.

b x a^2 = a^2b; b x (-ab) = -ab^2; b x b^2 = b^3.

Combine: a^3 + (-a^2b + a^2b) + (ab^2 - ab^2) + b^3 = a^3 +

b^3.

a^3 + b^3

(a - b)(a^2 - ab + b^2)

a x a^2 = a^3; a x (-ab) = -a^2b; a x b^2 = ab^2.

-b x a^2 = -a^2b; -b x (-ab) = ab^2; -b x b^2 = -b^3.

Combine: a^3 - 2a^2b + 2ab^2 - b^3.

a^3 - 2a^2b + 2ab^2 - b^3

(a + b)(a + b)

(a + b)^2 = a^2 + 2ab + b^2.

a^2 + 2ab + b^2

(x + y)(x - y)

(x + y)(x - y) = x^2 - y^2 by the identity (a + b)(a - b) = a^2 - b^2.

x^2 - y^2

Exercise 6.5 (p.123) - Simplification of Algebraic Expressions

Note: the bars (vinculums) in Exercise 6.5 group terms exactly like parentheses; each bar is read as a bracket.

Simplify: 2(4x - 7) - 5(3x - 1)

2(4x - 7) = 8x - 14.

-5(3x - 1) = -15x + 5.

8x - 14 - 15x + 5 = -7x - 9.

-7x - 9

Simplify: 6(2x - y) - 5(x + 2y)

6(2x - y) = 12x - 6y.

-5(x + 2y) = -5x - 10y.

12x - 6y - 5x - 10y = 7x - 16y.

7x - 16y

Simplify: x + (5x - (2x + 1))

The bar over 2x + 1 groups it: x + (5x - (2x + 1)).

5x - (2x + 1) = 5x - 2x - 1 = 3x - 1.

x + 3x - 1 = 4x - 1.

4x - 1

Simplify: 4x - (4x + (6x + 3))

The bar over 6x + 3 groups it: 4x - (4x + (6x + 3)).

4x + (6x + 3) = 10x + 3.

4x - (10x + 3) = -6x - 3.

-6x - 3

Simplify: 2x + (x - (4x - 3)) + 2x

The bar over 4x - 3 groups it: 2x + (x - (4x - 3)) + 2x.

x - (4x - 3) = x - 4x + 3 = -3x + 3.

2x + (-3x + 3) + 2x = x + 3.

x + 3

Simplify: 3x - 2{3(x - 2) - 2(3x - 4)}

Inside the braces: 3(x - 2) = 3x - 6 and 2(3x - 4) = 6x - 8.

3x - 6 - (6x - 8) = -3x + 2.

3x - 2(-3x + 2) = 3x + 6x - 4 = 9x - 4.

9x - 4

Simplify: 2(8x - 4x + 3x) - 6x

8x - 4x + 3x = 7x.

2(7x) - 6x = 14x - 6x = 8x.

8x

Simplify: 5x + {4(x - (2y + x)) - 2(3x + (y - 2))}

The bars group (2y + x) and (y - 2): 4(x - (2y + x)) = 4(-2y) = -8y.

-2(3x + (y - 2)) = -6x - 2y + 4.

Inside the braces: -8y - 6x - 2y + 4 = -6x - 10y + 4.

5x + (-6x - 10y + 4) = -x - 10y + 4.

-x - 10y + 4

Simplify: 5x + [3x + {2y - (4x - (18x - 6x))}]

The bar over 18x - 6x groups it: 18x - 6x = 12x.

4x - 12x = -8x.

2y - (-8x) = 2y + 8x.

3x + (2y + 8x) = 11x + 2y.

5x + 11x + 2y = 16x + 2y.

16x + 2y

Simplify: 2x + [y - 3{2y - (4x - (3x - 2y))}]

The bar over 3x - 2y groups it: 4x - (3x - 2y) = x + 2y.

2y - (x + 2y) = -x.

y - 3(-x) = y + 3x.

2x + y + 3x = 5x + y.

5x + y

Simplify: [2x - {x - (6x - (7y + 3x))}]

The bar over 7y + 3x groups it: 6x - (7y + 3x) = 3x - 7y.

x - (3x - 7y) = -2x + 7y.

2x - (-2x + 7y) = 4x - 7y.

4x - 7y

Simplify: [x + y - {2x - (5y - (9x - 3x))}]

The bar over 9x - 3x groups it: 9x - 3x = 6x.

5y - 6x stays.

2x - (5y - 6x) = 8x - 5y.

x + y - (8x - 5y) = -7x + 6y.

-7x + 6y

Exercise 6.6 (pp.127-128) - Algebraic Identities

1. Complete the following identities.

(a + b)^2 = ... + 2... + ...

(a + b)^2 = a^2 + 2ab + b^2.

(a - b)^2 = ... - 2... + ...

(a - b)^2 = a^2 - 2ab + b^2.

(a + b)(a - b) = ... - ...

(a + b)(a - b) = a^2 - b^2.

2. Expand the following by using identities.

(6x + 5)^2

(6x + 5)^2 = (6x)^2 + 2(6x)(5) + 5^2 = 36x^2 + 60x + 25.

36x^2 + 60x + 25

(3x + 7y)^2

(3x)^2 + 2(3x)(7y) + (7y)^2 = 9x^2 + 42xy + 49y^2.

9x^2 + 42xy + 49y^2

(3x + 4y)^2

(3x)^2 + 2(3x)(4y) + (4y)^2 = 9x^2 + 24xy + 16y^2.

$9x^2 + 24xy + 16y^2$

$(4x - 5y)^2$

$(4x)^2 - 2(4x)(5y) + (5y)^2 = 16x^2 - 40xy + 25y^2.$

$16x^2 - 40xy + 25y^2$

$(9x - 8y)^2$

$(9x)^2 - 2(9x)(8y) + (8y)^2 = 81x^2 - 144xy + 64y^2.$

$81x^2 - 144xy + 64y^2$

$(7x - 1)^2$

$(7x)^2 - 2(7x)(1) + 1^2 = 49x^2 - 14x + 1.$

$49x^2 - 14x + 1$

$(2x + 1)^2$

$(2x)^2 + 2(2x)(1) + 1^2 = 4x^2 + 4x + 1.$

$4x^2 + 4x + 1$

$(6x - y)^2$

$(6x)^2 - 2(6x)(y) + y^2 = 36x^2 - 12xy + y^2.$

$36x^2 - 12xy + y^2$

$(1 - 3x)^2$

1^2 - 2(1)(3x) + (3x)^2 = 1 - 6x + 9x^2.

1 - 6x + 9x^2

3. Evaluate the following by using identities.

(104)^2

104 = 100 + 4.

(100 + 4)^2 = 100^2 + 2(100)(4) + 4^2 = 10000 + 800 + 16 = 10816.

10816

(105)^2

105 = 100 + 5.

(100 + 5)^2 = 10000 + 1000 + 25 = 11025.

11025

(502)^2

502 = 500 + 2.

(500 + 2)^2 = 250000 + 2000 + 4 = 252004.

252004

(601)^2

601 = 600 + 1.

(600 + 1)^2 = 360000 + 1200 + 1 = 361201.

361201

(96)^2

96 = 100 - 4.

(100 - 4)^2 = 10000 - 800 + 16 = 9216.

9216

(95)^2

95 = 100 - 5.

(100 - 5)^2 = 10000 - 1000 + 25 = 9025.

9025

(49)^2

49 = 50 - 1.

(50 - 1)^2 = 2500 - 100 + 1 = 2401.

2401

(999)^2

999 = 1000 - 1.

(1000 - 1)^2 = 1000000 - 2000 + 1 = 998001.

998001

Note: Exercise 6.6 Q3 prints the number (ii) twice (105 squared and 96 squared); the items above are numbered in printed order.

4. Find the product of the following by using identities.

(x + 3)(x - 3)

By (a + b)(a - b) = a^2 - b^2, the product is x^2 - 9.

x^2 - 9

(2x + 5)(2x - 5)

By (a + b)(a - b) = a^2 - b^2, the product is 4x^2 - 25.

4x^2 - 25

(6x - 7y)(6x + 7y)

By (a + b)(a - b) = a^2 - b^2, the product is 36x^2 - 49y^2.

36x^2 - 49y^2

(5x - 10y)(5x + 10y)

By (a + b)(a - b) = a^2 - b^2, the product is 25x^2 - 100y^2.

25x^2 - 100y^2

(9 - 3y)(9 + 3y)

By (a + b)(a - b) = a^2 - b^2, the product is 81 - 9y^2.

81 - 9y^2

(1 + 8y)(1 - 8y)

By (a + b)(a - b) = a^2 - b^2, the product is 1 - 64y^2.

1 - 64y^2

5. Evaluate the following by using identities.

101 x 99

101 x 99 = (100 + 1)(100 - 1) = 100^2 - 1^2 = 10000 - 1.

9999

105 x 95

105 x 95 = (100 + 5)(100 - 5) = 10000 - 25.

9975

52 x 48

52 x 48 = (50 + 2)(50 - 2) = 2500 - 4.

2496

107 x 93

107 x 93 = (100 + 7)(100 - 7) = 10000 - 49.

9951

501 x 499

501 x 499 = (500 + 1)(500 - 1) = 250000 - 1.

249999

1001 x 999

1001 x 999 = (1000 + 1)(1000 - 1) = 1000000 - 1.

999999

6. Find the product of the following.

(a + b)(a - b)

By (a + b)(a - b) = a^2 - b^2, the product is a^2 - b^2.

a^2 - b^2

(x + 3y)(x - 3y)

By (a + b)(a - b) = a^2 - b^2, the product is x^2 - 9y^2.

x^2 - 9y^2

(3a - 2b)(3a + 2b)

By (a + b)(a - b) = a^2 - b^2, the product is 9a^2 - 4b^2.

9a^2 - 4b^2

(x - 5)(x + 5)

By (a + b)(a - b) = a^2 - b^2, the product is x^2 - 25.

x^2 - 25

(1 - x)(1 + x)

By (a + b)(a - b) = a^2 - b^2, the product is 1 - x^2.

1 - x^2

7. Evaluate the following by using identities.

108 x 108

108 x 108 = 108^2 = (100 + 8)^2 = 10000 + 1600 + 64.

11664

97 x 97

97 x 97 = 97^2 = (100 - 3)^2 = 10000 - 600 + 9.

9409

97 x 103

97 x 103 = (100 - 3)(100 + 3) = 10000 - 9.

9991

Exercise 6.7 (p.130) - Factorization of Algebraic Expressions

1. Factorize the following (take out the common factor).

4x + 16

Common factor 4: 4x + 16 = 4(x + 4).

4(x + 4)

55x - 25

Common factor 5: 55x - 25 = 5(11x - 5).

5(11x - 5)

6xy - 18y

Common factor 6y: 6xy - 18y = 6y(x - 3).

6y(x - 3)

8xy - 64x

Common factor 8x: 8xy - 64x = 8x(y - 8).

8x(y - 8)

x^3 - x^2

Common factor x^2: x^3 - x^2 = x^2(x - 1).

x^2(x - 1)

4x^4 - 24x^2

Common factor 4x^2: 4x^4 - 24x^2 = 4x^2(x^2 - 6).

4x^2(x^2 - 6)

9xy - 27y

Common factor 9y: 9xy - 27y = 9y(x - 3).

9y(x - 3)

3x + 6y + 9z

Common factor 3.

3(x + 2y + 3z)

5x - 20y - 30z

Common factor 5.

5(x - 4y - 6z)

3x^2y - 12xy - 18xy^2

Common factor 3xy: 3x^2y - 12xy - 18xy^2 = 3xy(x - 4 - 6y).

3xy(x - 4 - 6y)

5y^3 - 15y^2 + 25y

Common factor 5y.

5y(y^2 - 3y + 5)

7xy - 14x^2y^2 - 28x^3y^3

Common factor 7xy.

7xy(1 - 2xy - 4x^2y^2)

2. Factorize the following (grouping method).

3x^2 + 12xy + 16y^2 + 4yx

Group: 3x^2 + 12xy = 3x(x + 4y).

Group: 16y^2 + 4xy = 4y(4y + x).

Common factor (x + 4y): (x + 4y)(3x + 4y).

(x + 4y)(3x + 4y)

px + py + qx + qy

Group: px + py = p(x + y).

Group: qx + qy = q(x + y).

Common factor (x + y): (x + y)(p + q).

(x + y)(p + q)

m^2 - mn + 4m - 4n

Group: m^2 - mn = m(m - n).

Group: 4m - 4n = 4(m - n).

Common factor (m - n): (m - n)(m + 4).

(m - n)(m + 4)

6xy + 7yz + 6vx + 7vz

Group: 6xy + 6vx = 6x(y + v).

Group: 7yz + 7vz = 7z(y + v).

Common factor (y + v): (y + v)(6x + 7z).

(y + v)(6x + 7z)

5p^2 + 7pr - 21qr - 15pq

Regroup: 5p^2 - 15pq = 5p(p - 3q).

Regroup: 7pr - 21qr = 7r(p - 3q).

Common factor (p - 3q): (p - 3q)(5p + 7r).

(p - 3q)(5p + 7r)

kx + ly + ky + lx

Group: kx + lx = x(k + l).

Group: ly + ky = y(l + k).

Common factor (k + l): (k + l)(x + y).

(k + l)(x + y)

7xz + 5wx + 14zy + 10wy

Group: 7xz + 5wx = x(7z + 5w).

Group: 14zy + 10wy = 2y(7z + 5w).

Common factor (7z + 5w): (7z + 5w)(x + 2y).

(7z + 5w)(x + 2y)

6ab + 14dc + 12ad + 7bc

Group: 6ab + 12ad = 6a(b + 2d).

Group: 7bc + 14dc = 7c(b + 2d).

Common factor (b + 2d): (b + 2d)(6a + 7c).

(b + 2d)(6a + 7c)

x^2 - 15 + 3x - 5x

Combine like terms: 3x - 5x = -2x, so the expression is x^2 - 2x - 15.

Find two numbers with product -15 and sum -2: -5 and 3.

x^2 - 2x - 15 = (x - 5)(x + 3).

(x - 5)(x + 3)

x^3 - x^2 + x - 1

Group: x^3 - x^2 = x^2(x - 1).

Group: x - 1 = 1(x - 1).

Common factor (x - 1): (x - 1)(x^2 + 1).

(x - 1)(x^2 + 1)

2p - 4q - 8r

Common factor 2.

2(p - 2q - 4r)

3x^3y^3 - 12x^2y^2 - 18xy

Common factor 3xy.

3xy(x^2y^2 - 4xy - 6)

Exercise 6.8 (p.133) - Factorization of Quadratic Expressions

1. Factorize the following.

x^2 + 5x + 6

Find two numbers with the required product and sum; the

factorization is (x + 2)(x + 3).

(x + 2)(x + 3)

x^2 + 7x + 12

Find two numbers with the required product and sum; the factorization is (x + 3)(x + 4).

(x + 3)(x + 4)

x^2 - 7x + 10

Find two numbers with the required product and sum; the factorization is (x - 2)(x - 5).

(x - 2)(x - 5)

x^2 - 2x - 35

Find two numbers with the required product and sum; the factorization is (x - 7)(x + 5).

(x - 7)(x + 5)

x^2 + 5x - 14

Find two numbers with the required product and sum; the factorization is (x + 7)(x - 2).

(x + 7)(x - 2)

$x^2 + x - 6$

Find two numbers with the required product and sum; the factorization is $(x + 3)(x - 2)$.

$(x + 3)(x - 2)$

$x^2 - x - 30$

Find two numbers with the required product and sum; the factorization is $(x - 6)(x + 5)$.

$(x - 6)(x + 5)$

$x^2 - x - 12$

Find two numbers with the required product and sum; the factorization is $(x - 4)(x + 3)$.

$(x - 4)(x + 3)$

2. Factorize the following.

$2x^2 + 11x + 15$

Split the middle term and factor by grouping; the factorization is $(2x + 5)(x + 3)$.

$(2x + 5)(x + 3)$

$3x^2 + 7x + 4$

Split the middle term and factor by grouping; the factorization is $(3x + 4)(x + 1)$.

(3x + 4)(x + 1)

2x^2 - 5x - 3

Split the middle term and factor by grouping; the factorization is (2x + 1)(x - 3).

(2x + 1)(x - 3)

5x^2 - 4x - 1

Split the middle term and factor by grouping; the factorization is (5x + 1)(x - 1).

(5x + 1)(x - 1)

4x^2 + 4x - 3

Split the middle term and factor by grouping; the factorization is (2x - 1)(2x + 3).

(2x - 1)(2x + 3)

5x^2 + x - 4

Split the middle term and factor by grouping; the factorization is (5x - 4)(x + 1).

(5x - 4)(x + 1)

3. Factorize the following.

$x^2 + 6x - 7$

Split the middle term and factor by grouping; the factorization is $(x + 7)(x - 1)$.

$(x + 7)(x - 1)$

$6x^2 + 13x + 5$

Split the middle term and factor by grouping; the factorization is $(3x + 5)(2x + 1)$.

$(3x + 5)(2x + 1)$

$x^2 - 9x + 14$

Split the middle term and factor by grouping; the factorization is $(x - 2)(x - 7)$.

$(x - 2)(x - 7)$

$3x^2 + 4x - 7$

Split the middle term and factor by grouping; the factorization is $(3x + 7)(x - 1)$.

$(3x + 7)(x - 1)$

$x^2 - x - 42$

Split the middle term and factor by grouping; the factorization is $(x - 7)(x + 6)$.

(x - 7)(x + 6)

4x^2 - x - 5

Split the middle term and factor by grouping; the factorization is (4x - 5)(x + 1).

(4x - 5)(x + 1)

Review Exercise 6 (pp.133-134)

1. Tick the correct option.

i. The English letters or alphabets which are used to represent unknown quantities in algebra are called:

- constants
- coefficients
- exponents
- ✓ variables

ii. The number which represents repeated multiplication of the same variable is called:

- constant
- coefficient
- ✓ exponent
- variable

iii. A quantity which has a fixed value is called:

- ✓ constant
- coefficient
- exponent

variable

iv. In the algebraic expression 4x^2 - 5x + 2, the constant is:

1

✓ 2

-5

4

v. In the algebraic expression 3x^2 - 4, the coefficient is:

1

2

✓ 3

-4

vi. In a polynomial, the variable power/exponent is:

natural number

✓ whole number

integer

real number

vii. The greatest exponent of the variable, in a polynomial is:

coefficient

✓ degree

term

power

viii. A polynomial which contains only one term is called:

✓ monomial
binomial
trinomial
coefficient

ix. A polynomial which contains only two unlike terms is called:
monomial
✓ binomial
trinomial
coefficient

x. A polynomial which contains only three unlike terms is called:
monomial
binomial
✓ trinomial
coefficient

xi. The degree of the polynomial $6x^5 + 7x^4 - 2x^3 + x^2 - x - 3$ is:
2
3
4
✓ 5

xii. The number 7 is a polynomial of degree:
✓ 0
1
-1

2

xiii. The square of 3x + 2 is:

9x^2 - 12x - 4

9x^2 + 12x - 4

9x^2 - 12x + 4

✓ 9x^2 + 12x + 4

xiv. The square of 2x - 5 is:

4x^2 + 20x + 25

4x^2 - 20x - 25

✓ 4x^2 - 20x + 25

4x^2 + 20x - 25

xv. The factorization of 1 - 4x^2 is:

(1 - 2x)(1 - 2x)

(1 + 2x)(1 + 2x)

✓ (1 + 2x)(1 - 2x)

(2x + 1)(2x - 1)

2. Identify variables, constants and coefficients in each of the following algebraic expressions.

(i) 5x + 3

Variable: x. Constant: 3. Coefficient: 5.

(ii) 6x - 1

Variable: x. Constant: -1. Coefficient: 6.

(iii) x - 3

Variable: x. Constant: -3. Coefficient: 1.

(iv) 4x^2 - 5x + 7

Variable: x. Constant: 7. Coefficients: 4 (of x^2) and -5 (of x).

3. Write the number of terms of the following algebraic expressions.

2x - 1

2 terms.

x - 1

2 terms.

7x - 9y

2 terms.

5x - 7y + 3

3 terms.

4. Identify open and closed sentences.

Note: Review Exercise 6 Q4 prints items (i), (ii), (iii), (iv), (vi); no item (v) is printed.

(i) g - 5 = 4

Open sentence: it contains the variable g.

(ii) 3y > y

Open sentence: it contains the variable y.

(iii) $y - 3 = -3 + y$

Open sentence: it contains the variable y (it is true for every y, but it is still an open sentence).

(iv) $5x < 8x$

Open sentence: it contains the variable x.

(vi) $7x < 3x$

Open sentence: it contains the variable x.

Unit 7: Linear Equations and Coordinate System

Note: Unit 7 has no Exercise 7.2. Section 7.2 (solving methods) contains worked examples only; the exercises are 7.1, 7.3 and 7.4, plus the Review Exercise.

Exercise 7.1 (p.145) - Linear Equations

1. Solve the following.

x - 4 = 3

Add 4 to both sides: x = 3 + 4.

x = 7.

x + 2 = 5

Subtract 2 from both sides: x = 5 - 2.

x = 3.

2x + 1 = x + 2

Subtract x from both sides: x + 1 = 2.

Subtract 1 from both sides: x = 1.

x = 1.

3x = 12

Divide both sides by 3: x = 12/3.

x = 4.

4x - 5 = 3

Add 5 to both sides: 4x = 8.

Divide both sides by 4: x = 2.

x = 2.

3x - 2 = 7

Add 2 to both sides: 3x = 9.

Divide both sides by 3: x = 3.

x = 3.

(x + 1)/3 = 2

Multiply both sides by 3: x + 1 = 6.

Subtract 1 from both sides: x = 5.

x = 5.

(2x - 3)/5 = 3

Multiply both sides by 5: 2x - 3 = 15.

Add 3 to both sides: 2x = 18.

Divide both sides by 2: x = 9.

x = 9.

2. Identify the linear equation in two variables.

x + y = 2

Yes: two variables, each to power 1.

3x - y = 1

Yes: two variables, each to power 1.

3x + 1 = 2x

No: only one variable (x).

6x + 6y = 5

Yes: two variables, each to power 1.

2x^2 + 3x = 4

No: only one variable, and it is squared.

x - y = 4

Yes: two variables, each to power 1.

3x^2 - 4x = 1

No: only one variable, and it is squared.

5x - 4y = 0

Yes: two variables, each to power 1.

$3y^2 + 1 = 2y$

No: only one variable, and it is squared.

$5x - 1 = 3y$

Yes: two variables, each to power 1 (it can be written $5x - 3y = 1$).

3. Construct a linear equation in two variables for the following.

The price of 5 apples and 6 mangoes is Rs. 160.

Let the price of one apple be Rs. x and of one mango be Rs. y. Then $5x + 6y = 160$.

The price of 2 ink pens and 3 markers is Rs. 145.

Let the price of one ink pen be Rs. x and of one marker be Rs. y. Then $2x + 3y = 145$.

The sum of two natural numbers is 6.

Let the numbers be x and y. Then $x + y = 6$.

The difference of two odd numbers is 2.

Let the numbers be x and y. Then $x - y = 2$.

5 added to a number is equal to twice another number.

Let the numbers be x and y. Then $x + 5 = 2y$.

3 subtracted from a number is equal to 5 times another number.

Let the numbers be x and y. Then $x - 3 = 5y$.

Four times a number equals five times another number.
Let the numbers be x and y. Then 4x = 5y.

Three times the age of father is equal to 5 times and 4 years the age of his son.
Let the father's age be x years and the son's age be y years. Then 3x = 5y + 4. Flag: the printed wording is awkward ('5 times and 4 years'); it is read as five times the son's age plus 4 years.

The age of Alia and her mother is equal to 65 years.
Let Alia's age be x years and her mother's age be y years. Then x + y = 65.

7 added to three times a number is equal to 5 times a number reduced by 2.
Let the numbers be x and y. Then 3x + 7 = 5y - 2. Flag: the question asks for an equation in two variables, so the two 'numbers' are taken as two different variables; with one variable it would be 3x + 7 = 5x - 2.

4. The price of 2 books and 5 notebooks is Rs. 1000. Construct a linear equation in two variables.
Let the price of one book be Rs. x and of one notebook be Rs. y.

2x + 5y = 1000.

5. The length of red electric wire and black electric wire is 35 m.

Construct a linear equation in two variables.

Let the length of red wire be x metres and of black wire be y metres.

$x + y = 35$.

6. The volume of 5 tubs and 3 buckets is 150 liters. Construct a linear equation in two variables.

Let the volume of one tub be x liters and of one bucket be y liters.

$5x + 3y = 150$.

7. The price of 2 room colors and 5 solar ceiling fans is Rs. 15000. Construct a linear equation in two variables.

Let the price of one room color be Rs. x and of one solar ceiling fan be Rs. y.

$2x + 5y = 15000$.

Exercise 7.3 (pp.150-152) - Coordinate Plane

1. Draw the X-axis and Y-axis in the coordinate plane; also identify the quadrants and ordered pairs (x, y) in the given graph.

Draw a horizontal line for the X-axis (X to the right, X' to the left) and a vertical line for the Y-axis (Y upward, Y' downward), crossing at the origin O (0, 0). The four regions are: Quadrant I (top right): (x, y) with x positive, y positive, written (+, +). Quadrant II (top left): (-, +). Quadrant III (bottom left): (-, -). Quadrant IV (bottom right): (+, -). The given graph is a blank grid: draw the axes through

its centre and label the quadrants and their signs as above.

2. Identify the quadrant and find the ordered pairs in the given graph.

Point A (x, y) lies in Quadrant I: both coordinates positive, read from the graph as (5, 5). Point B (-x, y) lies in Quadrant II: x negative, y positive, read as (-7, 5). Point C (-x, -y) lies in Quadrant III: both coordinates negative, read as (-8, -7). Point D (x, -y) lies in Quadrant IV: x positive, y negative, read as (5, -7). The sign patterns are (+, +), (-, +), (-, -) and (+, -) in Quadrants I, II, III and IV.

3. Plot the following ordered pairs in the coordinate plane in the given graph: P(6, 4), Q(5, 3), R(-7, -6), S(8, 4), T(8, 6).

P(6, 4): from the origin count 6 units right on the X-axis and 4 units up, mark P (Quadrant I). Q(5, 3): 5 right, 3 up, mark Q (Quadrant I). R(-7, -6): 7 left, 6 down, mark R (Quadrant III). S(8, 4): 8 right, 4 up, mark S (Quadrant I). T(8, 6): 8 right, 6 up, mark T (Quadrant I).

4. Identify the abscissa, ordinate and quadrant.

A(3, 3)

Abscissa 3 (positive), ordinate 3 (positive), Quadrant I.

B(-4, 2)

Abscissa -4 (negative), ordinate 2 (positive), Quadrant II.

C(-2, -7)

Abscissa -2 (negative), ordinate -7 (negative), Quadrant III.

D(5, -8)

Abscissa 5 (positive), ordinate -8 (negative), Quadrant IV.

E(-6, 6)

Abscissa -6 (negative), ordinate 6 (positive), Quadrant II.

F(8, -7)

Abscissa 8 (positive), ordinate -7 (negative), Quadrant IV.

G(1, 4)

Abscissa 1 (positive), ordinate 4 (positive), Quadrant I.

H(-8, -3)

Abscissa -8 (negative), ordinate -3 (negative), Quadrant III.

5. Draw the coordinate plane with axes. Show quadrant and ordered pairs (x, y) along with signs in the given graph.

Draw the X-axis and Y-axis crossing at the origin in the given blank grid. Label the four quadrants: Quadrant I (top right) (+, +), Quadrant II (top left) (-, +), Quadrant III (bottom left) (-, -), Quadrant IV (bottom right) (+, -). Any point (x, y) is plotted by counting x units from the origin along the X-axis (right if positive, left if negative) and then y units parallel to the Y-axis (up if positive, down if negative).

6. Plot the following ordered pairs in the above graph: A(8, 5), B(-9, 6), C(-7, -4), D(10, -2), E(-5, 3), F(0, -5), G(6, 0), H(0, 3), I(-6, 0),

O(0, 0).

A(8, 5): Quadrant I. B(-9, 6): Quadrant II. C(-7, -4): Quadrant III. D(10, -2): Quadrant IV. E(-5, 3): Quadrant II. F(0, -5): on the Y-axis (below the origin), in no quadrant. G(6, 0): on the X-axis (right of the origin), in no quadrant. H(0, 3): on the Y-axis (above the origin), in no quadrant. I(-6, 0): on the X-axis (left of the origin), in no quadrant. O(0, 0): the origin, where the axes cross.

Exercise 7.4 (p.156) - Graphs of Linear Equations

1. Identify the vertical and horizontal lines of equations.

$x = 3$

Vertical line (x is fixed at 3).

$y = 4$

Horizontal line (y is fixed at 4).

$x + 5 = 0$, that is $x = -5$

Vertical line.

$x - 8 = 0$, that is $x = 8$

Vertical line.

$y + 6 = 0$, that is $y = -6$

Horizontal line.

$y - 7 = 0$, that is $y = 7$

Horizontal line.

4x - 12 = 0, that is x = 3

Vertical line.

3y - 15 = 0, that is y = 5

Horizontal line.

3x + 18 = 0, that is x = -6

Vertical line.

5y + 20 = 0, that is y = -4

Horizontal line.

2. Draw the graphs for the following equations of vertical lines.

4x + 12 = 0

x = -3: draw a vertical line through (-3, 0), parallel to the Y-axis.

x = 5

Draw a vertical line through (5, 0), parallel to the Y-axis.

x + 3 = 0

x = -3: draw a vertical line through (-3, 0), parallel to the Y-axis.

3x - 9 = 0

x = 3: draw a vertical line through (3, 0), parallel to the Y-axis.

7x - 35 = 0

x = 5: draw a vertical line through (5, 0), parallel to the Y-axis.

2x + 6 = 0

x = -3: draw a vertical line through (-3, 0), parallel to the Y-axis.

3. Draw the graphs for the following equations of horizontal lines.

y = 4

Draw a horizontal line through (0, 4), parallel to the X-axis.

y + 3 = 0

y = -3: draw a horizontal line through (0, -3), parallel to the X-axis.

2y + 8 = 0

y = -4: draw a horizontal line through (0, -4), parallel to the X-axis.

2y + 6 = 0

y = -3: draw a horizontal line through (0, -3), parallel to the X-axis.

4x - 16 = 0

Flag: this is NOT a horizontal line. 4x - 16 = 0 gives x = 4, which is a vertical line through (4, 0), parallel to the Y-axis; it is printed inside the horizontal-line question.

y = 3

Draw a horizontal line through (0, 3), parallel to the X-axis.

4. Plot the graphs for the following linear equations in two variables (tables of points; join the points with a straight line).

2x + y = 9

Points on the line: (0, 9), (1, 7), (2, 5), (4, 1): y = 9 - 2x. Plot these

points and join them with a straight line.

$x + 2y = 10$

Points on the line: (0, 5), (2, 4), (4, 3), (10, 0): $y = (10 - x)/2$. Plot these points and join them with a straight line.

$5x + y = 15$

Points on the line: (0, 15), (1, 10), (2, 5), (3, 0): $y = 15 - 5x$. Plot these points and join them with a straight line.

$x - y = 7$

Points on the line: (0, -7), (7, 0), (8, 1): $y = x - 7$. Plot these points and join them with a straight line.

$2x - y = 8$

Points on the line: (0, -8), (4, 0), (5, 2): $y = 2x - 8$. Plot these points and join them with a straight line.

$3x + 2y = 18$

Points on the line: (0, 9), (2, 6), (4, 3), (6, 0): $y = (18 - 3x)/2$. Plot these points and join them with a straight line.

$y = 2x - 5$

Points on the line: (0, -5), (1, -3), (2, -1), (3, 1). Plot these points and join them with a straight line.

$x = y - 5$

Points on the line: (0, 5), (1, 6), (-5, 0): $y = x + 5$. Plot these points

and join them with a straight line.

-5x + 4y = 8

Points on the line: (0, 2), (4, 7), (-4, -3): y = (5x + 8)/4. Plot these points and join them with a straight line.

Review Exercise 7 (pp.157-158)

1. Tick the correct option.

i. In a linear equation in two variables, the power of each variable is:

- 0
- ✓ 1
- 2
- 3

ii. 3x + 2 is an/a:

- ✓ expression
- equation
- true sentence
- false sentence

iii. The solution of a linear equation in one variable is:

- ✓ unique
- two
- infinite
- zero

iv. The solution of a linear equation in two variables is:

unique

two

✓ infinite

zero

v. The standard form of a linear equation in two variables is:

ax + b = 0

✓ ax + by = c

ax = b

ay = b

vi. Which of the following is a linear equation in two variables?

2x + 5 = 0

✓ 3x + y = 4

5y = 10

3x = 9

vii. Which of the following is an equation of a vertical line?

✓ x = 5

y = 6

x = y - z

y = x - 4

viii. Which of the following is an equation of a horizontal line?

x = 6

✓ y = 4

x = y + 3

y = x + 4

ix. The ordered pair (5, -5) lies in:

- Quadrant I
- Quadrant II
- Quadrant III
- ✓ Quadrant IV

x. The ordered pair (-4, -3) lies in:

- Quadrant I
- Quadrant II
- ✓ Quadrant III
- Quadrant IV

2. Write two linear equations in two variables.

Two examples: $2x + 3y = 12$ and $x - y = 5$. Both have two variables, each to power 1.

3. Construct a linear equation in two variables for the following.

The price of 6 bananas and 4 oranges is Rs. 160.

Let the price of one banana be Rs. x and of one orange be Rs. y.

Then $6x + 4y = 160$.

Sum of Manahil and her brother's ages is 36 years.

Let Manahil's age be x years and her brother's age be y years. Then $x + y = 36$.

The sum of two prime numbers is 13.

Let the prime numbers be x and y. Then $x + y = 13$.

The difference of two even numbers is 8.

Let the even numbers be x and y. Then x - y = 8.

A number decreased by 5 equals 3 times another number.

Let the numbers be x and y. Then x - 5 = 3y.

6 added to twice a number equals 4 times another number.

Let the numbers be x and y. Then 2x + 6 = 4y.

4. Find the solution of the following linear equations in two variables.

(i) 2x - y = 6

A linear equation in two variables has infinitely many solutions.

Three of them: (3, 0) since 2(3) - 0 = 6; (4, 2) since 8 - 2 = 6; (0, -6) since 0 - (-6) = 6.

(ii) x + 3y = 8

Three solutions: (8, 0) since 8 + 0 = 8; (2, 2) since 2 + 6 = 8; (5, 1) since 5 + 3 = 8.

5. The sum of two consecutive even numbers is 90. Find the numbers.

Let the first even number be x; the next consecutive even number is x + 2.

x + (x + 2) = 90, so 2x + 2 = 90.

2x = 88, so x = 44.

The second number is 44 + 2 = 46.

The numbers are 44 and 46.

6. The mass of 3 salt bags and 4 sugar bags is 50 kg. Construct a linear equation in two variables for this situation.

Let the mass of one salt bag be x kg and of one sugar bag be y kg.

3x + 4y = 50.

7. Solve the following linear equations and find at least 6 solutions for each. Also plot the graphs for each.

(i) x + y = 4

Six solutions: (0, 4), (1, 3), (2, 2), (3, 1), (4, 0), (-1, 5). Plot these points and join them with a straight line.

(ii) x - y = 3

Six solutions: (3, 0), (4, 1), (5, 2), (0, -3), (1, -2), (2, -1). Plot these points and join them with a straight line.

8. Draw the coordinate plane, identify the quadrant and plot the following ordered pairs in the given graph.

(i) P(-3, -4)

Quadrant III: 3 units left on the X-axis, 4 units down.

(ii) Q(-4, 2)

Quadrant II: 4 units left on the X-axis, 2 units up. Flag: the OCR text reads Q(4, 2), but the printed page (p.158) reads Q(-4, 2); the printed page is followed.

(iii) R(5, 4)

Quadrant I: 5 units right on the X-axis, 4 units up.

(iv) S(6, 5)

Quadrant I: 6 units right on the X-axis, 5 units up.

Unit 8: Distance and Speed

Exercise 8.1 (p.164) - Length Conversions

Conversion facts used throughout: 1 km = 1000 m; 1 m = 100 cm; 1 cm = 10 mm; 1 km = 100000 cm; 1 km = 1000000 mm.

1. (i) Convert 5 km into metres.

5 km = 5 x 1000 m.

5000 m.

1. (ii) Convert 16 km into metres.

16 km = 16 x 1000 m.

16000 m.

1. (iii) Convert 20 km into metres.

20 km = 20 x 1000 m.

20000 m.

2. (i) Convert 3000 m into kilometres.

3000 m = 3000/1000 km.

3 km.

2. (ii) Convert 25000 m into kilometres.

25000 m = 25000/1000 km.

25 km.

2. (iii) Convert 35000 m into kilometres.

35000 m = 35000/1000 km.

35 km.

3. (i) Convert 13 km 300 m into metres.

13 km = 13000 m; 13000 + 300 = 13300 m.

13300 m.

3. (ii) Convert 15 km 150 m into metres.

15 km = 15000 m; 15000 + 150 = 15150 m.

15150 m.

3. (iii) Convert 20 km 575 m into metres.

20 km = 20000 m; 20000 + 575 = 20575 m.

20575 m.

4. (i) Convert 1570 m into kilometres and metres.

1570 m = 1000 m + 570 m = 1 km 570 m.

1 km 570 m.

4. (ii) Convert 2360 m into kilometres and metres.

2360 m = 2000 m + 360 m = 2 km 360 m.

2 km 360 m.

4. (iii) Convert 4385 m into kilometres and metres.

4385 m = 4000 m + 385 m = 4 km 385 m.

4 km 385 m.

5. (i) Convert 12 m into centimetres.

12 m = 12 x 100 cm.

1200 cm.

5. (ii) Convert 18 m into centimetres.

18 m = 18 x 100 cm.

1800 cm.

5. (iii) Convert 23 m into centimetres.

23 m = 23 x 100 cm.

2300 cm.

6. (i) Convert 3600 cm into metres.

3600 cm = 3600/100 cm.

36 m.

6. (ii) Convert 1800 cm into metres.

1800 cm = 1800/100 cm.

18 m.

6. (iii) Convert 2500 cm into metres.

2500 cm = 2500/100 cm.

25 m.

7. (i) Convert 16 m 68 cm into centimetres.

16 m = 1600 cm; 1600 + 68 = 1668 cm.

1668 cm.

7. (ii) Convert 32 m 75 cm into centimetres.

32 m = 3200 cm; 3200 + 75 = 3275 cm.

3275 cm.

7. (iii) Convert 5 m 20 cm into centimetres.

5 m = 500 cm; 500 + 20 = 520 cm.

520 cm.

8. (i) Convert 1365 cm into metres and centimetres.

1365 cm = 1300 cm + 65 cm = 13 m 65 cm.

13 m 65 cm.

8. (ii) Convert 2576 cm into metres and centimetres.

2576 cm = 2500 cm + 76 cm = 25 m 76 cm.

25 m 76 cm.

8. (iii) Convert 3984 cm into metres and centimetres.

3984 cm = 3900 cm + 84 cm = 39 m 84 cm.

39 m 84 cm.

9. (i) Convert 17 cm into millimetres.

17 cm = 17 x 10 mm.

170 mm.

9. (ii) Convert 26 cm into millimetres.

26 cm = 26 x 10 mm.

260 mm.

9. (iii) Convert 45 cm into millimetres.

45 cm = 45 x 10 mm.

450 mm.

10. (i) Convert 350 mm into centimetres.

350 mm = 350/10 cm.

35 cm.

10. (ii) Convert 560 mm into centimetres.

560 mm = 560/10 cm.

56 cm.

10. (iii) Convert 850 mm into centimetres.

850 mm = 850/10 cm.

85 cm.

11. (i) Convert 18 cm 8 mm into millimetres.

18 cm = 180 mm; 180 + 8 = 188 mm.

188 mm.

11. (ii) Convert 28 cm 2 mm into millimetres.

28 cm = 280 mm; 280 + 2 = 282 mm.

282 mm.

11. (iii) Convert 13 cm 6 mm into millimetres.

13 cm = 130 mm; 130 + 6 = 136 mm.

136 mm.

12. (i) Convert 4365 mm into metres and millimetres.

4365 mm = 4000 mm + 365 mm = 4 m 365 mm.

4 m 365 mm.

12. (ii) Convert 6535 mm into metres and millimetres.

6535 mm = 6000 mm + 535 mm = 6 m 535 mm.

6 m 535 mm.

12. (iii) Convert 8725 mm into metres and millimetres.

8725 mm = 8000 mm + 725 mm = 8 m 725 mm.

8 m 725 mm.

13. (i) Convert 7 km into centimetres.

7 km = 7 x 100000 cm.

700000 cm.

13. (ii) Convert 14 km into centimetres.

14 km = 14 x 100000 cm.

1400000 cm.

13. (iii) Convert 27 km into centimetres.

27 km = 27 x 100000 cm.

2700000 cm.

14. (i) Convert 600000 cm into kilometres.

600000 cm = 600000/100000 km.

6 km.

14. (ii) Convert 1800000 cm into kilometres.

1800000 cm = 1800000/100000 km.

18 km.

14. (iii) Convert 3200000 cm into kilometres.

3200000 cm = 3200000/100000 km.

32 km.

15. (i) Convert 8 km 35678 cm into centimetres.

8 km = 800000 cm; 800000 + 35678 = 835678 cm.

835678 cm.

15. (ii) Convert 12 km 26568 cm into centimetres.

12 km = 1200000 cm; 1200000 + 26568 = 1226568 cm.

1226568 cm.

15. (iii) Convert 23 km 36508 cm into centimetres.

23 km = 2300000 cm; 2300000 + 36508 = 2336508 cm.

2336508 cm.

16. (i) Convert 273125 cm into kilometres and centimetres.

273125 cm = 200000 cm + 73125 cm = 2 km 73125 cm.

2 km 73125 cm.

16. (ii) Convert 300060 cm into kilometres and centimetres.

300060 cm = 300000 cm + 60 cm = 3 km 60 cm.

3 km 60 cm.

16. (iii) Convert 375600 cm into kilometres and centimetres.

375600 cm = 300000 cm + 75600 cm = 3 km 75600 cm.

3 km 75600 cm.

Exercise 8.2 (p.170) - Time and Speed Conversions

1. Convert the given time into the required time format (12-hour clock to 24-hour clock, and 24-hour clock to 12-hour clock).

5:00 a.m. (12-hour) to 24-hour clock

05:00.

8:30 (24-hour) to 12-hour clock

8:30 a.m.

11:45 a.m. (12-hour) to 24-hour clock

11:45.

11:20 (24-hour) to 12-hour clock

11:20 a.m.

11:59 a.m. (12-hour) to 24-hour clock

11:59.

11:59 (24-hour) to 12-hour clock

11:59 a.m.

12:00 p.m. (12-hour) to 24-hour clock

12:00.

12:00 (24-hour) to 12-hour clock

12:00 p.m.

3:15 p.m. (12-hour) to 24-hour clock

15:15.

16:00 (24-hour) to 12-hour clock

4:00 p.m.

11:59 p.m. (12-hour) to 24-hour clock

23:59.

21:00 (24-hour) to 12-hour clock

9:00 p.m.

12:00 a.m. (12-hour) to 24-hour clock

00:00.

23:59 (24-hour) to 12-hour clock

11:59 p.m.

12:01 a.m. (12-hour) to 24-hour clock

00:01.

24:00 (24-hour) to 12-hour clock

12:00 a.m. (midnight).

2. Convert the following 12-hour clock format to 24-hour clock format.

11:15 p.m.

23:15.

09:30 a.m.

09:30.

11:59 a.m.

11:59.

11:59 p.m.

23:59.

12:00 a.m.

00:00.

3. Convert the following 24-hour clock format to 12-hour clock format.

6:40

6:40 a.m.

11:59

11:59 a.m.

12:00

12:00 p.m.

18:00

6:00 p.m.

00:00

12:00 a.m.

4. Convert the following hours to minutes.

5 h

5 h = 300 x 60 min.

300 min.

6 h

6 h = 360 x 60 min.

360 min.

12 h

12 h = 720 x 60 min.

720 min.

5. Convert the following km/h to m/s (multiply by 1000/3600 = 5/18).

12 km/h

12 x 1000/3600 = 12000/3600 = 3.33 m/s.

3.33 m/s

25 km/h

25 x 1000/3600 = 25000/3600 = 6.94 m/s.

6.94 m/s

33 km/h

33 x 1000/3600 = 33000/3600 = 9.17 m/s.

9.17 m/s

6. Convert the following m/s to km/h (multiply by 3600/1000 = 18/5).

3000 m/s

3000 x 3600/1000 = 10800 km/h.

10800 km/h

250 m/s

250 x 3600/1000 = 900 km/h.

900 km/h

12 m/s

12 x 3600/1000 = 43.2 km/h.

43.2 km/h

7. A car travels at a speed of 36 km/h. Convert the speed into m/s.

36 km/h = 36 x 1000/3600 m/s = 10 m/s.

10 m/s.

Exercise 8.3 (p.173) - Journey Time

1. Complete the table (departure time, journey time, arrival time).

(i) Departure 12:30 p.m., arrival 02:15 a.m.

Journey time: from 12:30 p.m. to 12:00 midnight is 11 h 30 min, plus 2 h 15 min to 02:15 a.m. = 13 h 45 min.

(ii) Journey time 7 h 20 min, arrival 03:45 a.m.

Departure time: 03:45 a.m. minus 7 h 20 min = 8:25 p.m. on the previous day.

(iii) Departure 08:30 p.m., arrival 08:15 a.m.

Journey time: from 08:30 p.m. to 12:00 midnight is 3 h 30 min, plus 8 h 15 min to 08:15 a.m. = 11 h 45 min.

(iv) Departure 09:25 a.m., journey time 5 h 20 min

Arrival time: 09:25 a.m. + 5 h 20 min = 2:45 p.m.

(v) Departure 10:00 p.m., arrival 08:00 a.m.

Journey time: from 10:00 p.m. to 12:00 midnight is 2 h, plus 8 h to 08:00 a.m. = 10 h. Flag: the printed table is cramped; row (v) is read as departure 10:00 p.m. and arrival 08:00 a.m.

2. The journey time from Quetta to Islamabad by car is 11 hours 30 min. Alia departs at 4:30 a.m. Find her time of arrival in Islamabad.

4:30 a.m. + 11 h 30 min = 4:00 p.m.

4:00 p.m.

3. Waseem went to Karachi from Quetta; the journey time is 10 hr 30 min. The departure time is 4:15 a.m. Find his arrival time in Karachi.

4:15 a.m. + 10 h 30 min = 2:45 p.m.

2:45 p.m.

4. Malaika went from Quetta to Rawalpindi in the Balochistan Express Train. Her departure time is 7:30 a.m., which takes 12 hours 45 minutes to reach Rawalpindi. Find the arrival time of the train in Rawalpindi.
7:30 a.m. + 12 h 45 min = 8:15 p.m.

8:15 p.m.

5. Atif went to Islamabad from Quetta in his car. His departure time from Quetta is 4:30 a.m. and his arrival time in Islamabad is 7:15 p.m. Calculate the journey time.
From 4:30 a.m. to 12:00 noon is 7 h 30 min.

From 12:00 noon to 7:15 p.m. is 7 h 15 min.

Total: 7 h 30 min + 7 h 15 min = 14 h 45 min.

14 hours 45 minutes.

6. Shazia went to Ziarat Residency with her classmates on a picnic tour. The time of arrival at Ziarat Residency was 8:15 a.m. and the journey time from Quetta to Ziarat Residency was 3 hours 15 minutes. Find her departure time.
8:15 a.m. minus 3 h 15 min = 5:00 a.m.

5:00 a.m.

Exercise 8.4 (p.176) - Speed, Distance and Time

1. Shazia covered 150 km distance in 2 hours by car. Find the average speed of the car.

Average speed = total distance / total time = 150/2.

75 km/h.

2. A train covered a distance of 260 km for the first 2 hours and then the speed of the train changed and it covered a distance of 160 km for the next 4 hours. Find the average speed of the train.

Total distance = 260 + 160 = 420 km.

Total time = 2 + 4 = 6 hours.

Average speed = 420/6 = 70 km/h.

70 km/h.

3. An aeroplane covers 1600 km in one hour. What is the speed of the aeroplane? Also convert the speed to m/s.

Speed = 1600/1 = 1600 km/h.

In m/s: 1600 x 1000/3600 = 444.44 m/s.

1600 km/h = 444.44 m/s.

4. Usman covers 6 km distance in 40 minutes. Find his speed in

km/h.

40 minutes = 40/60 hours = 2/3 hours.

Speed = 6 / (2/3) = 6 x 3/2 = 9 km/h.

9 km/h.

5. A 2-D car speed is 9 m/s. How far will it travel in 3 hours?

3 hours = 3 x 3600 = 10800 seconds.

Distance = speed x time = 9 x 10800 = 97200 m = 97.2 km.

97.2 km.

6. A truck covers 150 km in 5 hours. How much distance will it cover in 10 hours?

Speed = 150/5 = 30 km/h.

Distance in 10 hours = 30 x 10 = 300 km.

300 km.

7. A container covers a distance in 10 hours with a speed of 30 km/h. Find the distance.

Distance = speed x time = 30 x 10 = 300 km.

300 km.

8. Asif covered 12 km by walking for 4 hours. Find the speed of Asif.

Speed = 12/4 = 3 km/h.

3 km/h.

Review Exercise 8 (p.177)

1. Tick the correct option.

i. One kilometre is equal to metres:

- 10 m
- 100 m
- ✓ 1000 m
- 10000 m

ii. One metre is equal to centimetres:

- 10 cm
- ✓ 100 cm
- 1000 cm
- 10000 cm

iii. One centimetre is equal to millimetres:

- 1 mm
- 5 mm
- ✓ 10 mm
- 20 mm

iv. One hour has minutes:

- 10 min
- 30 min

45 min

✓ 60 min

v. One hour has seconds:

1600

2600

✓ 3600

4600

vi. The distance covered in a unit time is called:

distance

time

✓ speed

kilometre

vii. The same distance covered in the same time is called:

distance

time

✓ uniform speed

average speed

viii. The total distance travelled by an object in a particular time interval is called:

distance

time

uniform speed

✓ average speed

ix. The total length of the actual path followed by an object is called:

- ✓ distance
- time
- uniform speed
- average speed

x. The formula of speed is:

- S = d x t
- ✓ S = d/t
- S = t/d
- S = d + t

2. Convert the following units.

8 km to m

By the conversion facts, 8 km to m = 8000 m.

8000 m.

4 km to cm

By the conversion facts, 4 km to cm = 400000 cm.

400000 cm.

3 km to mm

By the conversion facts, 3 km to mm = 3000000 mm.

3000000 mm.

8 m to cm

By the conversion facts, 8 m to cm = 800 cm.

800 cm.

15 cm to mm

By the conversion facts, 15 cm to mm = 150 mm.

150 mm.

9 m to mm

By the conversion facts, 9 m to mm = 9000 mm.

9000 mm.

1 cm to mm

By the conversion facts, 1 cm to mm = 10 mm.

10 mm.

10 m to mm

By the conversion facts, 10 m to mm = 10000 mm.

10000 mm.

3. Convert the following units.

3 h to min

By the conversion facts, 3 h to min = 180 min.

180 min.

1 h to s

By the conversion facts, 1 h to s = 3600 s.

3600 s.

12 min to s

By the conversion facts, 12 min to s = 720 s.

720 s.

7200 s to min

By the conversion facts, 7200 s to min = 120 min.

120 min.

360 min to h

By the conversion facts, 360 min to h = 6 h.

6 h.

14400 s to h

By the conversion facts, 14400 s to h = 4 h.

4 h.

180 min to h

By the conversion facts, 180 min to h = 3 h.

3 h.

10800 s to h

By the conversion facts, 10800 s to h = 3 h.

3 h.

Unit 9: Polygon and Circle

Exercise 9.1 (p.186) - Quadrilaterals

1. Name the following figures and write the names of their sides which are equal in length.

Figure ABCD

Square: it has four right angles and all four sides equal in length, so AB = BC = CD = DA.

Figure EFGH

Rectangle: it has four right angles and opposite sides equal, so EF = GH and EG = FH.

Figure IJKL

Parallelogram: opposite sides are equal, so IJ = LK and IL = JK.

Figure MNOP

Parallelogram (rhombus-type): opposite sides are equal, so MN = PO and MP = NO.

Figure QRST

Trapezium with QR parallel to TS: the two sides QR and TS are the parallel sides; the non-parallel sides are QT and RS.

Figure UVWX

Kite: the angles at X and V are marked equal, and the diagonals are

perpendicular, with UW as the axis of symmetry. Its pairs of adjacent equal sides are UX = VX and UW-side pairs: UX = UV and XW = VW (adjacent sides meeting at the symmetry axis ends are equal).

2. Find the unknown angles and which angles are equal of the following figures.

Figure ABCD (square)

All angles are right angles: A = B = C = D = 90 degrees. All four angles are equal.

Figure EFGH (rectangle)

All angles are right angles: E = F = G = H = 90 degrees. All four angles are equal.

Figure IJKL (parallelogram) with angle L = 75 degrees

Opposite angles of a parallelogram are equal, so angle J = angle L = 75 degrees.

Adjacent angles are supplementary, so angle I = 180 - 75 = 105 degrees and angle K = 180 - 75 = 105 degrees.

Angles: L = J = 75 degrees, I = K = 105 degrees. Equal pairs: L = J and I = K.

Figure MNOP (parallelogram) with angle P = 70 degrees

Opposite angles are equal, so angle N = angle P = 70 degrees.

Adjacent angles are supplementary, so angle M = 180 - 70 = 110 degrees and angle O = 180 - 70 = 110 degrees.

Angles: P = N = 70 degrees, M = O = 110 degrees. Equal pairs: P = N and M = O.

Figure QRST (trapezium, QR parallel to TS) with angle T = 120 degrees and angle S = 110 degrees

Angles on the same side between parallel lines are supplementary: angle Q + angle T = 180, so angle Q = 180 - 120 = 60 degrees.

Angle R + angle S = 180, so angle R = 180 - 110 = 70 degrees.

Angles: Q = 60 degrees, R = 70 degrees, T = 120 degrees, S = 110 degrees. No two of these angles are equal.

Figure UVWX (kite) with angle U = 50 degrees and angle X = 115 degrees

In a kite the angles at X and V (marked equal in the figure) are equal, so angle V = 115 degrees.

The angles of a quadrilateral sum to 360 degrees, so angle W = 360 - 50 - 115 - 115 = 80 degrees.

Angles: U = 50 degrees, X = V = 115 degrees, W = 80 degrees. Equal pair: X = V.

3. Identify the parallel and equal sides of the following figures.

Figure ABCD (square, side 4 cm)

Parallel sides: AB parallel to DC, and AD parallel to BC. Equal sides: all four sides are equal, each 4 cm, so AB = BC = CD = DA = 4 cm.

Figure EFGH (parallelogram, EF = 3 cm)

Parallel sides: EF parallel to HG, and EH parallel to FG. Equal sides: opposite sides are equal, so EF = HG = 3 cm and EH = FG.

Figure IJKL (rectangle, 6 cm by 2 cm)

Parallel sides: IJ parallel to LK, and IL parallel to JK. Equal sides: opposite sides are equal, so IJ = LK = 6 cm and IL = JK = 2 cm.

Exercise 9.2 (p.196) - Polygons

1. Identify regular and irregular polygons in the following.

(i) Triangle (equilateral in appearance)

Regular polygon: all sides and all angles equal.

(ii) Pentagon

Regular polygon: all sides and all angles equal.

(iii) Concave quadrilateral

Irregular polygon: its sides and angles are not all equal.

(iv) Elongated hexagon

Irregular polygon: its sides are not all equal.

(v) Concave polygon

Irregular polygon: its sides and angles are not all equal.

(vi) Octagon

Regular polygon: all sides and all angles equal.

2. Identify convex and concave polygons.

(i) Hexagon

Convex polygon: all its interior angles are less than 180 degrees and it has no dent.

(ii) Octagon

Convex polygon: all its interior angles are less than 180 degrees.

(iii) Heptagon

Convex polygon: all its interior angles are less than 180 degrees.

(iv) Arrowhead (dart)

Concave polygon: it has a reflex interior angle (greater than 180 degrees) at the dent.

(v) Polygon with a dent

Concave polygon: it has a reflex interior angle at the dent.

(vi) Polygon with a dent

Concave polygon: it has a reflex interior angle at the dent.

3. Differentiate between convex and concave polygons with examples.

A convex polygon has all interior angles less than 180 degrees, no

dent, and all its diagonals lie inside it; examples are a square, a regular pentagon and a regular hexagon. A concave polygon has at least one interior angle greater than 180 degrees (a reflex angle), a dent pushed inward, and at least one diagonal lying outside it; examples are an arrowhead (dart) and the dented polygons in Question 2 (iv)-(vi).

4. Calculate the sum of interior angles and each interior angle of a regular polygon.

(i) Triangle (regular, 3 sides)

Sum of interior angles = (n - 2) x 180 = (3 - 2) x 180 = 180 degrees.

Each interior angle = 180/3 = 60 degrees.

Sum = 180 degrees, each angle = 60 degrees.

(ii) Rectangle (4 sides)

Sum of interior angles = (4 - 2) x 180 = 360 degrees.

Each interior angle = 360/4 = 90 degrees.

Sum = 360 degrees, each angle = 90 degrees. Flag: a rectangle is not a regular polygon unless it is a square (its sides need not be equal); its angles are still all 90 degrees, so the same numbers apply.

(iii) Heptagon (regular, 7 sides)

Sum of interior angles = (7 - 2) x 180 = 5 x 180 = 900 degrees.

Each interior angle = 900/7 = 128.57 degrees (approximately).

Sum = 900 degrees, each angle = 128.57 degrees (approximately).

(iv) Octagon (regular, 8 sides)

Sum of interior angles = (8 - 2) x 180 = 6 x 180 = 1080 degrees.

Each interior angle = 1080/8 = 135 degrees.

Sum = 1080 degrees, each angle = 135 degrees.

5. Define convex and concave polygon.

A convex polygon is a polygon in which all interior angles are less than 180 degrees, so that the whole polygon bulges outward and every diagonal lies inside it. A concave polygon is a polygon in which at least one interior angle is greater than 180 degrees (a reflex angle), so that part of the polygon is pushed inward (a dent) and at least one diagonal lies outside it.

6. Calculate the sum of interior angles and each interior angle of the following regular polygons.

(i) 9 sides (regular nonagon)

Sum = (9 - 2) x 180 = 7 x 180 = 1260 degrees.

Each interior angle = 1260/9 = 140 degrees.

Sum = 1260 degrees, each angle = 140 degrees.

(ii) 12 sides (regular dodecagon)

Sum = (12 - 2) x 180 = 10 x 180 = 1800 degrees.

Each interior angle = 1800/12 = 150 degrees.

Sum = 1800 degrees, each angle = 150 degrees.

(iii) 20 sides (regular icosagon)

Sum = (20 - 2) x 180 = 18 x 180 = 3240 degrees.

Each interior angle = 3240/20 = 162 degrees.

Sum = 3240 degrees, each angle = 162 degrees.

Exercise 9.3 (p.200) - Terms of the Circle

1. Match the names of terms of the circle with the given figure.

In the given figure: the centre is the point C. The diameter is the line segment AB passing through the centre C and joining two points of the circle. A radius is the line segment from the centre C to any point on the circle, for example CA or CB. The circumference is the whole outer boundary (perimeter) of the circle. A chord is a line segment joining two points of the circle; in the figure the vertical segment to the right of the centre is a chord. The minor segment is the smaller region cut off by that chord (the shaded region in the figure). The major segment is the larger remaining region. A semi-circle is half of the circle, cut by the diameter AB.

2. Describe the following terms of the circle with the help of figures.

i. Circle

A circle is the set of all points in a plane that are at the same fixed distance from a fixed point.

ii. Centre of the circle

The centre is the fixed point inside the circle from which every point of the circle is at the same distance.

iii. Radius

The radius is the line segment (or its length) joining the centre to any point on the circle. All radii of the same circle are equal.

iv. Diameter

The diameter is a chord passing through the centre; it joins two points of the circle through the centre and is twice the radius.

v. Semi-circle

A semi-circle is half of a circle, formed when a diameter divides the circle into two equal halves.

i. Arc

An arc is a part of the circumference of a circle between two points on the circle.

ii. Segment

A segment is the region of a circle cut off by a chord: the region

between the chord and its arc.

iii. Minor segment

The minor segment is the smaller of the two regions cut off by a chord.

iv. Major segment

The major segment is the larger of the two regions cut off by a chord.

Unit 10: Perimeter, Area and Volume

Note: throughout Unit 10, pi is taken as 22/7, as in the book's worked examples. Flag: the book's Example 10.15 (p.220) contains print errors (its surface-area line uses 12 x 6 instead of 12 x 7, and its volume line is arithmetically wrong); the exercise solutions below use the correct formulas and arithmetic.

Exercise 10.1 (p.205) - Perimeter and Area of Composite Figures

(i) Shaded square of side 12 m inside a 20 m square frame: find the perimeter and area of the shaded portion.

Perimeter = 4 x side = 4 x 12 = 48 m.

Area = side x side = 12 x 12 = 144 m^2.

Perimeter = 48 m, area = 144 m^2.

(ii) L-shaped figure (top bar 10 cm wide and 5 cm tall; stem 5 cm wide; total height 15 cm): find the perimeter and area of the shaded portion.

The stem height is 15 - 5 = 10 cm.

Perimeter (going around the outline): 10 + 15 + 5 + 10 + 5 + 5 = 50 cm.

Area = area of top bar + area of stem = (10 x 5) + (5 x 10) = 50 + 50 = 100 cm^2.

Perimeter = 50 cm, area = 100 cm^2.

(iii) Two shaded diamonds of side 4 cm inside a 20 cm by 12 cm rectangle: find the perimeter and area of the shaded portion.

Perimeter of one diamond = 4 x 4 = 16 cm; two diamonds = 32 cm.

Area of one diamond (taken as a square of side 4 cm, as labelled) = 4 x 4 = 16 cm^2; two diamonds = 32 cm^2.

Perimeter = 32 cm, area = 32 cm^2. Flag: the figure is not drawn to scale (two 4 cm diamonds cannot span the 20 cm frame); the area is computed from the labelled side length.

Exercise 10.2 (p.211) - Circumference and Area of Circles

Circle of radius 7 cm

Circumference = 2 x (22/7) x 7 = 44 cm.

Area = (22/7) x 7 x 7 = 154 cm^2.

Circumference = 44 cm, area = 154 cm^2.

Circle of radius 21 m

Circumference = 2 x (22/7) x 21 = 132 m.

Area = (22/7) x 21 x 21 = 1386 m^2.

Circumference = 132 m, area = 1386 m^2.

Circle of radius 35 m

Circumference = 2 x (22/7) x 35 = 220 m.

Area = (22/7) x 35 x 35 = 3850 m^2.

Circumference = 220 m, area = 3850 m^2.

Circle of radius 10.5 cm

Circumference = 2 x (22/7) x 10.5 = 66 cm.

Area = (22/7) x 10.5 x 10.5 = 346.5 cm^2.

Circumference = 66 cm, area = 346.5 cm^2.

Circle of radius 56 m

Circumference = 2 x (22/7) x 56 = 352 m.

Area = (22/7) x 56 x 56 = 9856 m^2.

Circumference = 352 m, area = 9856 m^2.

Circle of radius 70 cm

Circumference = 2 x (22/7) x 70 = 440 cm.

Area = (22/7) x 70 x 70 = 15400 cm^2.

Circumference = 440 cm, area = 15400 cm^2.

Exercise 10.3 (p.217) - Cylinders

(i) Cylinder of length 36 cm and diameter 2.8 cm: find the total surface area and volume.

Radius r = 2.8/2 = 1.4 cm, height h = 36 cm.

Total surface area = 2 x (22/7) x r x (r + h) = 2 x (22/7) x 1.4 x 37.4 = 8.8 x 37.4 = 329.12 cm^2.

Volume = (22/7) x r^2 x h = (22/7) x 1.96 x 36 = 221.76 cm^3.

Total surface area = 329.12 cm^2, volume = 221.76 cm^3.

(ii) Cylinder of height 18 cm and diameter 9 cm: find the total surface area and volume.

Radius r = 9/2 = 4.5 cm (the 9 cm arrow spans the full width of the cylinder, so it is the diameter), height h = 18 cm.

Total surface area = 2 x (22/7) x 4.5 x (4.5 + 18) = (198/7) x 22.5 = 636.43 cm^2.

Volume = (22/7) x 4.5 x 4.5 x 18 = 8019/7 = 1145.57 cm^3.

Total surface area = 636.43 cm^2, volume = 1145.57 cm^3.

(iii) Cylinder of radius 0.7 cm and height 5.6 cm: find the total surface area and volume.

Total surface area = 2 x (22/7) x 0.7 x (0.7 + 5.6) = 4.4 x 6.3 =

27.72 cm^2.

Volume = (22/7) x 0.7 x 0.7 x 5.6 = 1.54 x 5.6 = 8.624 cm^3.

Total surface area = 27.72 cm^2, volume = 8.624 cm^3.

Exercise 10.4 (p.220) - Right Triangular Prisms

Formulas used: total surface area = b x h + l x (b + h + s), where b and h are the base and height of the triangular face, s is its slant side and l is the prism length; volume = (1/2) x b x h x l.

(i) Right triangular prism with b = 8 cm, h = 6 cm, s = 10 cm, l = 4 cm: find the surface area and volume.

Check: 8^2 + 6^2 = 64 + 36 = 100 = 10^2, so the triangular face is right-angled.

Surface area = (8 x 6) + 4 x (8 + 6 + 10) = 48 + 96 = 144 cm^2.

Volume = (1/2) x 8 x 6 x 4 = 96 cm^3.

Surface area = 144 cm^2, volume = 96 cm^3.

(ii) Right triangular prism with b = 12 cm, h = 5 cm, s = 13 cm, l = 6 cm: find the surface area and volume.

Check: 12^2 + 5^2 = 144 + 25 = 169 = 13^2, so the triangular face is right-angled.

Surface area = (12 x 5) + 6 x (12 + 5 + 13) = 60 + 180 = 240 cm^2.

Volume = (1/2) x 12 x 5 x 6 = 180 cm^3.

Surface area = 240 cm^2, volume = 180 cm^3.

2. A right triangular mosquito net (camping tent) made of nylon is 8 m long. The lengths of the triangular face are 3 m, 4 m and 5 m. Find the total surface area and volume of the tent.
Check: 3^2 + 4^2 = 9 + 16 = 25 = 5^2, so the triangular face is right-angled with base 3 m and height 4 m.

Area of one triangular face = (1/2) x 3 x 4 = 6 m^2.

Total surface area = 2 x 6 + 8 x (3 + 4 + 5) = 12 + 96 = 108 m^2.

Volume = 6 x 8 = 48 m^3.

Total surface area = 108 m^2, volume = 48 m^3. Flag: the figure labels one side '4cm'; the question text gives 4 m, which is used.

Exercise 10.5 (p.225) - Area Conversions
Conversion facts: 1 m^2 = 10000 cm^2; 1 cm^2 = 100 mm^2; 1 m^2 = 1000000 mm^2.
13 m^2 to cm^2
By the conversion facts, 13 m^2 to cm^2 = 130000 cm^2.

130000 cm^2.

14 m^2 to mm^2

By the conversion facts, 14 m^2 to mm^2 = 14000000 mm^2.

14000000 mm^2.

35 cm^2 to mm^2

By the conversion facts, 35 cm^2 to mm^2 = 3500 mm^2.

3500 mm^2.

6000 cm^2 to m^2

By the conversion facts, 6000 cm^2 to m^2 = 0.6 m^2.

0.6 m^2.

900000 mm^2 to m^2

By the conversion facts, 900000 mm^2 to m^2 = 0.9 m^2.

0.9 m^2.

160 mm^2 to cm^2

By the conversion facts, 160 mm^2 to cm^2 = 1.6 cm^2.

1.6 cm^2.

16 m^2 to cm^2

By the conversion facts, 16 m^2 to cm^2 = 160000 cm^2.

160000 cm^2.

19 cm^2 to mm^2

By the conversion facts, 19 cm^2 to mm^2 = 1900 mm^2.

1900 mm^2.

4500000 cm^2 to m^2

By the conversion facts, 4500000 cm^2 to m^2 = 450 m^2.

450 m^2.

3400000000 mm^2 to m^2

By the conversion facts, 3400000000 mm^2 to m^2 = 3.4 m^2.

3.4 m^2.

7200 mm^2 to cm^2

By the conversion facts, 7200 mm^2 to cm^2 = 72 cm^2.

72 cm^2.

Review Exercise 10 (pp.226-227)

1. Tick the correct option.

i. The total length of the boundary of a circle is called:

- area
- ✓ circumference
- diameter

volume

ii. The symbol of pi is:

A

d

✓ pi

r

iii. Circumference of a circle is:

pi x r

✓ 2 x pi x r

pi x r^2

2 x pi x r^2

iv. The area of the circular region is:

pi x r

2 x pi x r

✓ pi x r^2

2 x pi x r^2

v. The surface area of a cylinder is:

✓ 2 x pi x r x (r + h)

2 x pi x (r + h)

2 x pi x r x h

pi x r x h

vi. The volume of a cylinder is:

2 x pi x r x (r + h)

2 x pi x (r + h)

2 x pi x r x h

✓ pi x r^2 x h

vii. 16 m^2 is equal to:

160 cm^2

1600 cm^2

16000 cm^2

✓ 160000 cm^2

viii. 26000000 mm^2 is equal to:

2.6 m^2

✓ 26 m^2

260 m^2

2600 m^2

ix. Surface area of a right triangular prism is:

✓ b x h + l x (b + h + s)

b x h + (b + h + l)

b x h + l x (b + h + l)

b x h + (b + h + s)

Flag: Review Exercise 10 Q1 is numbered (i)-(vii), then (vii) again, then (ix); the second (vii) is treated as (viii) above. Stray fractions 1/2 and 1/3 printed under options (a) and (c) of Q1 (ix) are a layout quirk and are ignored; the formula in option (a) matches the unit's formula.

2. Find the total surface area and volume of a plastic pipe whose

diameter is 2.1 m and length is 7 m.

Radius r = 2.1/2 = 1.05 m, height (length) h = 7 m.

Total surface area = 2 x (22/7) x 1.05 x (1.05 + 7) = 6.6 x 8.05 = 53.13 m^2.

Volume = (22/7) x 1.05 x 1.05 x 7 = 3.465 x 7 = 24.255 m^3.

Total surface area = 53.13 m^2, volume = 24.255 m^3.

3. Find the perimeter of the shaded portion and the area of the unshaded portion of the given composite figure (outer square 4 cm, inner unshaded square 2 cm).

The shaded portion is the border between the two squares; its perimeter is the outer perimeter plus the inner perimeter = (4 x 4) + (4 x 2) = 16 + 8 = 24 cm.

Area of the unshaded (inner) square = 2 x 2 = 4 cm^2.

Perimeter of shaded portion = 24 cm, area of unshaded portion = 4 cm^2. (The shaded area itself is 16 - 4 = 12 cm^2.)

4. Find the circumference and area of the given circle (radius 3.6 cm).

Circumference = 2 x (22/7) x 3.6 = 158.4/7 = 22.63 cm (approximately).

Area = (22/7) x 3.6 x 3.6 = 285.12/7 = 40.73 cm^2

(approximately).

Circumference = 22.63 cm (approximately), area = 40.73 cm^2 (approximately).

5. Find the circumference of the shaded region and the area of the unshaded region (shaded circle of radius 7 cm inside a square).
Circumference of the shaded circle = 2 x (22/7) x 7 = 44 cm.

The square side equals the circle diameter = 14 cm, so the square area = 14 x 14 = 196 cm^2.

Circle area = (22/7) x 7 x 7 = 154 cm^2.

Unshaded area = 196 - 154 = 42 cm^2.

Circumference of shaded region = 44 cm, area of unshaded region = 42 cm^2.

6. Find the surface area and volume of the given right prism (b = 12 cm, h = 9 cm, s = 15 cm, l = 3 cm).
Check: 12^2 + 9^2 = 144 + 81 = 225 = 15^2, so the triangular face is right-angled.

Surface area = (12 x 9) + 3 x (12 + 9 + 15) = 108 + 108 = 216 cm^2.

Volume = (1/2) x 12 x 9 x 3 = 162 cm^3.

Surface area = 216 cm^2, volume = 162 cm^3.

7. Farooq made a right triangular plastic tent as shown in the figure. Find the volume and surface area of this tent (b = 16 m, h = 12 m, s = 20 m, l = 4 m).
Check: 16^2 + 12^2 = 256 + 144 = 400 = 20^2, so the triangular face is right-angled.

Surface area = (16 x 12) + 4 x (16 + 12 + 20) = 192 + 192 = 384 m^2.

Volume = (1/2) x 16 x 12 x 4 = 384 m^3.

Surface area = 384 m^2, volume = 384 m^3.

8. Convert.
8 m^3 to cm^3 and mm^3
By the conversion facts (1 m^3 = 1000000 cm^3 = 1000000000 mm^3), 8 m^3 to cm^3 and mm^3 = 8000000 cm^3 and 8000000000 mm^3.

8000000 cm^3 and 8000000000 mm^3.

900000 mm^2 to cm^2 and m^2
By the conversion facts (1 m^3 = 1000000 cm^3 = 1000000000 mm^3), 900000 mm^2 to cm^2 and m^2 = 9000 cm^2 and 0.9 m^2.

9000 cm^2 and 0.9 m^2.

8 m^2 to cm^2

By the conversion facts (1 m^3 = 1000000 cm^3 = 1000000000 mm^3), 8 m^2 to cm^2 = 80000 cm^2.

80000 cm^2.

25 m^2 to mm^2

By the conversion facts (1 m^3 = 1000000 cm^3 = 1000000000 mm^3), 25 m^2 to mm^2 = 25000000 mm^2.

25000000 mm^2.

5 m^3 to cm^3 and mm^3

By the conversion facts (1 m^3 = 1000000 cm^3 = 1000000000 mm^3), 5 m^3 to cm^3 and mm^3 = 5000000 cm^3 and 5000000000 mm^3.

5000000 cm^3 and 5000000000 mm^3.

Unit 11: Practical Geometry and Symmetry

Exercise 11.1 (p.233) - Angles in Triangles

1. Calculate the unknown angles in the following.

(i) Right triangle ABC, right angle at B, interior angle at C = 45 degrees, exterior angle at C = x, angle at A = a.

Exterior angle at C: x = 180 - 45 = 135 degrees.

Angle a = 180 - 90 - 45 = 45 degrees.

a = 45 degrees, x = 135 degrees.

(ii) Triangle ABC, angle at A = 50 degrees, interior angle at C = 65 degrees, right angle at B = b, exterior angle at C = x.

Angle b is marked with a right-angle square, so b = 90 degrees.

Exterior angle at C: x = 180 - 65 = 115 degrees.

b = 90 degrees, x = 115 degrees. Flag: as printed, the three interior angles sum to 50 + 65 + 90 = 205 degrees, which is impossible for a triangle, so the figure is inconsistent. The exterior angle above follows the marked interior angle at C (65 degrees); if instead the right angle and 50 degrees are taken as correct, the interior angle at C would be 40 degrees and x would be 140 degrees.

(iii) Triangle ABC with angle at A = 50 degrees, angle at B = 80 degrees, interior angle at C = c, exterior angle at C = x.
Interior angle c = 180 - 50 - 80 = 50 degrees.

Exterior angle x = 180 - 50 = 130 degrees.

c = 50 degrees, x = 130 degrees.

(iv) Triangle with angle at B = 30 degrees, interior angle at C = 30 degrees, angle at A = y.
The angles of a triangle sum to 180 degrees, so y = 180 - 30 - 30 = 120 degrees.

y = 120 degrees.

(v) Triangle ABZ with all three interior angles marked 'a', and exterior angles x (at B), y (at A) and c (at Z).
All three interior angles are marked equal, so each is a = 180/3 = 60 degrees.

Each exterior angle is 180 - 60 = 120 degrees.

a = 60 degrees, x = y = c = 120 degrees.

2. Calculate the unknown angles in a triangle.
(i) Isosceles triangle ABC with AB = AC (marked equal) and angle at B = 60 degrees.
Base angles of an isosceles triangle are equal, so angle at C =

angle at B = 60 degrees.

Angle at A = 180 - 60 - 60 = 60 degrees.

Angles: A = 60 degrees, B = 60 degrees, C = 60 degrees (the triangle is equilateral in effect).

(ii) Right triangle XYZ, right angle at Y, angle at X = 50 degrees, angle at Z = x.
x = 180 - 90 - 50 = 40 degrees.

x = 40 degrees.

(iii) Triangle with angle at the top vertex = 30 degrees and angle at the bottom-right vertex = 20 degrees; find the angle at the bottom-left vertex.
The unknown angle = 180 - 30 - 20 = 130 degrees.

The bottom-left angle = 130 degrees (obtuse).

(iv) Right triangle PQR, right angle at Q, angle at R = 60 degrees; find the angle at P.
Angle at P = 180 - 90 - 60 = 30 degrees.

Angle at P = 30 degrees.

(v) Triangle with angle at the top vertex = 40 degrees, angle at the right vertex = 110 degrees, interior angle at C = c degrees, exterior

angle at C = x. (Printed as a second item (ii).)

Interior angle c = 180 - 40 - 110 = 30 degrees.

Exterior angle x = 180 - 30 = 150 degrees.

c = 30 degrees, x = 150 degrees. Flag: this item is printed with the number (ii) a second time.

3. Calculate the unknown angles in a triangle.

(i) Triangle ABC with AB = BC = CA = 3 cm and all angles marked x.

All three sides are equal, so the triangle is equilateral and all angles are equal: x = 180/3 = 60 degrees.

Each angle x = 60 degrees.

(ii) Isosceles triangle PQR with PR = PQ = 4 cm, RQ = 3 cm, vertex angle at P = 40 degrees, base angles marked x.

Base angles are equal: x = (180 - 40)/2 = 140/2 = 70 degrees.

Each base angle x = 70 degrees.

(iii) Right triangle ABC, right angle at C, angle at B = 2x, angle at A = x.

x + 2x + 90 = 180, so 3x = 90 and x = 30 degrees.

Angle at B = 2x = 60 degrees.

x = 30 degrees, angle at B = 60 degrees.

Exercise 11.2 (p.237) - Construction of Triangles

General method: draw the given side with a ruler, construct the given angles with a compass (or mark the given side lengths with arcs from the endpoints), and join the intersection point to the endpoints.

1. Construct an equilateral triangle PQR when PQ = QR = RP = 4 cm.

Draw PQ = 4 cm. With the compass open to 4 cm, draw an arc from P and an arc from Q; they intersect at R. Join PR and QR. Triangle PQR is equilateral with all sides 4 cm.

2. Construct an equilateral triangle ABC when AB = BC = CA = 5 cm.

Draw AB = 5 cm. With the compass open to 5 cm, draw an arc from A and an arc from B; they intersect at C. Join AC and BC. Triangle ABC is equilateral with all sides 5 cm.

3. Construct an isosceles triangle XYZ when XZ = YZ = 4 cm and XY = 3 cm.

Draw XY = 3 cm. With the compass open to 4 cm, draw an arc from X and an arc from Y; they intersect at Z. Join XZ and YZ. Triangle XYZ is isosceles with XZ = YZ = 4 cm.

4. Construct an isosceles triangle ABC when AB = 4 cm, angle A = 30 degrees and angle B = 30 degrees.

Draw AB = 4 cm. Construct an angle of 30 degrees at A and an angle of 30 degrees at B with the compass. The two arms intersect

at C. Triangle ABC is isosceles with AC = BC and angle at C = 180 - 30 - 30 = 120 degrees.

5. Construct a scalene triangle PQR when PQ = 5 cm, angle P = 120 degrees and PR = 6 cm.
Draw PQ = 5 cm. Construct an angle of 120 degrees at P with the compass. Mark R on the arm so that PR = 6 cm. Join QR. Triangle PQR is scalene (all sides different).

6. Construct a scalene triangle XYZ when XY = 6 cm, XZ = 5 cm and YZ = 7 cm.
Draw XY = 6 cm. With the compass open to 5 cm, draw an arc from X; with the compass open to 7 cm, draw an arc from Y; they intersect at Z. Join XZ and YZ. Triangle XYZ is scalene.

7. Construct an acute-angled triangle PQR when PQ = 4 cm, angle P = 45 degrees and angle Q = 60 degrees.
Draw PQ = 4 cm. Construct an angle of 45 degrees at P and an angle of 60 degrees at Q with the compass. The two arms intersect at R. Angle at R = 180 - 45 - 60 = 75 degrees, so all angles are acute and triangle PQR is acute-angled.

Exercise 11.3 (p.242) - Lines of Symmetry

1. Identify and draw a line of symmetry of the following figures.
Green parallelogram
No line of symmetry: a general parallelogram (not a rectangle or rhombus) cannot be folded onto itself along any line.

Letter D

One line of symmetry: a horizontal line through the middle of the letter, dividing it into identical upper and lower halves.

Division symbol (a horizontal bar with one dot above and one dot below)

Two lines of symmetry: one vertical line through the middle and one horizontal line through the middle.

Wheel (circle with spokes)

Infinitely many lines of symmetry: any line through the centre of the circle divides it into two identical halves.

Cat face (photograph)

One line of symmetry: a vertical line through the middle of the face, between the eyes and ears.

Aeroplane (top view)

One line of symmetry: a vertical line along the fuselage, dividing the wings into identical left and right halves.

Light bulb

One line of symmetry: a vertical line through the centre of the bulb and its base.

Exercise 11.4 (p.248) - Translation

(i) Triangle ABC with A(-7, 3), B(-5, 7), C(-3, 3) is translated to A'(4, -7), B'(6, -3), C'(8, -7). Describe the translation and write the

coordinate equation.

A to A': x changes from -7 to 4 (11 units right), y changes from 3 to -7 (10 units down).

Check B: -5 + 11 = 6 and 7 - 10 = -3. Check C: -3 + 11 = 8 and 3 - 10 = -7.

The triangle is translated 11 units to the right and 10 units down. Coordinate equation of translation: (x, y) -> (x + 11, y - 10).

(ii) Parallelogram ABCD with A(3, 6), B(8, 6), C(2, 3), D(7, 3) is translated to the parallelogram in the lower grid. Describe the translation and write the coordinate equation.

A(3, 6) to A'(-7, -5): x changes by -10 (10 units left), y changes by -11 (11 units down).

Check B(8, 6) to B'(-2, -5): 8 - 10 = -2 and 6 - 11 = -5. Check the bottom vertices: (2, 3) to (-8, -8) and (7, 3) to (-3, -8), both also 10 left and 11 down.

The parallelogram is translated 10 units to the left and 11 units down. Coordinate equation of translation: (x, y) -> (x - 10, y - 11).

Flag: in the printed image the bottom vertices are labelled D' (bottom-left) and C' (bottom-right), swapped relative to the original C (bottom-left) and D (bottom-right); the translation itself is consistent for all four vertices.

Review Exercise 11 (pp.252-253)

Flag: the review is printed with the heading 'Review Exercise 4'; it is the Unit 11 review (it follows Unit 11 and precedes Unit 12).

1. Tick the correct option.

i. The equilateral triangle is:

- ✓ triangle with all three sides marked equal (double ticks)
- triangle with two sides double-ticked and the base single-ticked
- triangle with all sides marked differently
- right triangle with a 30 degree angle

ii. The acute-angled triangle is:

- ✓ equilateral triangle (all angles 60 degrees)
- right triangle with a 30 degree angle
- obtuse triangle
- parallelogram (not a triangle)

iii. The isosceles triangle is:

- triangle with all three sides marked equal
- ✓ triangle with two sides double-ticked and the base single-ticked
- triangle with all sides marked differently
- right triangle with a 30 degree angle

iv. The right-angled triangle is:

- equilateral triangle
- ✓ right triangle with a right-angle mark and a 30 degree angle
- obtuse triangle
- parallelogram (not a triangle)

v. The scalene triangle is:

triangle with all three sides marked equal

triangle with two sides double-ticked and the base single-ticked

✓ triangle with all sides marked differently (single, double and triple ticks)

right triangle with a 45 degree angle

vi. The obtuse-angled triangle is:

equilateral triangle

isosceles triangle

✓ very flat triangle with one obtuse angle

right triangle with a 45 degree angle

vii. The symmetrical figure is:

✓ isosceles triangle (two equal sides)

parallelogram

scalene obtuse triangle

isosceles trapezium

viii. The non-symmetrical figure is:

isosceles triangle

rectangle

isosceles trapezium

✓ general parallelogram (slanted quadrilateral)

ix. The symmetrical figure is (letters):

✓ B

F

G

J

x. The symmetrical figure is (objects):

mug with a handle on one side

✓ aeroplane (top view)

pressure cooker

spiral bulb

Flag: Review Q1 (vii) has two symmetrical options as drawn: the isosceles triangle (a) and the isosceles trapezium (d) both have a line of symmetry. Option (a) is ticked as the primary answer; (d) is also symmetrical.

2. Identify and draw a line of symmetry of the following figures.

Star (sun-like figure with pointed rays)

Many lines of symmetry: one line through the centre and each point of the star (a vertical line, a horizontal line and the diagonal lines through opposite points).

Letter W

One line of symmetry: a vertical line through the middle of the letter.

Rectangle

Two lines of symmetry: one vertical line through the middle and one horizontal line through the middle.

Ring (green circle with a circular hole)

Infinitely many lines of symmetry: any line through the centre of

the ring divides it into two identical halves.

Strawberry

One line of symmetry: a vertical line through the middle of the strawberry, from the leaves to the tip.

Bottle

One line of symmetry: a vertical line through the centre of the bottle.

Aeroplane (top view)

One line of symmetry: a vertical line along the fuselage.

Egg

One line of symmetry: a vertical line through the centre of the egg, from top to bottom.

Unit 12: Data Management and Probability

Exercise 12.1 (p.266) - Bar Graph

Note: the bar heights in Exercise 12.1 fall between labelled gridlines, so the numbers below are readings from the graph (to the nearest 100 students), not exact printed values. The question text spells the spots 'Kharwari-Baba' and 'Hannah Urak', while the graph labels them 'Kharwari' and 'Hanna Urk'.

1. What is the title of the vertical bar graph?

The visit of favorite picnic spot of students.

2. Which picnic spot was visited by the maximum number of students?

Ziarat, with about 8600 students (the tallest bar).

3. Which picnic spot was visited by the minimum number of students?

Kharwari, with about 4500 students (the shortest bar).

4. How many students visited Kharwari-Baba?

About 4500 students (the Kharwari bar reaches about halfway between 4000 and 5000).

5. How many students visited Hanna Urak?

About 6600 students (the Hanna Urk bar reaches about two-thirds of the way between 6000 and 7000).

Other bars in the graph (for reference).

Peer Ghaib: about 7500 students. Hanna Lake: about 8000 students.

Draw a simple conclusion from the graph.

Ziarat is the students' favourite picnic spot and Kharwari the least visited; most students prefer the Ziarat and Hanna Lake spots, while far fewer visit Kharwari.

Exercise 12.2 (p.272) - Line Graph

1. What is the title of the line graph?

Monsoon rainfall in Barkhan.

2. In which month is the rainfall maximum?

September, with 30 mm of rainfall.

3. In which month is the rainfall minimum?

June, with 5 mm of rainfall.

4. In August, what is the measure of rainfall?

25 mm.

5. Draw a simple conclusion from the given graph.

The monsoon rainfall in Barkhan increased steadily through the season: 5 mm in June, 15 mm in July, 25 mm in August and 30 mm in September.

Exercise 12.3 (p.280) - Discrete, Continuous, Grouped and Ungrouped Data

1. Identify the discrete and continuous data in the following.

(i) The height of five students (in feet): 5.3, 5.4, 3.5, 5.6, 6.1.

Continuous data: heights are measured quantities that can take decimal (fractional) values.

(ii) The top 5 students' marks in Maths test out of 50: 49, 20, 48, 40, 45.

Discrete data: marks are whole-number counts that cannot be subdivided into fractions.

(iii) Waqar's wickets in one-day cricket matches: 7, 9, 4, 5.

Discrete data: wickets are whole-number counts. Flag: the question says 'last five' matches but only four values are printed.

(iv) The number of rooms in 6 students' homes: 6, 7, 8, 2, 9, 3.

Discrete data: numbers of rooms are whole-number counts.

(v) The weights of 4 teachers (in kg): 74.4, 80.5, 84.5, 90.6.

Continuous data: weights are measured quantities that can take decimal values.

(vi) The number of pages in different notebooks of Grade VII: 230, 120, 180, 80.

Discrete data: numbers of pages are whole-number counts.

2. Identify the grouped and ungrouped data.

(i) Grade VII, top 5 students' marks in Geography test out of 50 marks: 40, 35, 45, 36, 39.
Ungrouped data: the individual marks are listed directly, not arranged in class intervals.

(ii) Grade VII students' marks in Science test out of 40 marks, given in the table (1-10: 4, 11-20: 1, 21-30: 3, 31-40: 2).
Grouped data: the marks are arranged in class intervals with frequencies.

(iii) The result of the mid term Science paper of Grade VII.
Flag: no data is printed for this item in the book, so it cannot be identified as grouped or ungrouped from the page.

(iv) The result of the mid term of Grade VII.
Flag: no data is printed for this item in the book, so it cannot be identified as grouped or ungrouped from the page.

(v) Final assessment report of Musfira.
Flag: no data is printed for this item in the book, so it cannot be identified as grouped or ungrouped from the page.

3. Write the difference between continuous and discrete data with the help of three examples.
Discrete data takes only whole-number, countable values that cannot be subdivided: for example, the number of students in a class (50), the number of wickets in a match (4), and marks out of 50 (49). Continuous data takes any value within a range, including

decimals, because it comes from measurement: for example, the height of a student (5.3 feet), the weight of a teacher (74.4 kg), and the rainfall in a month (25.5 mm).

Exercise 12.4 (p.288) - Histogram

1. Observe the given histogram, answer the following questions and make the frequency distribution table.

(i) What is the total number of class intervals?

12 class intervals.

(ii) What is the size of the class interval?

5 years (each interval covers 5 years of age).

(iii) What is the total number of visitors?

300 + 400 + 500 + 600 + 700 + 900 + 1000 + 900 + 800 + 600 + 400 + 100 = 7200 visitors.

(iv) Which class interval has maximum frequency?

30-35 years, with 1000 visitors.

(v) Which class interval has minimum frequency?

55-60 years, with 100 visitors.

(vi) Make the frequency distribution table.

Age (years): 0-5: 300; 5-10: 400; 10-15: 500; 15-20: 600; 20-25: 700; 25-30: 900; 30-35: 1000; 35-40: 900; 40-45: 800; 45-50: 600; 50-55: 400; 55-60: 100. Total: 7200.

2. Draw a histogram to represent the frequency distribution table of the weights (in kg) of employees of a departmental store (40-45: 5, 45-50: 8, 50-55: 12, 55-60: 13, 60-65: 14, 65-70: 10, 70-75: 6).

Draw the X-axis for weights (kg) and the Y-axis for frequency. Draw 7 bars of equal width 5 kg, side by side with no gaps between them, with heights 5, 8, 12, 13, 14, 10 and 6 for the intervals 40-45, 45-50, 50-55, 55-60, 60-65, 65-70 and 70-75 respectively. Title the histogram 'Weights of employees of a departmental store'.

Exercise 12.5 (p.295) - Mean, Median and Mode

1. (i) Calculate the mean, median and mode of 25, 30, 40, 35, 45, 50, 60.

Mean: sum = 25 + 30 + 40 + 35 + 45 + 50 + 60 = 285; mean = 285/7 = 40.71 (approximately).

Median: arrange in ascending order: 25, 30, 35, 40, 45, 50, 60; the middle value (4th of 7) is 40.

Mode: no value repeats, so there is no mode.

Mean = 40.71 (approximately), median = 40, no mode.

1. (ii) Calculate the mean, median and mode of 70, 72, 71, 72, 73, 74, 75.

Mean: sum = 70 + 72 + 71 + 72 + 73 + 74 + 75 = 507; mean = 507/7 = 72.43 (approximately).

Median: arrange in ascending order: 70, 71, 72, 72, 73, 74, 75; the

middle value is 72.

Mode: 72 occurs twice, more than any other value.

Mean = 72.43 (approximately), median = 72, mode = 72.

1. (iii) Calculate the mean, median and mode of 110, 120, 130, 140, 150, 160, 150, 160.

Mean: sum = 110 + 120 + 130 + 140 + 150 + 160 + 150 + 160 = 1120; mean = 1120/8 = 140.

Median: arrange in ascending order: 110, 120, 130, 140, 150, 150, 160, 160; the two middle values are 140 and 150, so the median is (140 + 150)/2 = 145.

Mode: 150 and 160 each occur twice, so there are two modes.

Mean = 140, median = 145, modes = 150 and 160.

2. Calculate the mean of Hassan's marks obtained in 8 subject tests: 5, 7, 8, 5, 6, 9, 10, 7.

Sum = 5 + 7 + 8 + 5 + 6 + 9 + 10 + 7 = 57.

Mean = 57/8 = 7.125.

Mean = 7.125.

Exercise 12.6 (p.304) - Probability

1. Explain the following terms with an example.

(i) Certain event

An event that is sure to happen: for example, the sun rising in the east tomorrow. Its probability is 1.

(ii) Impossible event

An event that can never happen: for example, rolling a 7 on an ordinary six-faced die. Its probability is 0.

(iii) Complement of an event

The complement of an event is the event that the original event does NOT happen: for example, if the event is 'getting a head' on a coin toss, its complement is 'getting a tail'. The probabilities of an event and its complement add to 1.

2. Categorize the following events (certain event, impossible event, complement of an event).

(i) The Pakistan football team win the toss against Iran.

An ordinary uncertain event: it may or may not happen. Its complement is 'the Pakistan football team do not win the toss'.

(ii) Afghanistan won the first one-day cricket match against Scotland.

An ordinary event (already decided in the past): its complement is 'Afghanistan did not win the match'.

(iii) Naseem won the gold medal in javelin throw in the Olympic games.

An ordinary event: its complement is 'Naseem did not win the gold medal'.

(iv) Sun is the main source of energy in the universe.
Certain event: it is always true.

(v) Brazil and Germany football match was tied in the world cup football tournament.
An ordinary event: its complement is 'the match was not tied'.

(vi) During rolling a die, Sidra missed the chance to roll the die.
Flag: no die outcome occurs here at all, so as an event of the die roll it is an impossible event (none of the outcomes 1-6 occurs).

(vii) Toss between India and Pakistan became tied.
Impossible event: a coin toss has only two outcomes, head or tail, and cannot end in a tie.

3. Muhammad Huzaifa is tossing a coin. Write the sample space, events and outcome for it.
Sample space S = {Head, Tail}. Events: getting a head {Head} and getting a tail {Tail}. Outcomes: Head or Tail (each toss gives one of these two outcomes).

4. Musfira is rolling a die. Write the sample space, events and outcomes for it.
Sample space S = {1, 2, 3, 4, 5, 6}. Events include: getting an even number {2, 4, 6}, getting an odd number {1, 3, 5}, and

getting a prime number {2, 3, 5}. Outcomes: 1, 2, 3, 4, 5 or 6.

5. Waseem and Waqar spin the given spinner (three equal sectors: red, green and blue). Find the probability of green and red colors.
The spinner has 3 equal sectors, so each colour has probability 1/3.

Probability of green = 1/3.

Probability of red = 1/3.

P(green) = 1/3, P(red) = 1/3.

6. Two friends Manahil and Alia tossed a coin. Alia chose the head and Manahil chose the tail. Find the probability.
Sample space = {Head, Tail}, two equally likely outcomes.

Probability that Alia (head) wins = 1/2.

Probability that Manahil (tail) wins = 1/2.

P(Alia wins) = 1/2, P(Manahil wins) = 1/2.

7. Three friends Naeem, Kaleem and Kareem rolled a die and got the following events: A = even numbers, B = prime numbers, C = not even nor prime. Find the probability of events A, B, C.
Sample space = {1, 2, 3, 4, 5, 6} (6 outcomes).

A = {2, 4, 6}, so P(A) = 3/6 = 1/2.

B = {2, 3, 5}, so P(B) = 3/6 = 1/2.

C = {1} (1 is the only number from 1 to 6 that is neither even nor prime), so P(C) = 1/6.

P(A) = 1/2, P(B) = 1/2, P(C) = 1/6.

Review Exercise 12 (pp.306-308)

1. Choose the correct option.

i. A graph in which data is represented by vertical or horizontal bars of equal width is called:

- picto-graph
- ✓ bar-graph
- pie-graph
- line graph

ii. The extension of a simple bar graph in which the data are given corresponding to a single quantity is called:

- pie-graph
- line graph
- ✓ multiple bar graph
- histogram

iii. Such a multiple bar graph in which we represent the multiple bars in vertical form is called:

- vertical bar-graph
- horizontal bar-graph

✓ vertical multiple bar-graph

horizontal multiple bar-graph

iv. Such a multiple bar graph in which we represent the multiple bars in horizontal form is called:

vertical bar-graph

horizontal bar-graph

vertical multiple bar-graph

✓ horizontal multiple bar-graph

v. A graph representing data in a circular form, with each slice or sector of the circle representing a proportionate part of the whole, is called:

picto-graph

bar-graph

✓ pie-graph

line-graph

vi. When a graph is drawn using lines only on specific points, it is called:

picto-graph

bar-graph

pie-graph

✓ line-graph

vii. The values/quantities that are whole numbers, finite, countable and cannot be subdivided into further parts are called:

✓ discrete data

continuous data

ungrouped data

grouped data

viii. The values/quantities that can be presented in fractions/decimals, are not countable, and can be subdivided into smaller parts according to measurement/precision are called:

discrete data

✓ continuous data

ungrouped data

grouped data

ix. Data collected directly, not arranged or processed, obtained first time from the field and not arranged, is called:

discrete data

continuous data

✓ ungrouped data

grouped data

x. Data obtained in the same order, processed or arranged in different groups, is called:

discrete data

continuous data

ungrouped data

✓ grouped data

xi. The data 1, 2, 3, 4, 4, 5, 6, 7, 8, 10: mean is:

4

✓ 5

6

8

xii. The data 1, 2, 3, 4, 5, 6, 7, 8, 9, 10: median is:

3.5

4.5

✓ 5.5

6.5

xiii. The data 1, 2, 3, 3, 4, 5, 5, 6, 6, 7, 8: mode is:

3

5

3, 5

✓ 3, 5, 6

xiv. The lower class limit of the class (31-35) is:

30.5

✓ 31

35

35.5

xv. An action which produces different results/outcomes is called:

✓ experiment

outcomes

sample space

event

xvi. A specific result of a random experiment is called:

- experiment
- ✓ outcomes
- sample space
- event

xvii. The set of all possible outcomes of a random experiment is called:

- experiment
- outcomes
- ✓ sample space
- event

xviii. Any subset of a sample space is called:

- experiment
- outcome
- sample space
- ✓ event

xix. An event which has no outcomes is called:

- ✓ impossible event
- sure event
- simple event
- compound event

xx. If you toss a coin, it can either land on head or not land on head. These two events are called:

impossible events

sure events

✓ simple events

compound events

Flags on Review Exercise 12 Q1: (xi) the exact mean of 1, 2, 3, 4, 4, 5, 6, 7, 8, 10 is 48/10 = 4.8, which is not among the options; the nearest option (b) 5 is ticked. (xx) 'head' and 'not head' are complementary events, but that term is not offered among the options; each of the two events contains exactly one outcome, so option (c) simple events is ticked. Q1 (ii) is worded confusingly ('single quantity'); option (c) multiple bar graph is the extension of a simple bar graph.

2. Construct a frequency table for the given data: 11, 12, 13, 11, 12, 14, 15, 13, 14, 15, 16, 14, 15, 14, 10, 10, 11, 10, 11, 12, 16, 14, 15, 16, 12, 11, 12, 13, 14, 15.

Count each value: 10 occurs 3 times, 11 occurs 5 times, 12 occurs 5 times, 13 occurs 3 times, 14 occurs 6 times, 15 occurs 5 times, 16 occurs 3 times.

Check: 3 + 5 + 5 + 3 + 6 + 5 + 3 = 30 values, matching the data.

Frequency table: 10: 3; 11: 5; 12: 5; 13: 3; 14: 6; 15: 5; 16: 3. Total = 30.

3. Hasnain's cricket team scored the following runs in a cricket match: 50, 30, 40, 25, 15, 35, 25, 20, 30, 30. Calculate the mean, median and mode.

Mean: sum = 50 + 30 + 40 + 25 + 15 + 35 + 25 + 20 + 30 + 30 = 300; mean = 300/10 = 30.

Median: arrange in ascending order: 15, 20, 25, 25, 30, 30, 30, 35, 40, 50; the two middle values are both 30, so the median is 30.

Mode: 30 occurs three times, more than any other value.

Mean = 30, median = 30, mode = 30.

4. Sidra rolled a die. Find the probability of getting (i) a number less than six, (ii) a number that is a factor of six, (iii) prime numbers.
Sample space = {1, 2, 3, 4, 5, 6} (6 outcomes).

(i) Numbers less than six: {1, 2, 3, 4, 5}, so probability = 5/6.

(ii) Factors of six: {1, 2, 3, 6}, so probability = 4/6 = 2/3.

(iii) Prime numbers: {2, 3, 5}, so probability = 3/6 = 1/2.

P(less than six) = 5/6, P(factor of six) = 2/3, P(prime) = 1/2.